

UC San Diego

JACOBS SCHOOL OF ENGINEERING

Electrical and Computer Engineering

From Algorithms to Cognitive Partners: AI-Mediated Communication and Human Intelligence

Songlin Xu

Background: AI helps humans everywhere.

ChatGPT for problem solving (i.e., writing)

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ChatGPT for problem solving (i.e., writing)

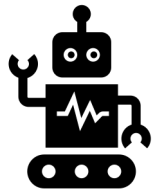
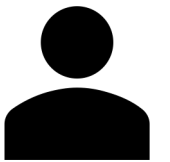
Polish my manuscript
below...



Background: AI helps humans everywhere.

ChatGPT for problem solving (i.e., writing)

Polish my manuscript
below...



Here is the polished file
below...



Background

AI solves humans' tasks,
rather than **teaching humans how to do better.**

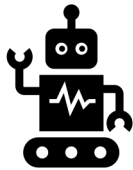
ChatGPT solves your tasks $\neq$ Improve your own intelligence^[1].



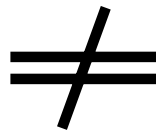
Student



Homework



ChatGPT



Learning Performance

Reference

[1]. Does Using ChatGPT Result in Human Cognitive Augmentation? International Conference on Human-Computer Interaction. 2025

Research Question

~~Asking AI to solve human tasks directly.~~

How to use AI to regulate human cognitive behaviors and augment human intelligence?

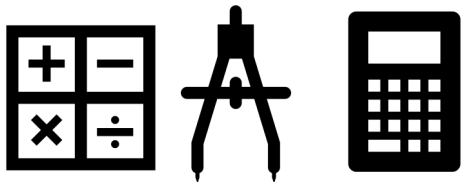
Research Question

~~Asking AI to solve human tasks directly.~~

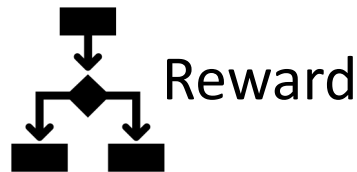
How to use AI to regulate human cognitive behaviors and augment human intelligence?

augment human cognition

Logical Reasoning



Decision Making



Planning



Attention

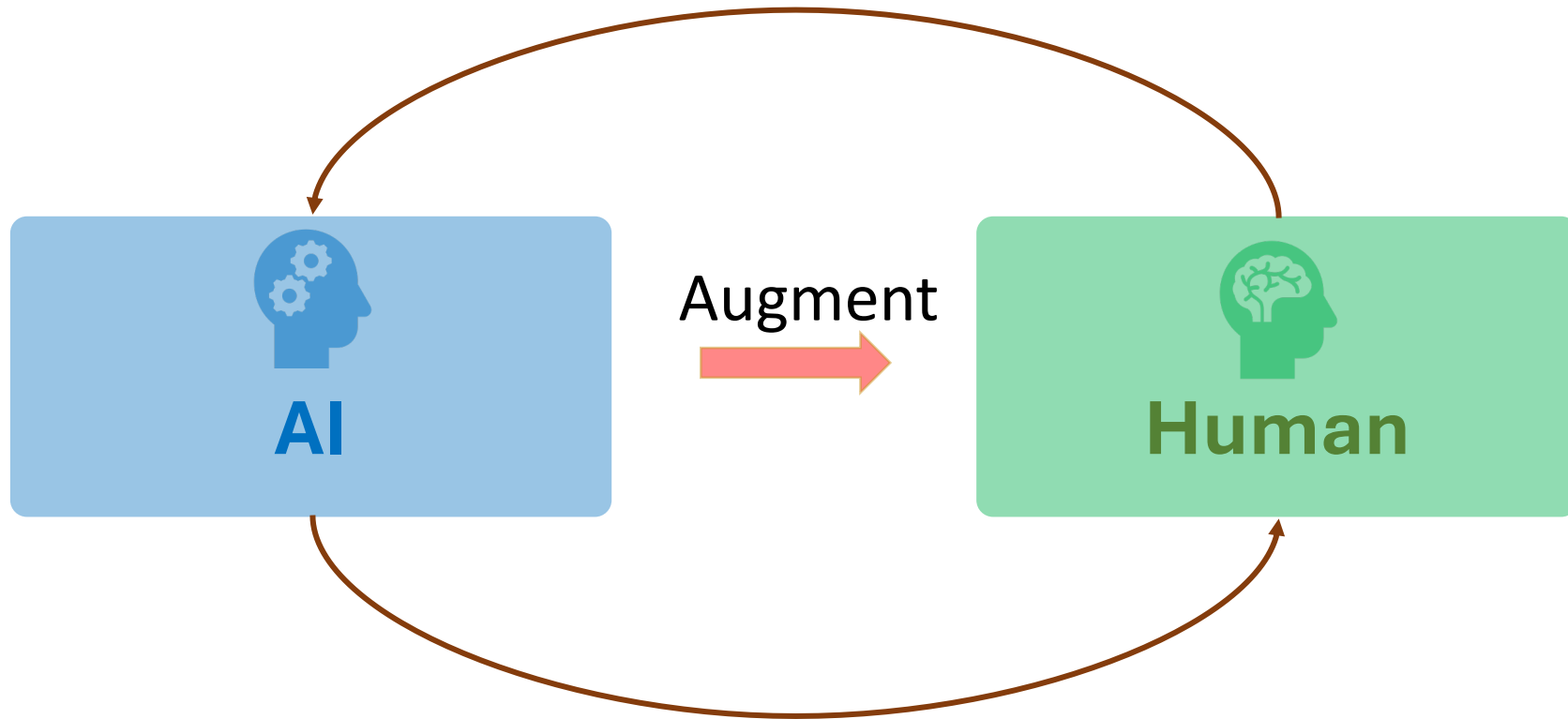


Fundamental Human Abilities

Human-AI Communication System

Observe human behaviors via **sensors**.

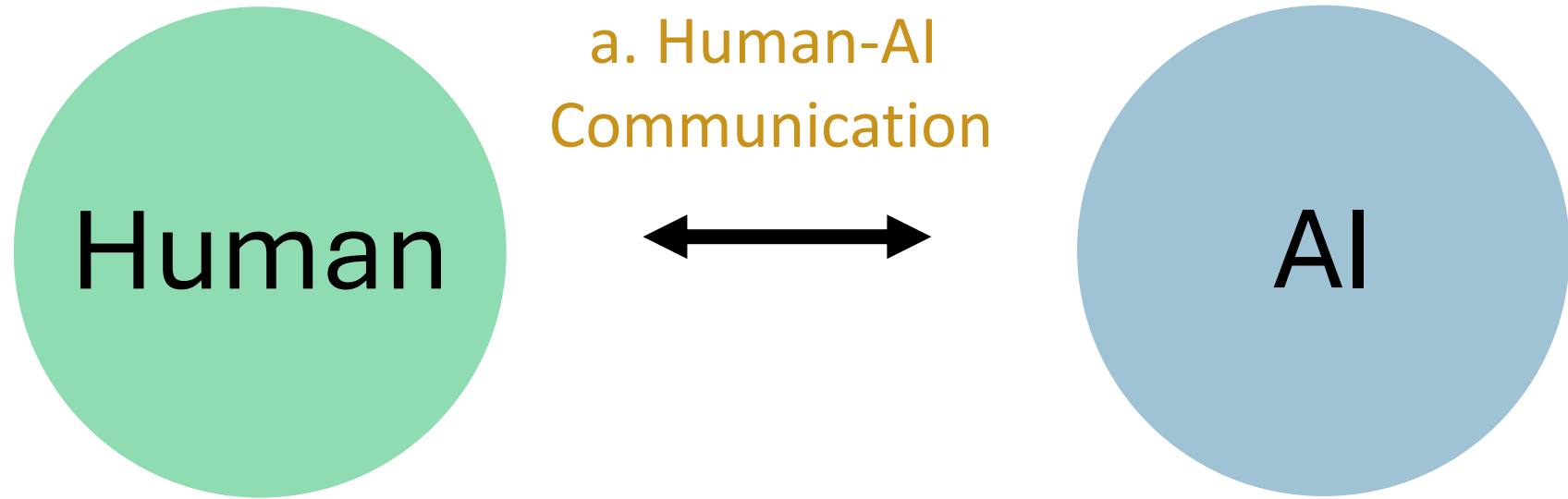
[Physical Sensors (Camera), Contextual / Social Sensing (Social Media)]



Modulate human behaviors via **adaptive feedback**.

[Textual (ChatGPT), Visual, Haptic, Auditory, etc.]

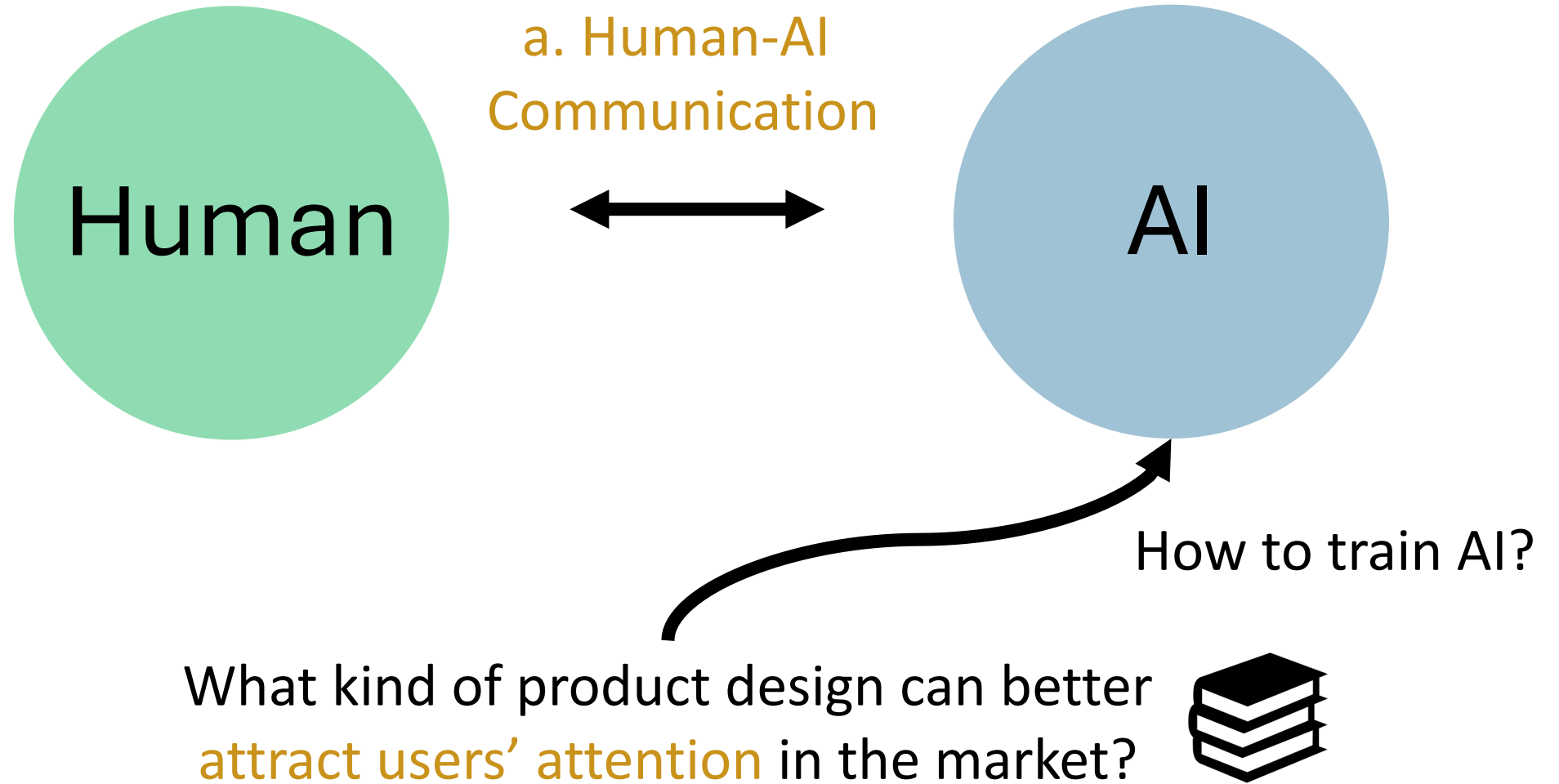
Research Overview



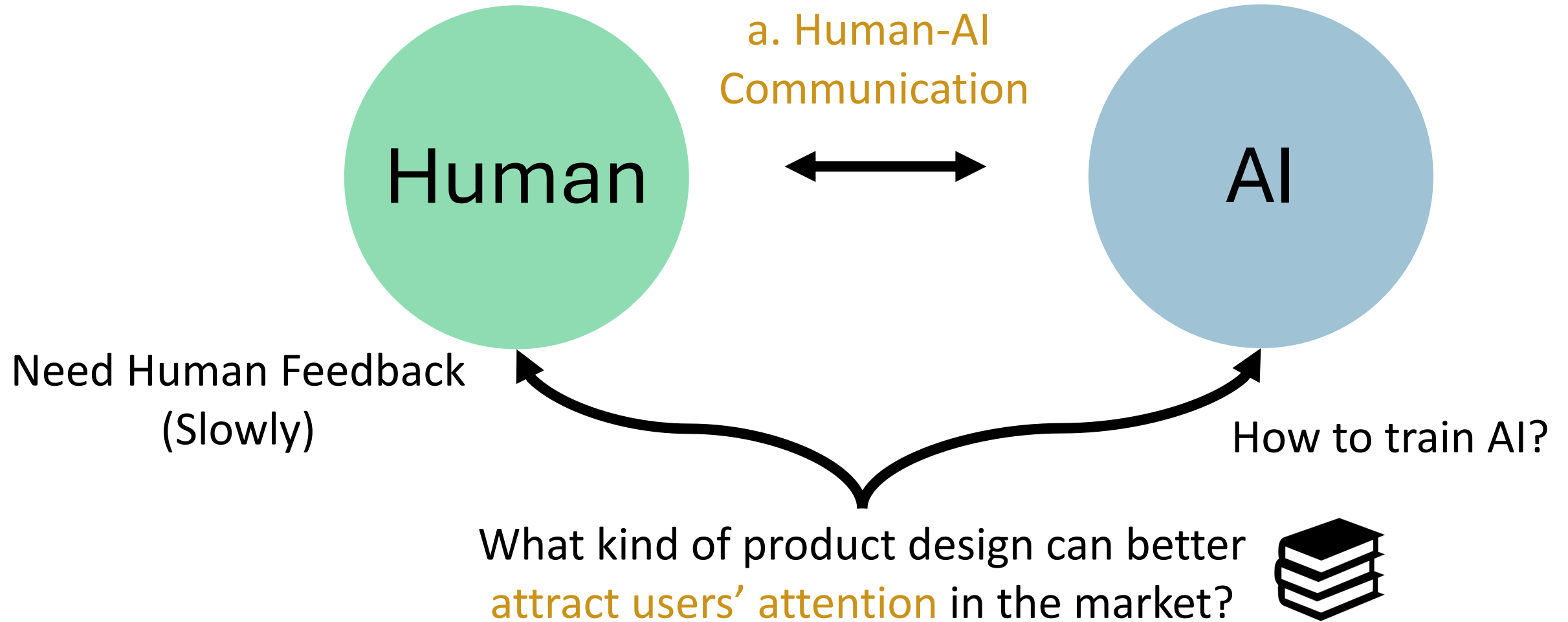
What kind of product design can better
attract users' attention in the market?



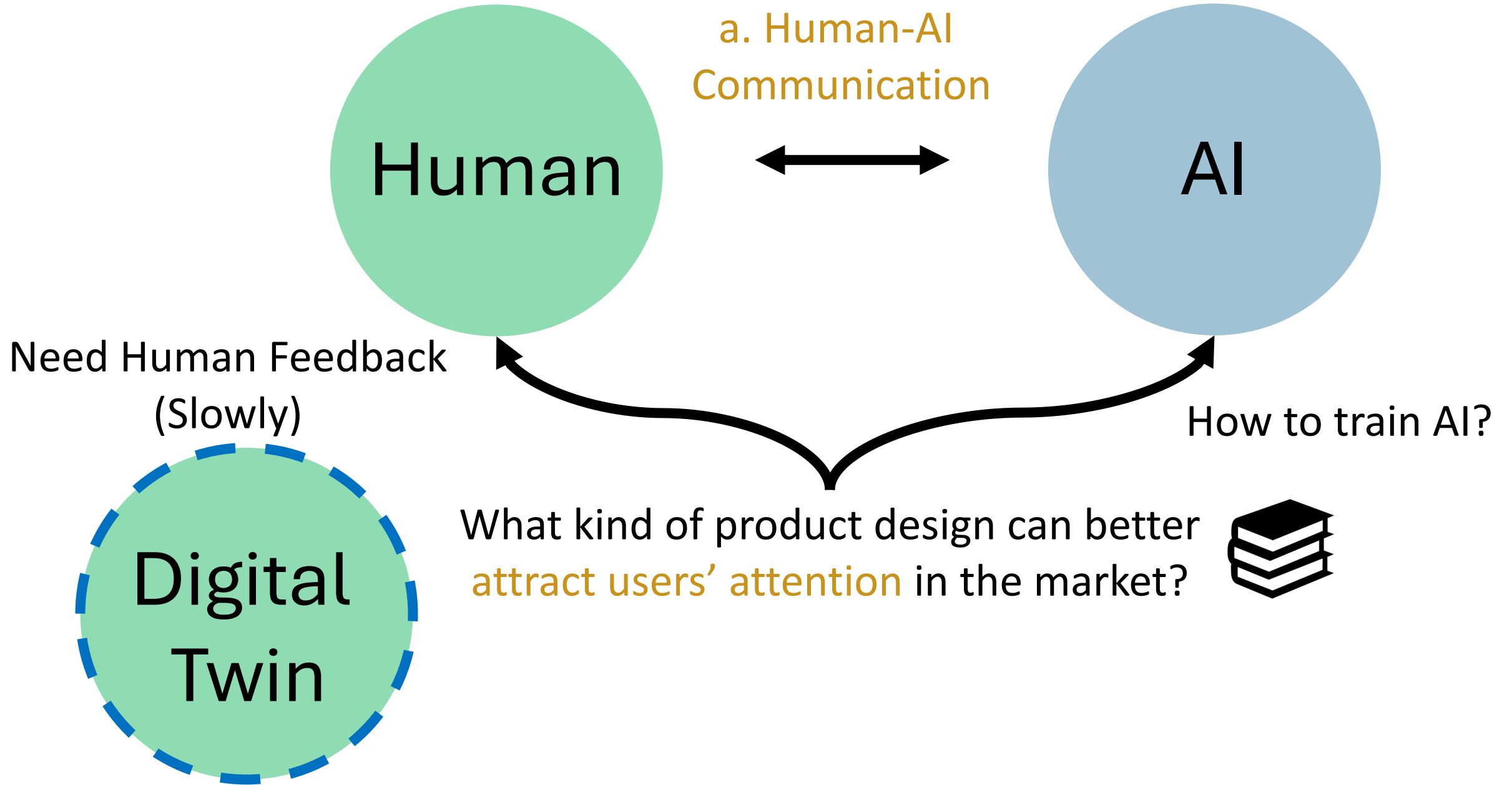
Research Overview



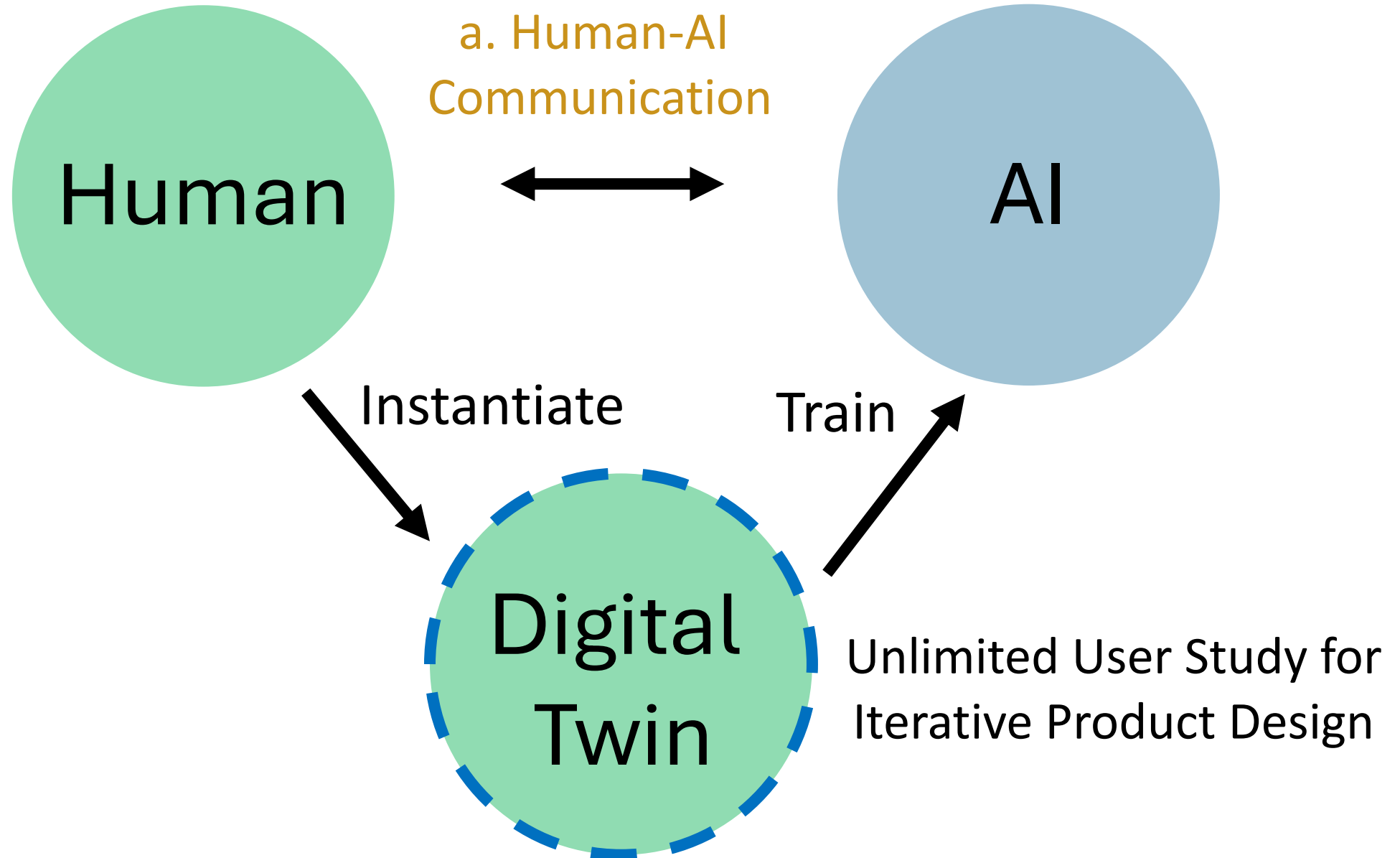
Research Overview



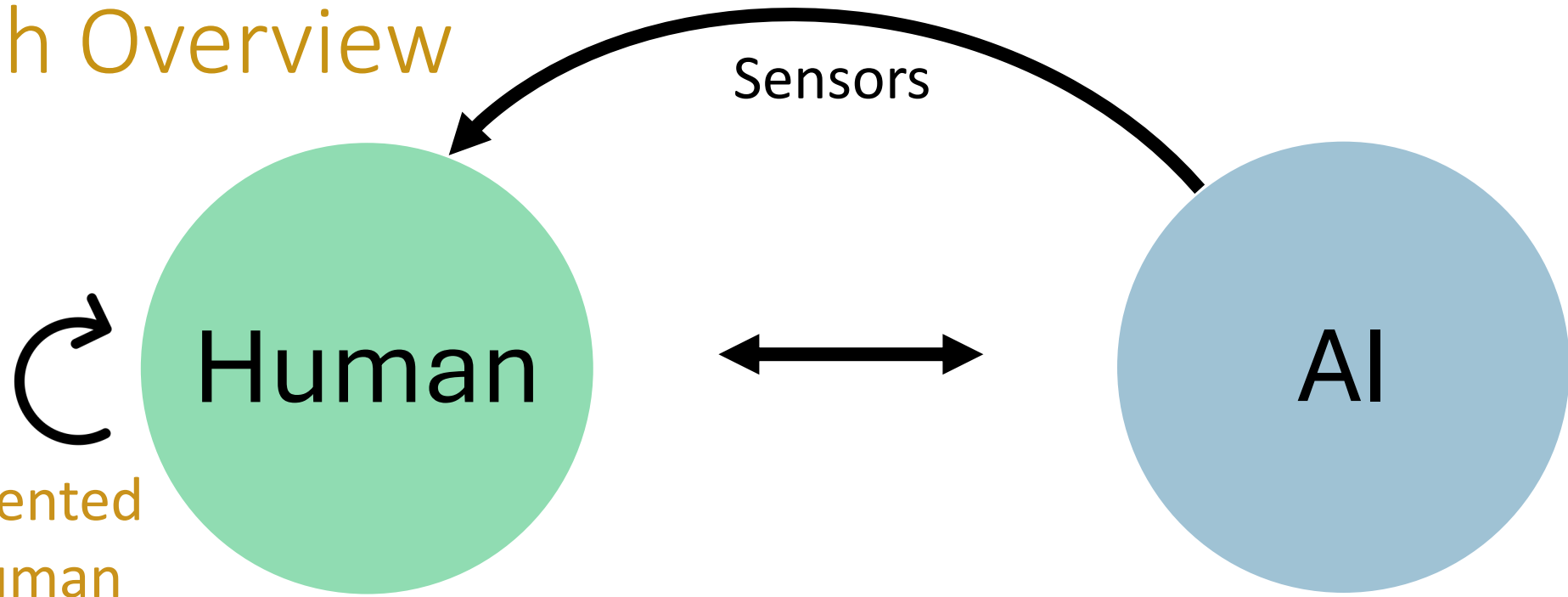
Research Overview



Research Overview

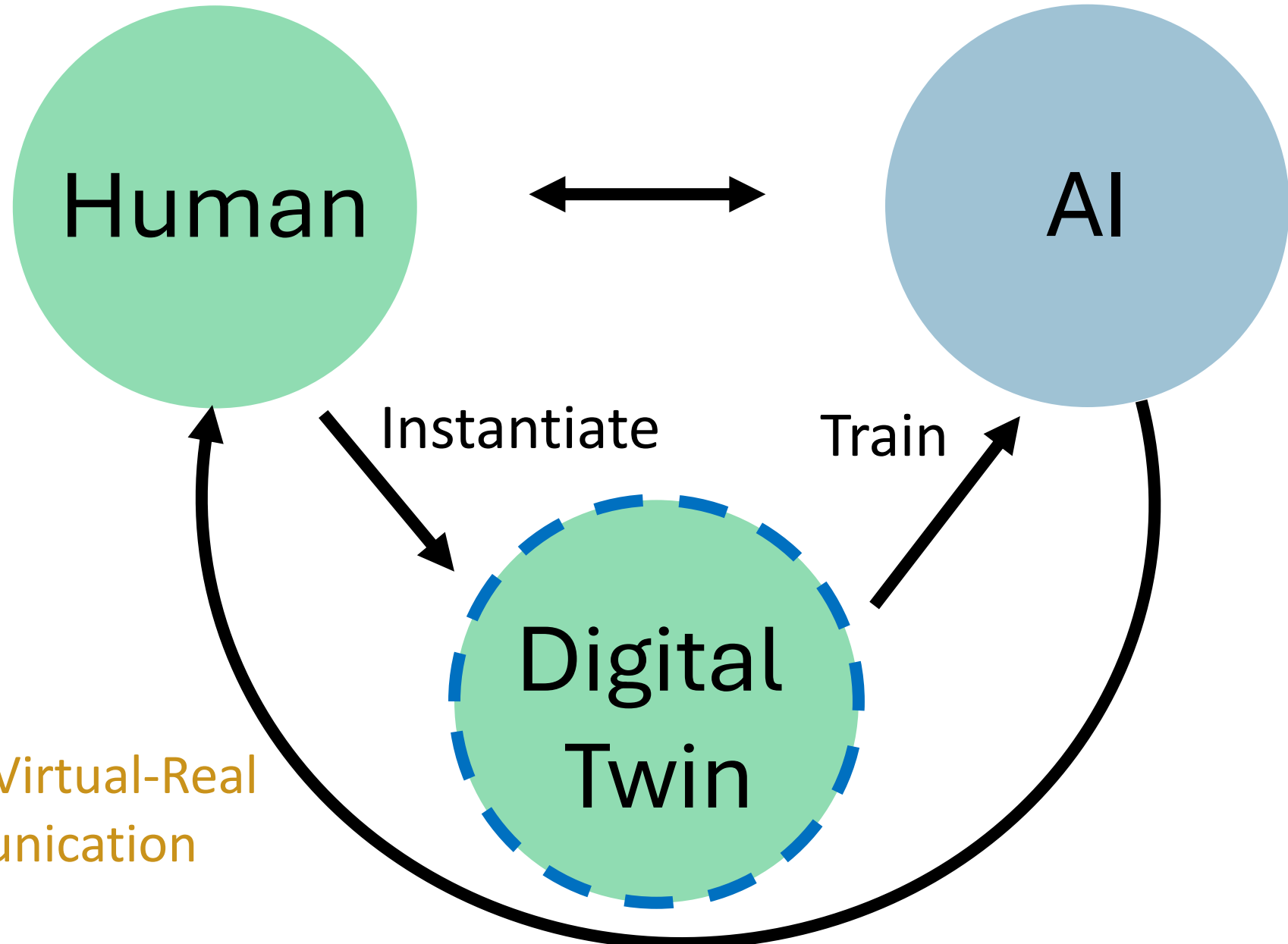


Research Overview



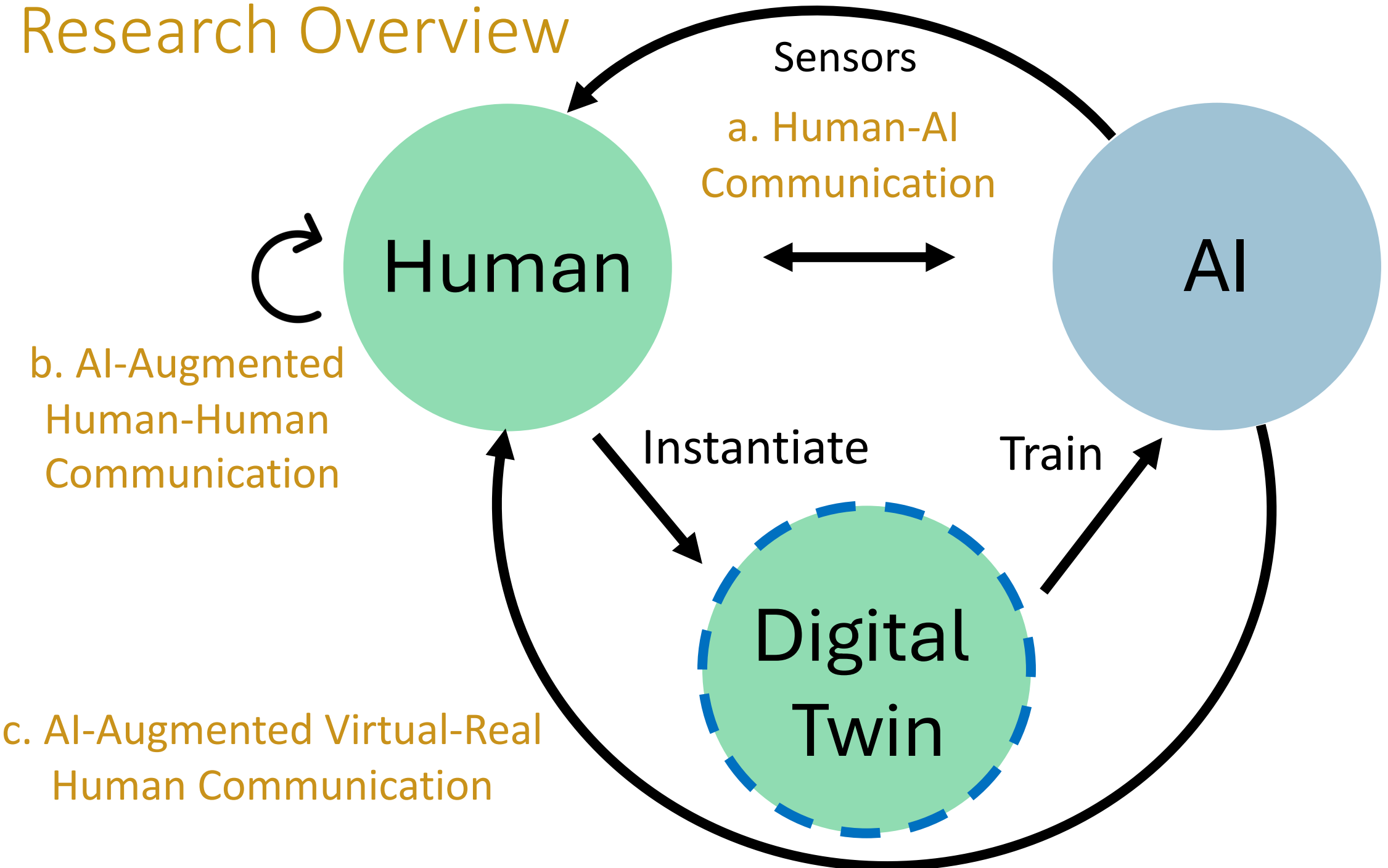
b. AI-Augmented
Human-Human
Communication

Research Overview

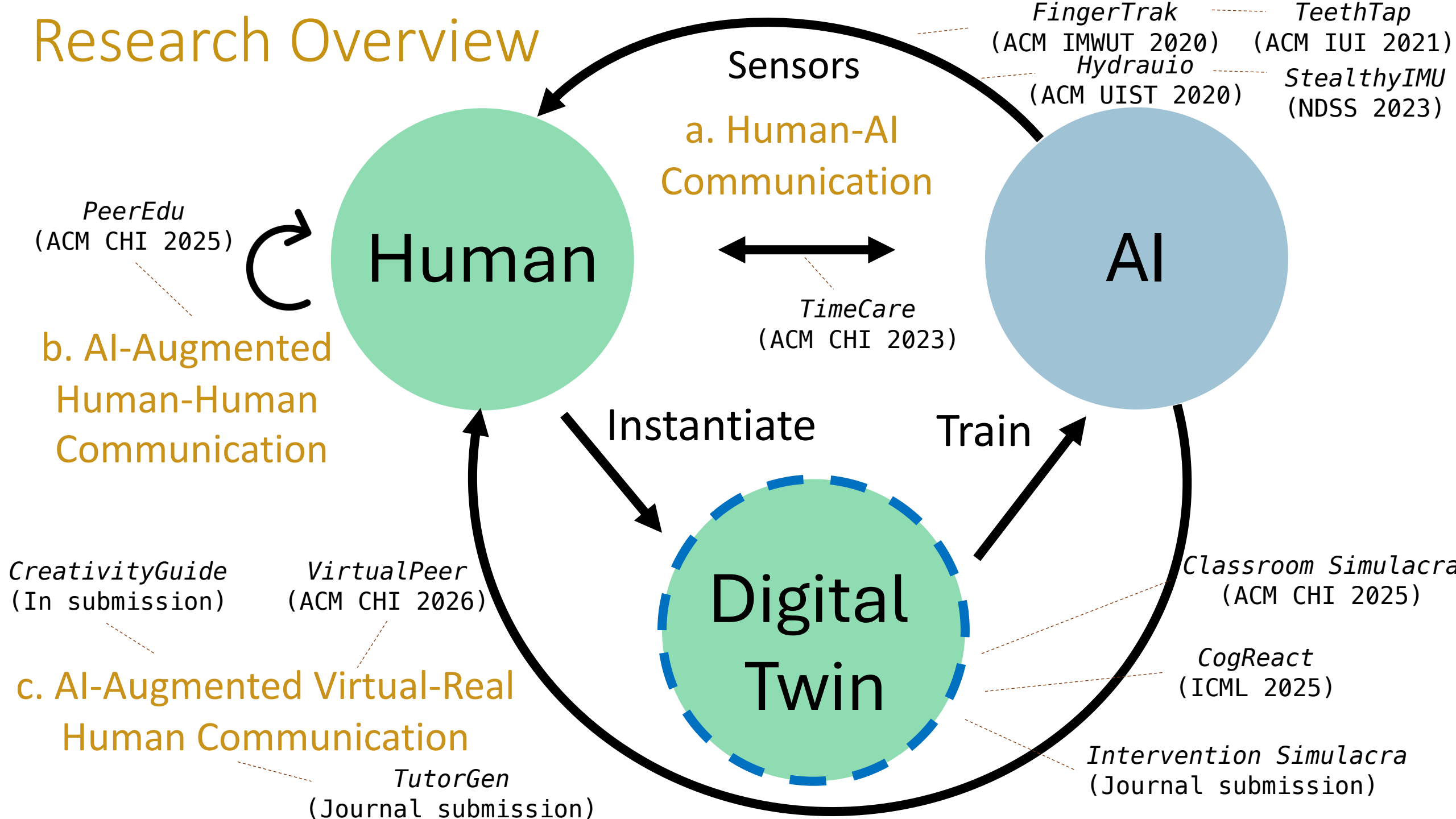


c. AI-Augmented Virtual-Real
Human Communication

Research Overview



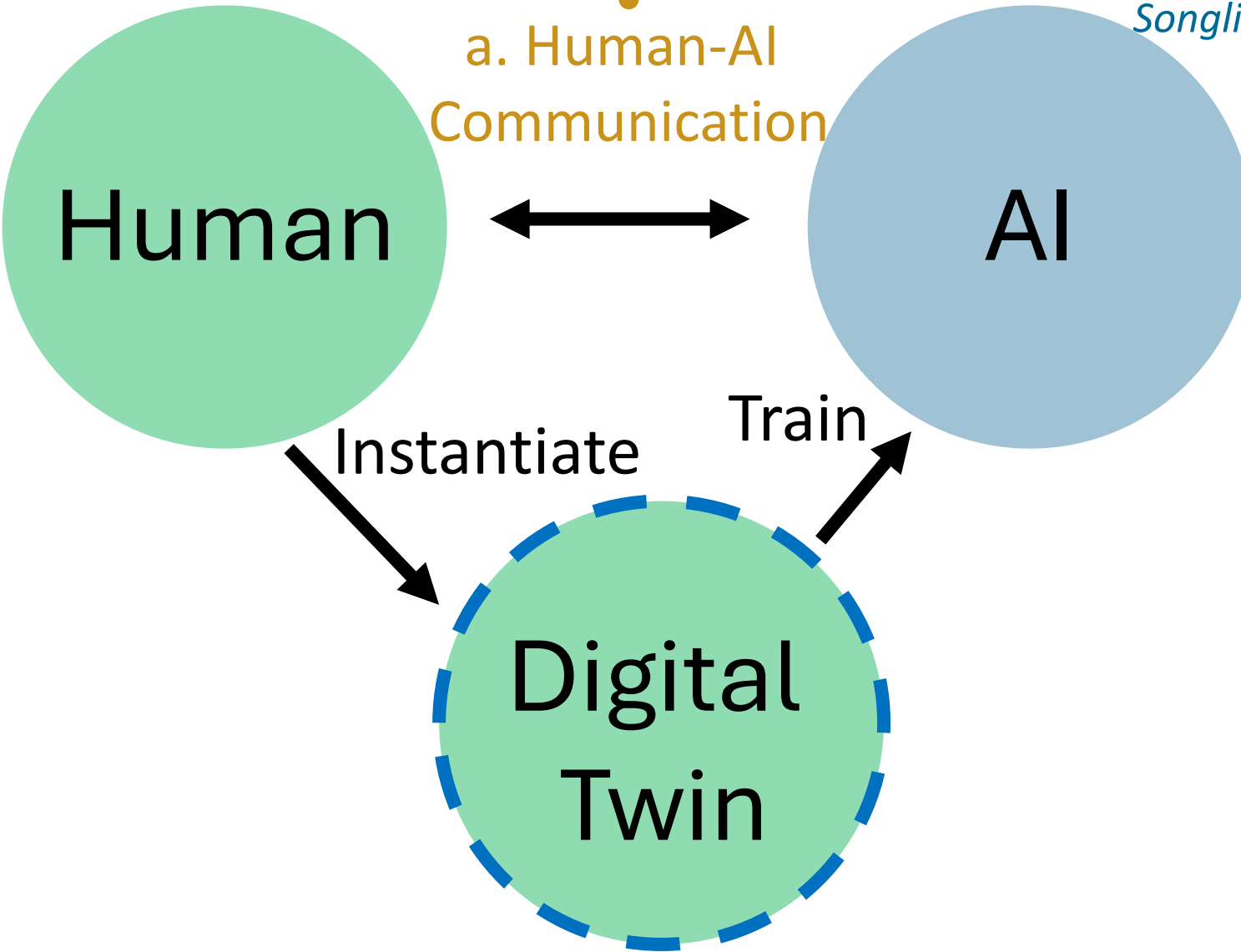
Research Overview



Cognitive Augmentation

Augmenting Human Cognition with an
AI-Mediated Intelligent Visual Feedback

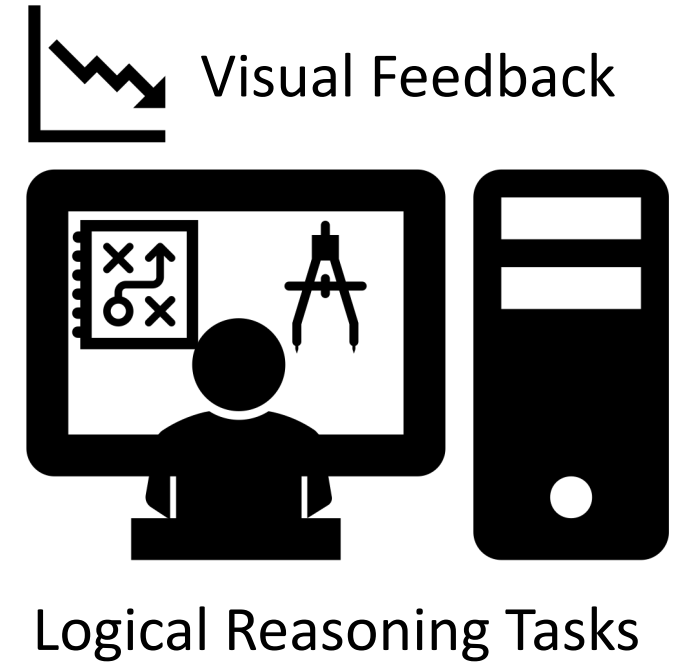
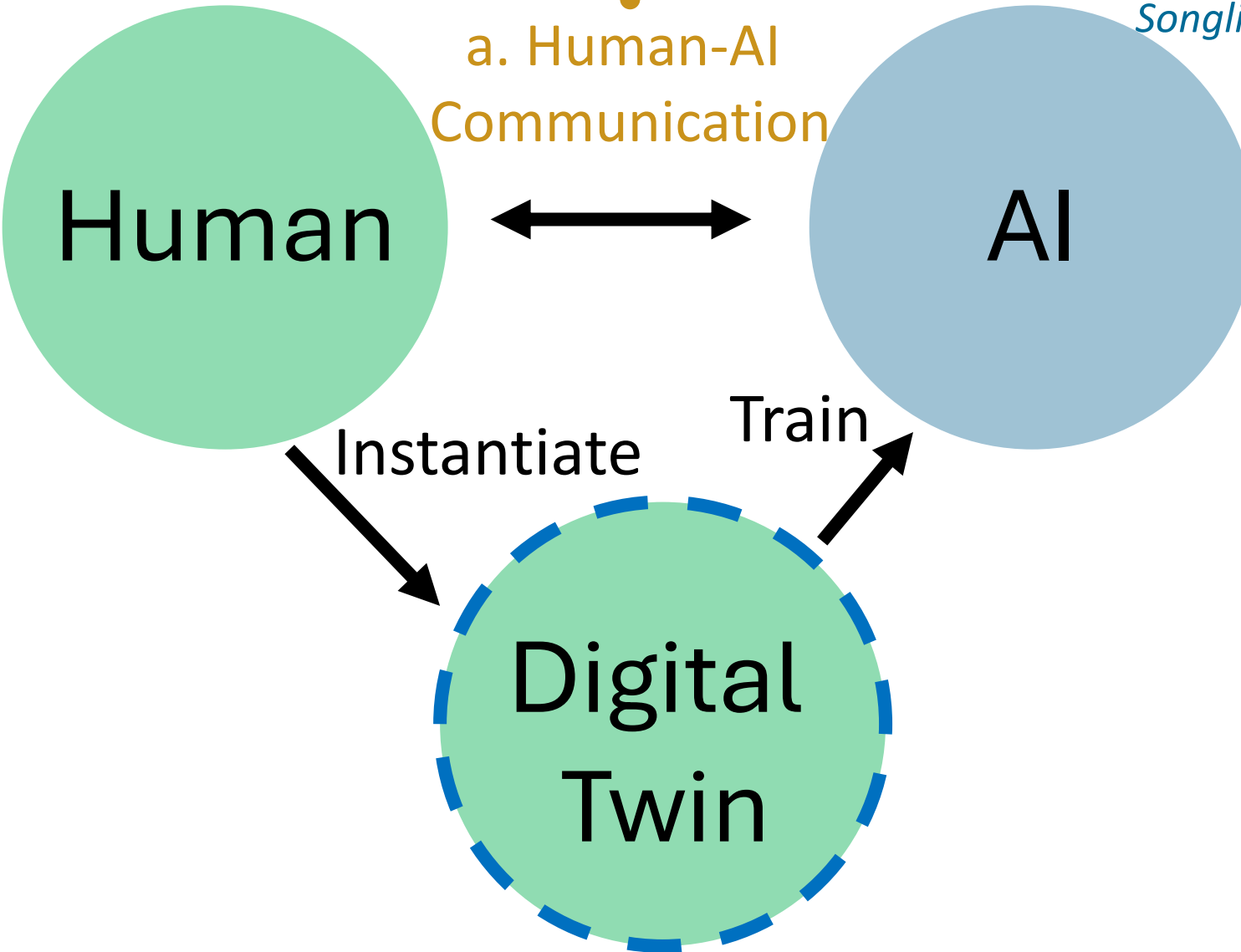
Songlin Xu and Xinyu Zhang. ACM CHI 2023.



Cognitive Augmentation

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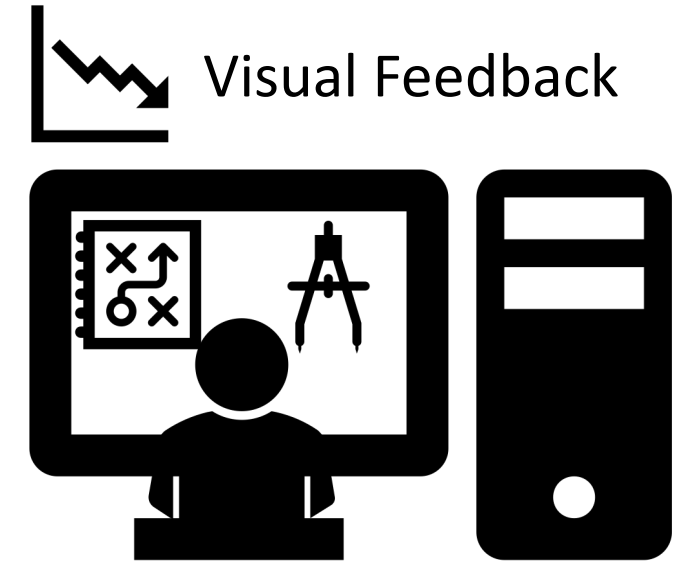
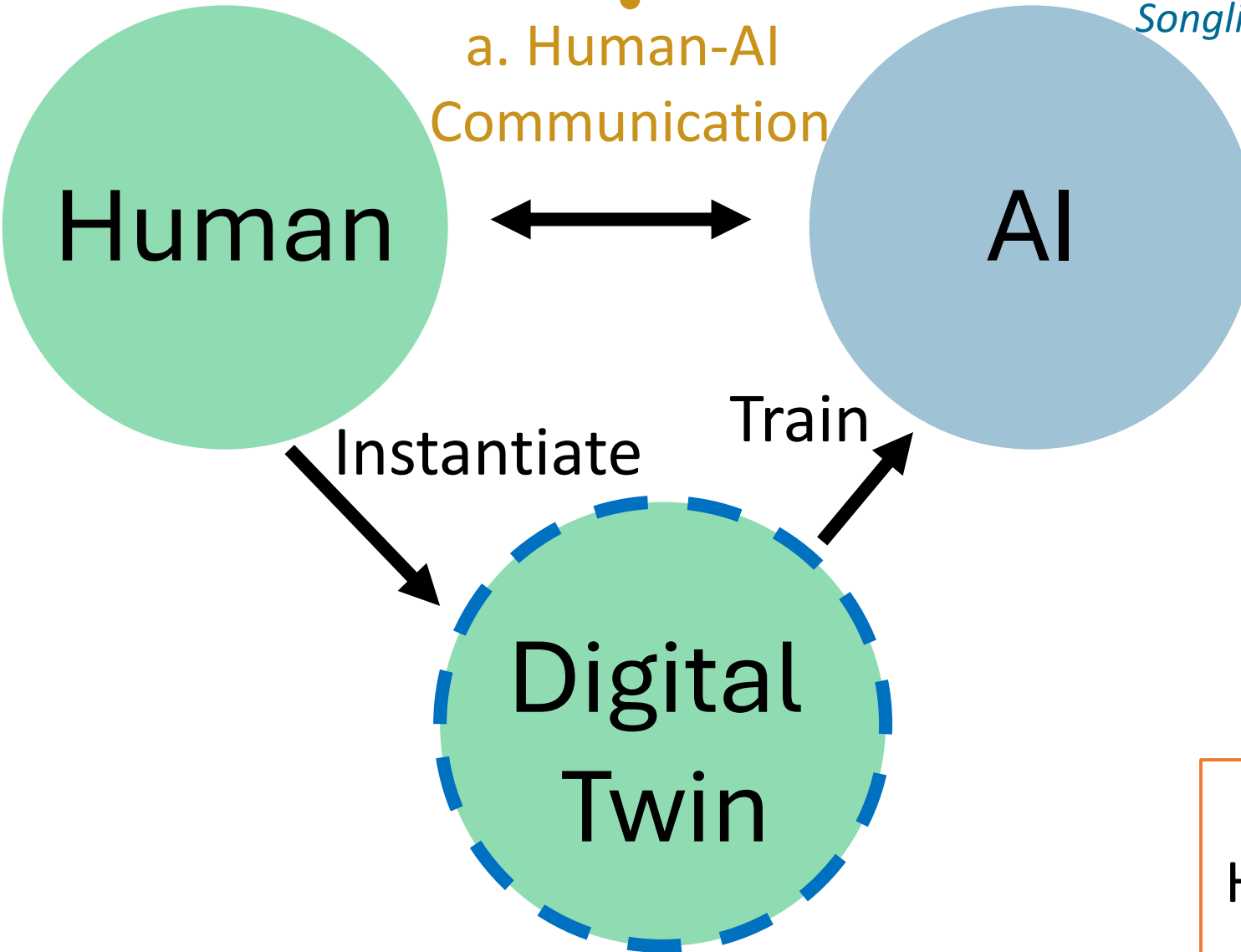
Songlin Xu and Xinyu Zhang. ACM CHI 2023.



Cognitive Augmentation

Augmenting Human Cognition with an AI-Mediated Intelligent Visual Feedback

Songlin Xu and Xinyu Zhang. ACM CHI 2023.



Logical Reasoning Tasks

Research Question:

How to use AI to augment human logical reasoning?

Existing Approach: Visual feedback to Increase Human Arousal

nature human behaviour

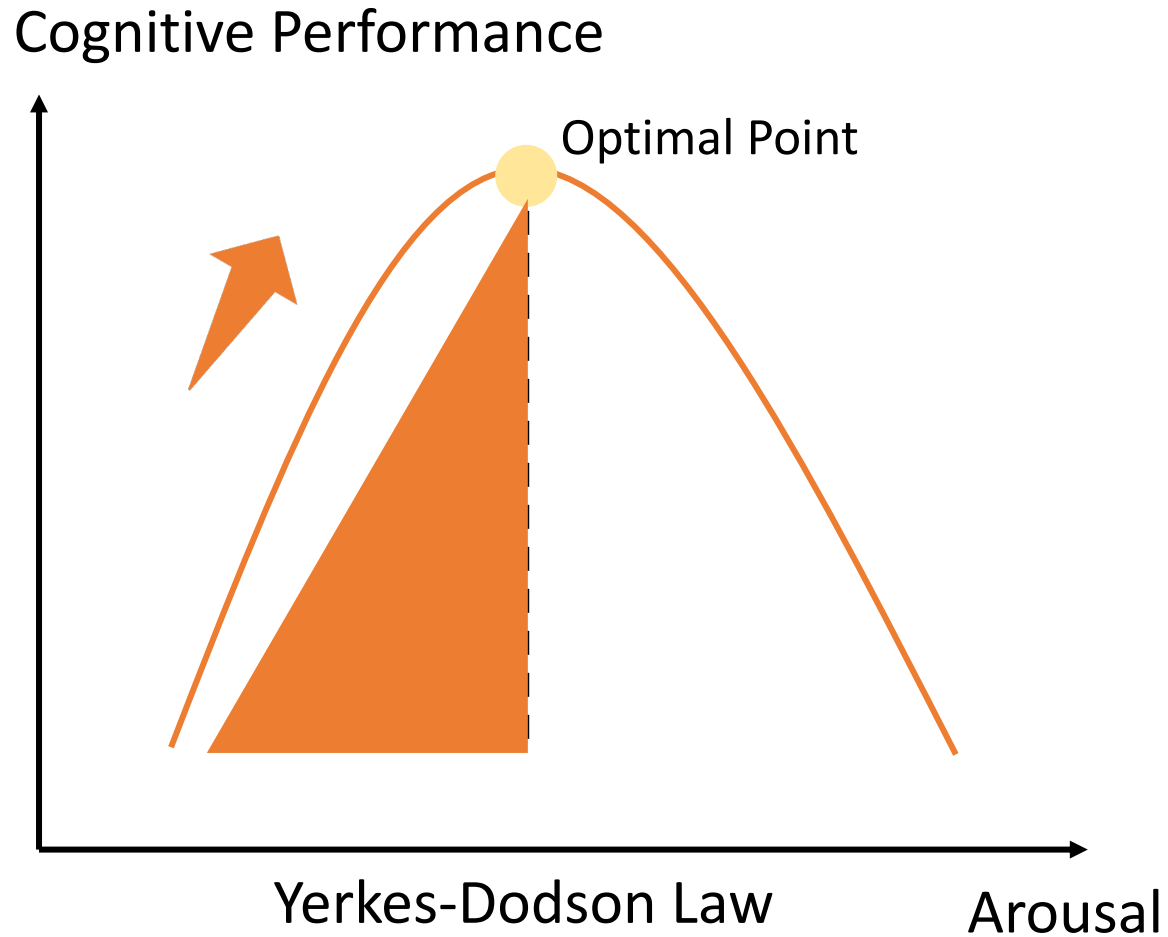
Article

<https://doi.org/10.1038/>

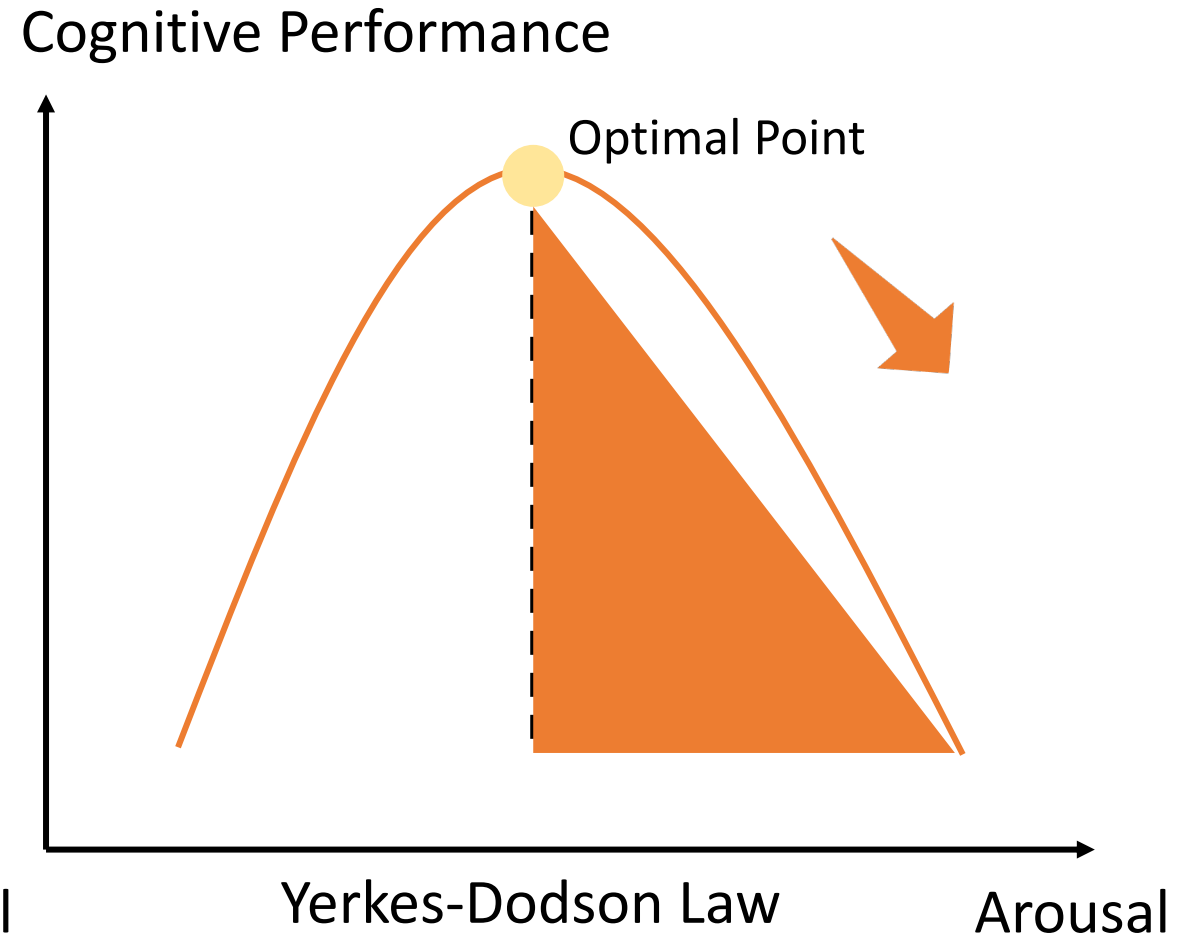
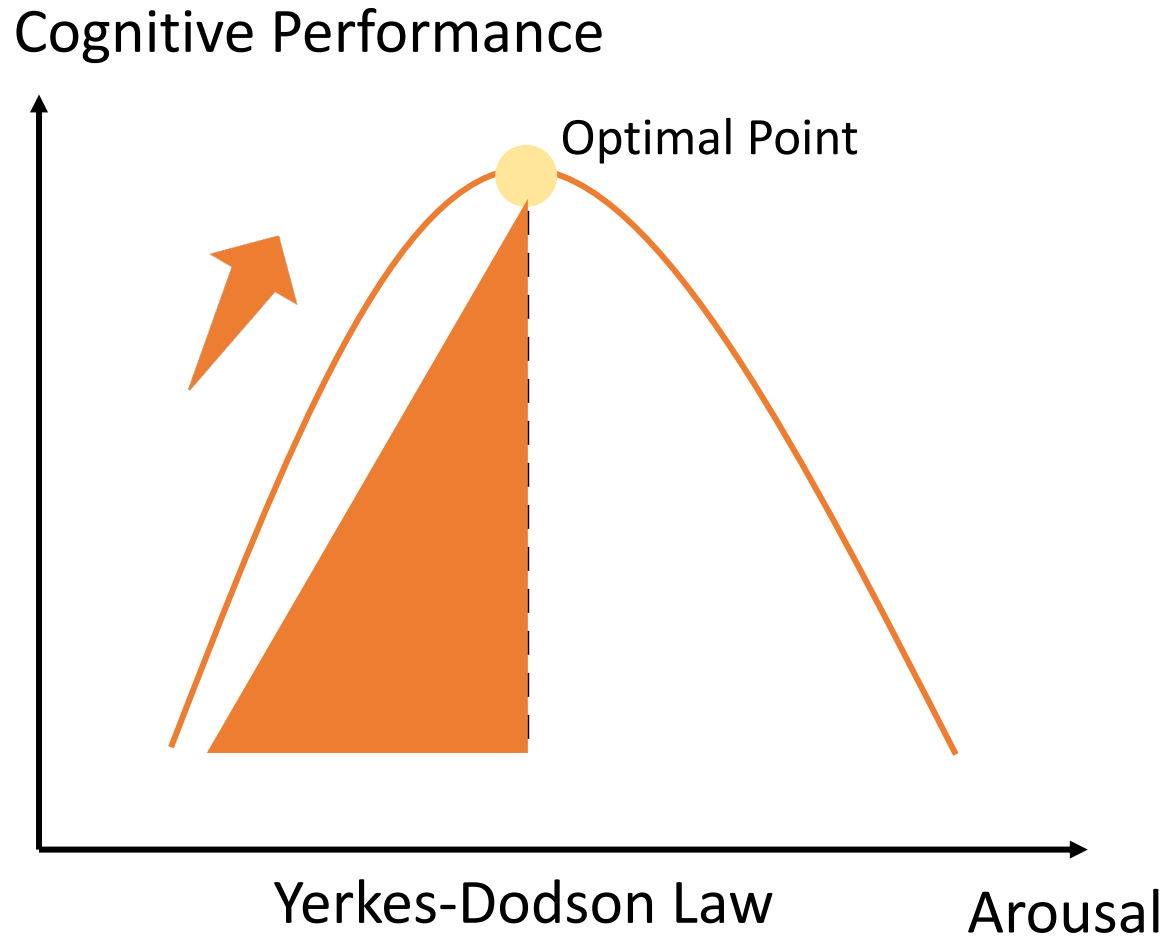
Self-regulating arousal via pupil-based biofeedback

Meissner et al. 2023

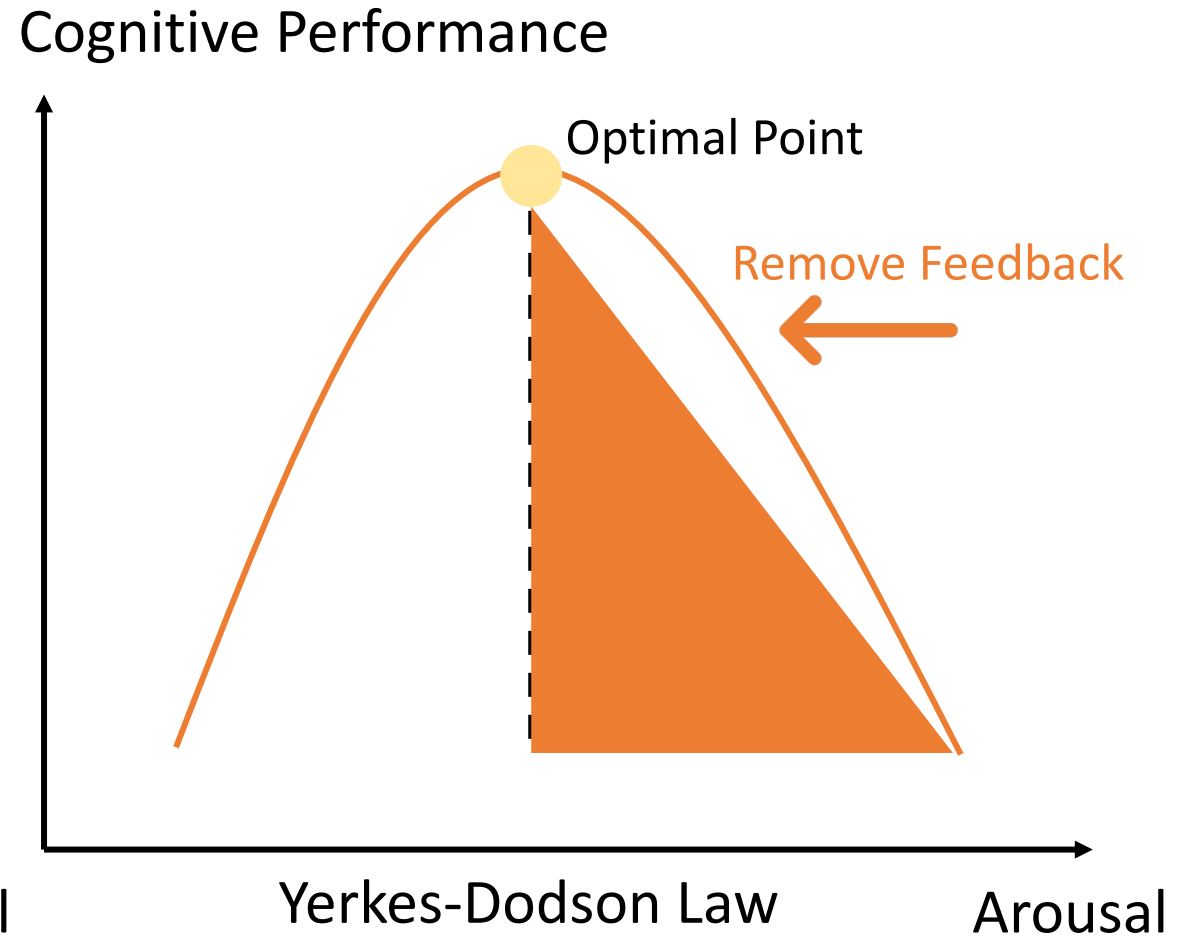
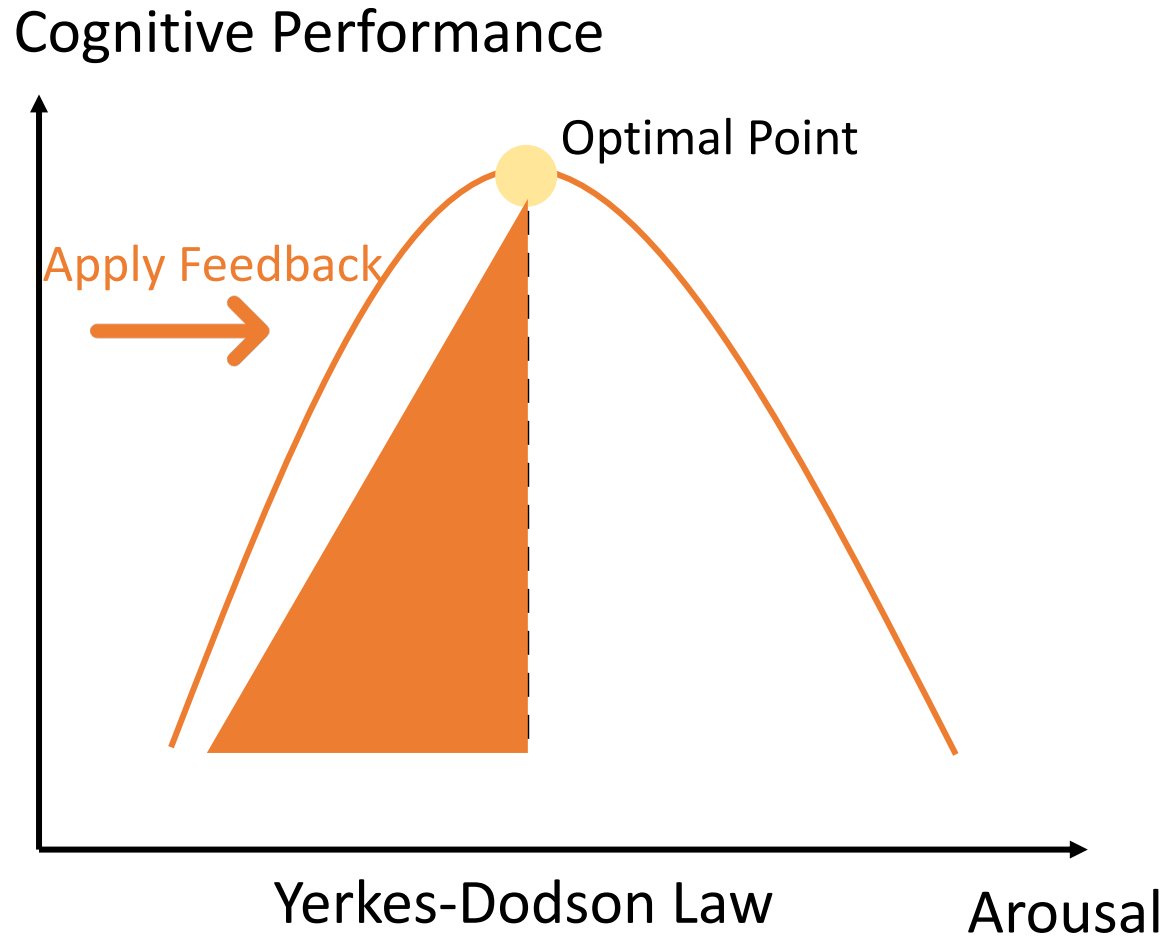
Existing Approach: Visual feedback to Increase Human Arousal



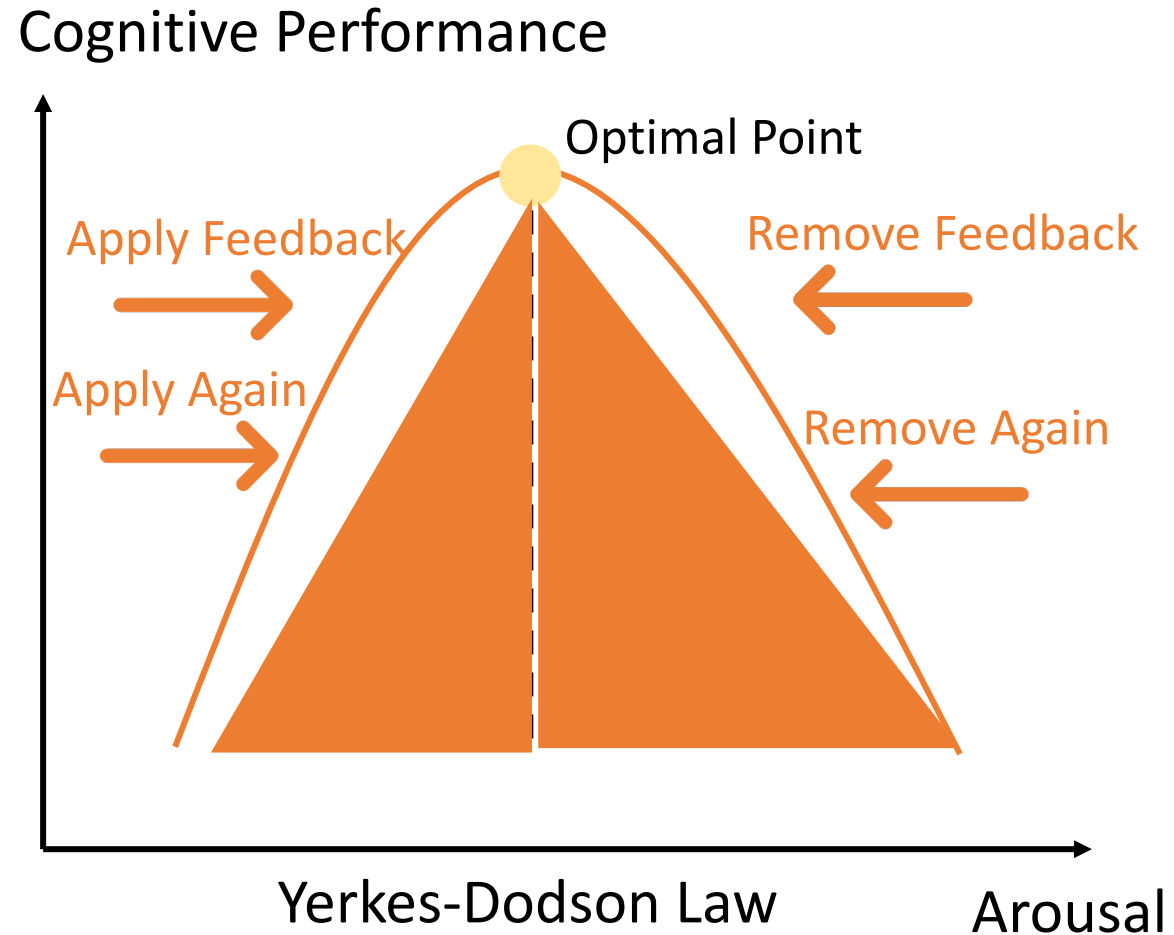
Existing Approach: Visual feedback to Increase Human Arousal



Adaptive Feedback Control: Rule-Based



Rule-Based Feedback Control



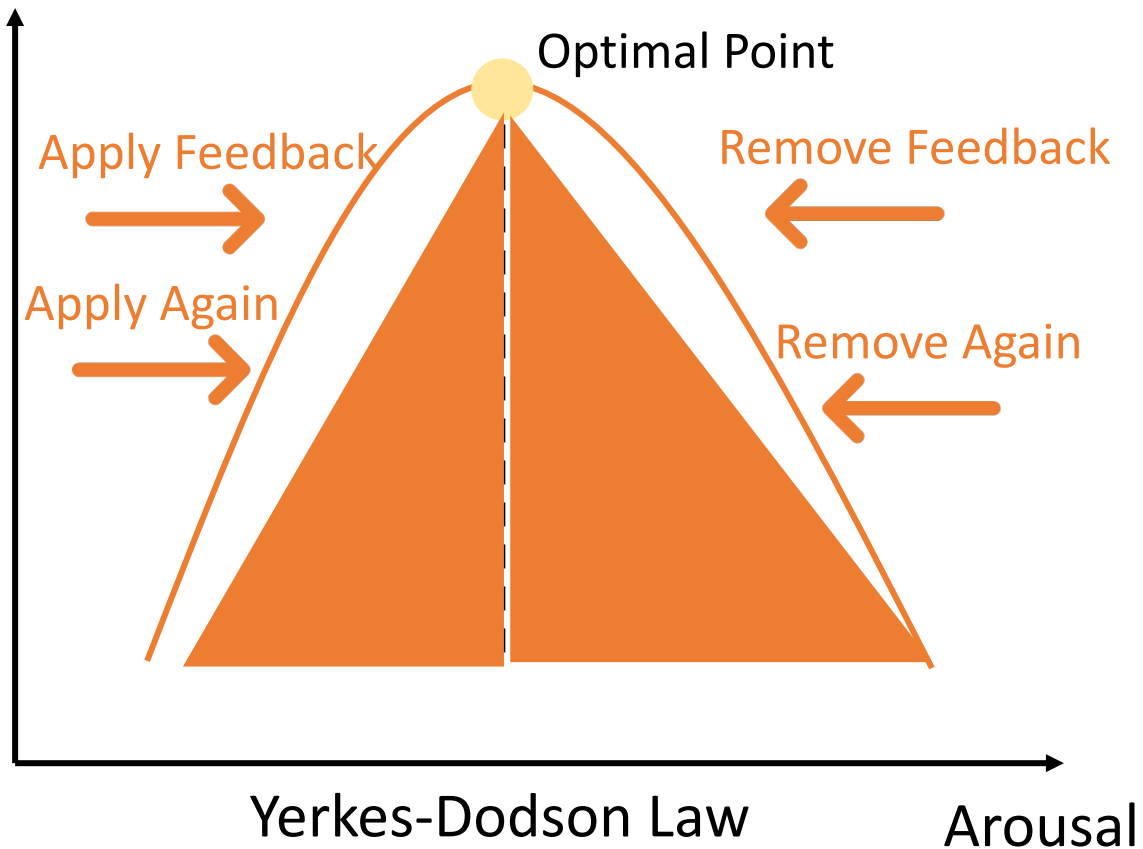
Short-Sighted;

Local Optimization;

Hard to Handle Such Complexity

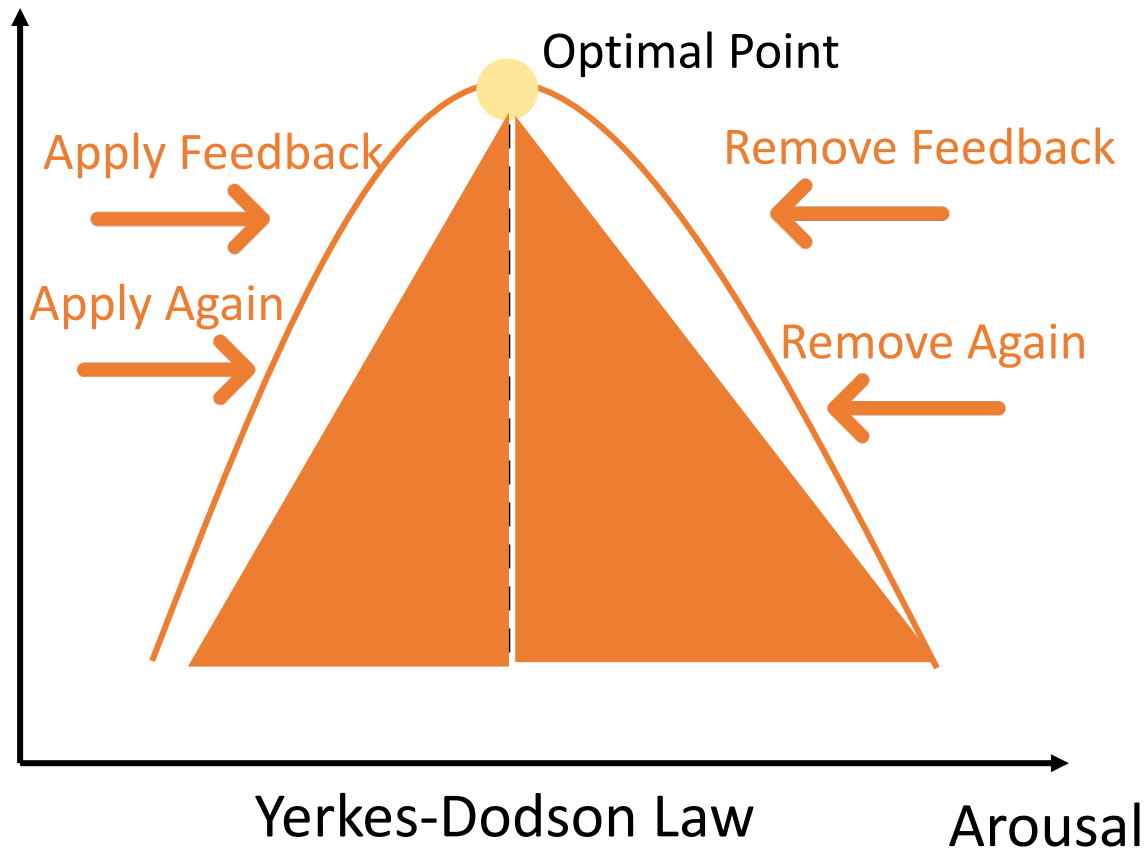
AI Feedback Controller: Human-in-the-Loop Reinforcement Learning (RL)

Cognitive Performance



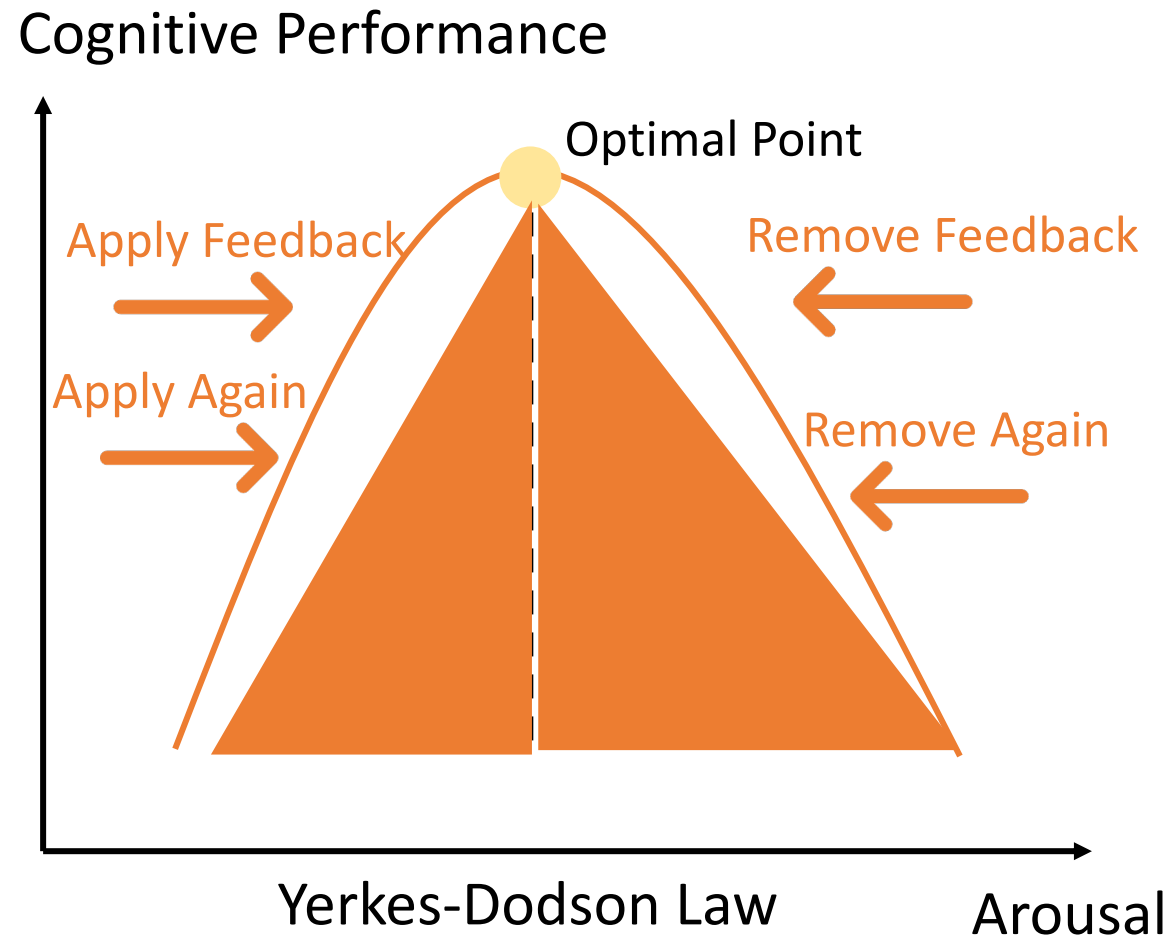
AI Feedback Controller: Human-in-the-Loop Reinforcement Learning (RL)

Cognitive Performance



Challenge: Data Scarcity

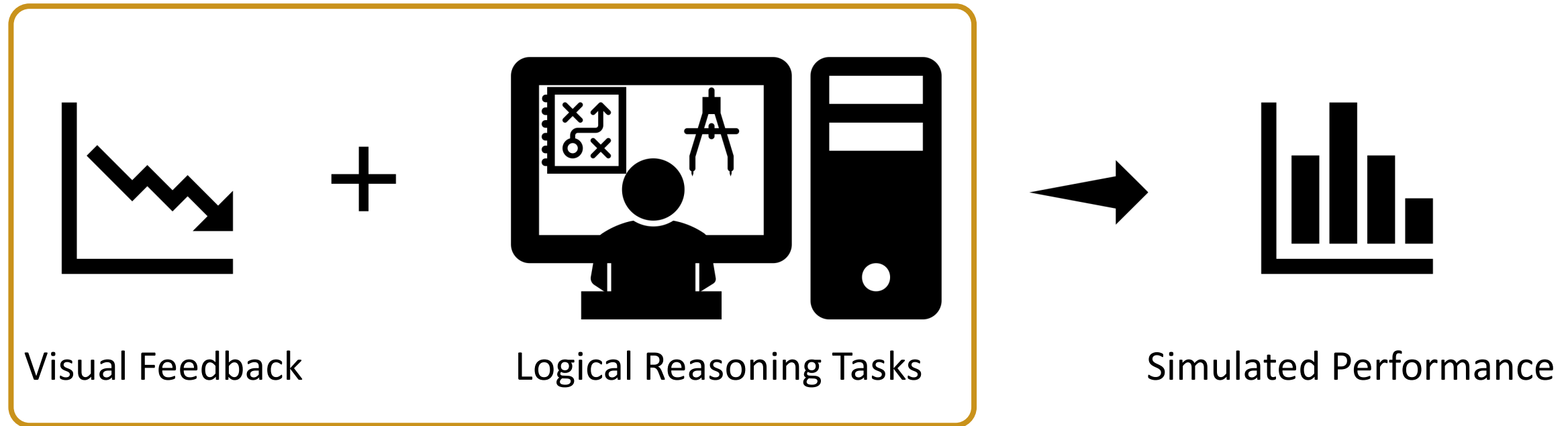
AI Feedback Controller: Human-in-the-Loop Reinforcement Learning (RL)



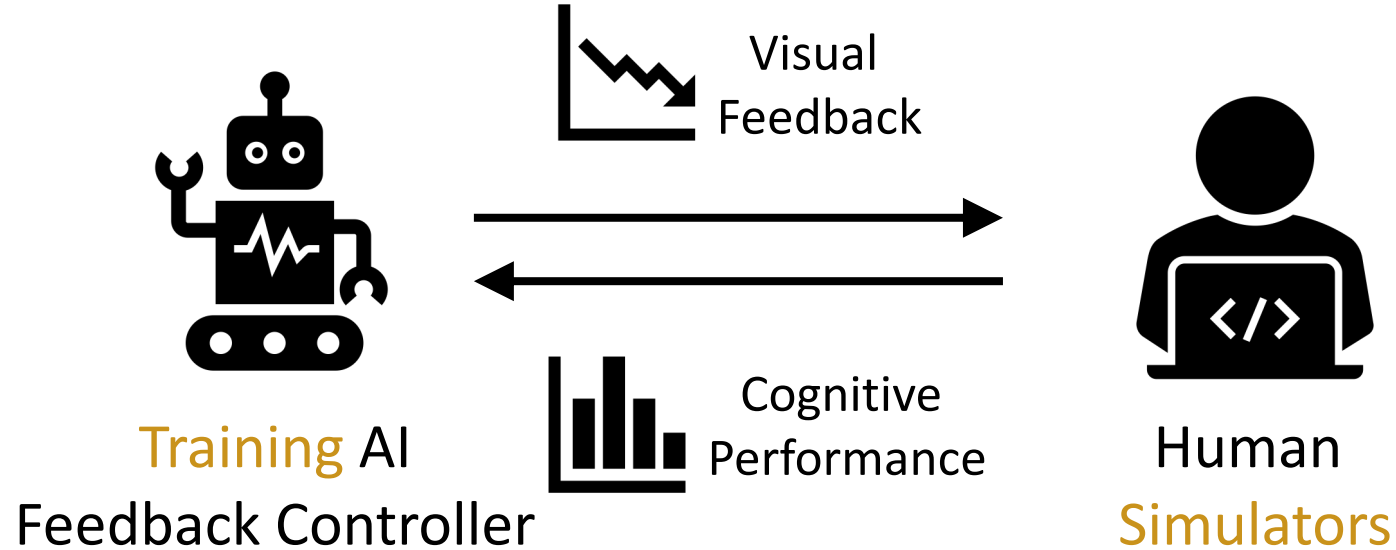
Challenge: Data Scarcity

Solution:
Human Cognitive Digital Twin

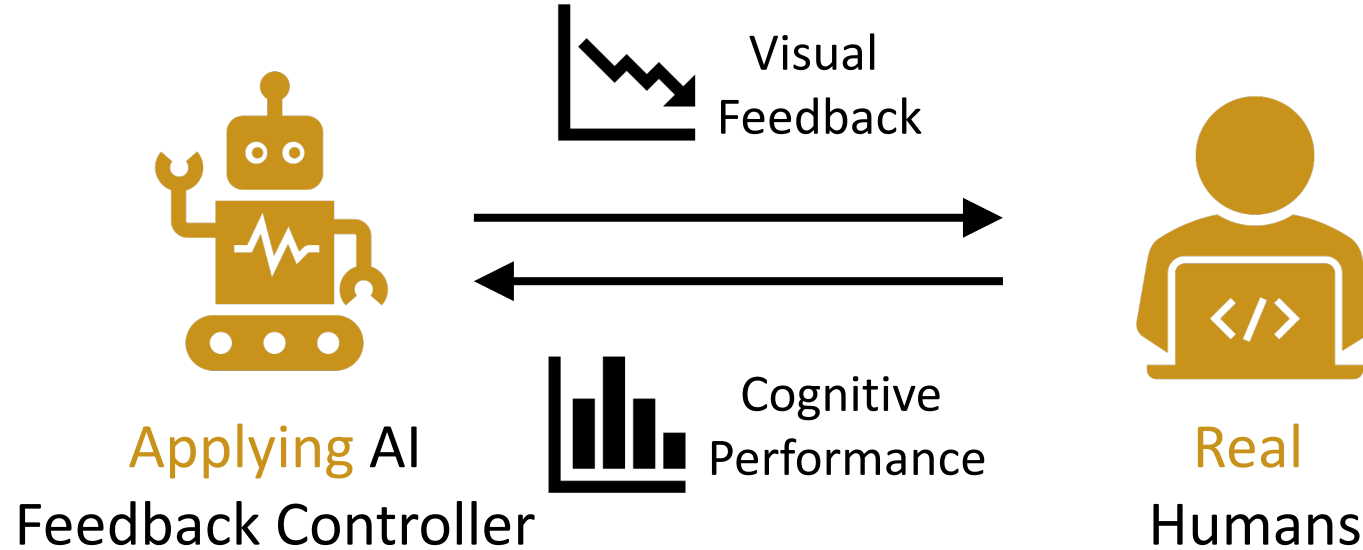
Digital Twin to Simulate Human Cognitive Performance



AI Feedback Controller: Human-in-the-Loop Reinforcement Learning (RL)

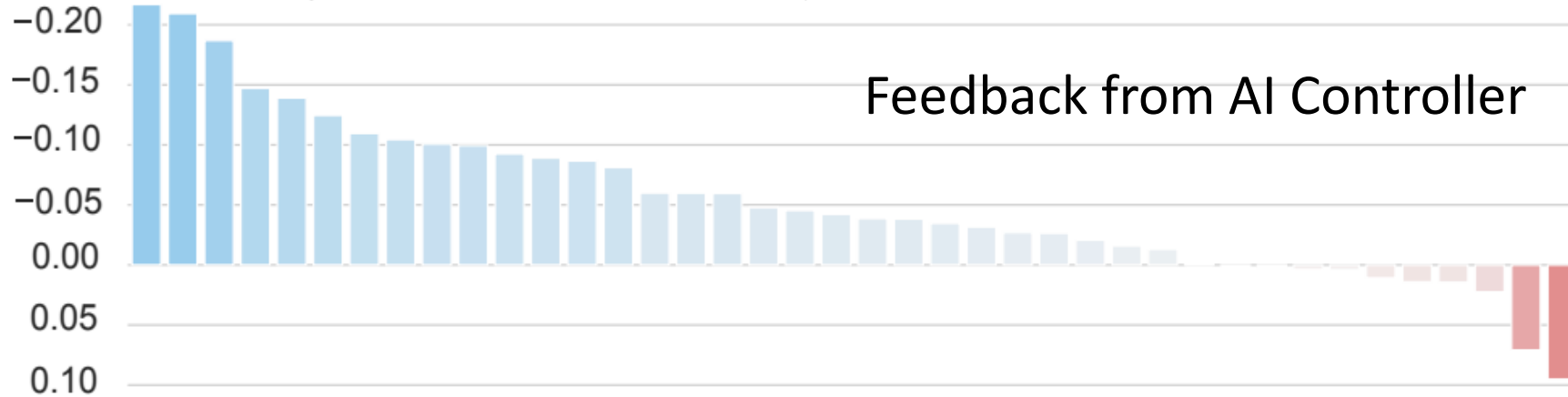


AI Feedback Controller: Human-in-the-Loop Reinforcement Learning (RL)



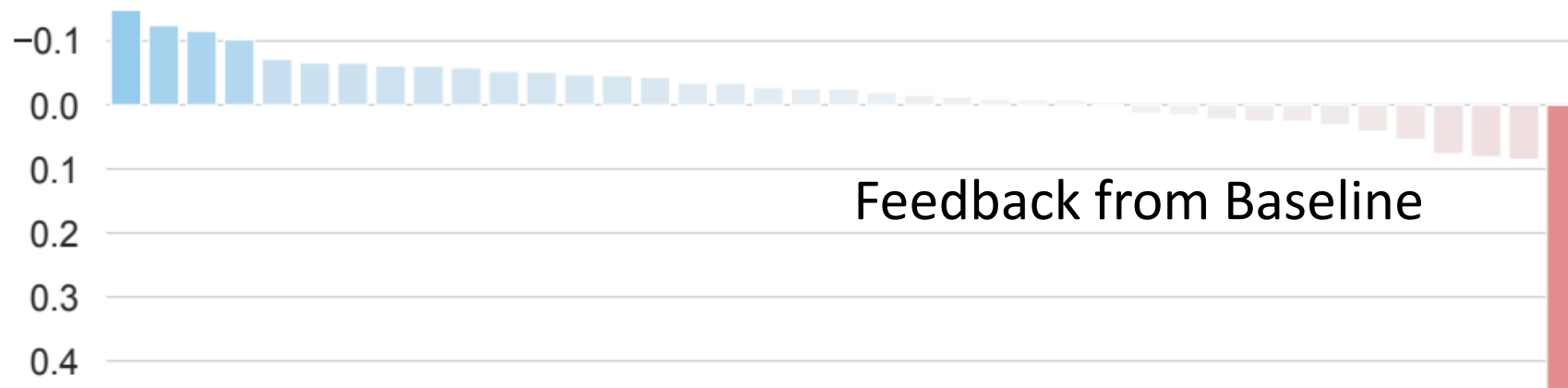
Evaluation: N = 80 between-subject study

Response Time Change to Solve Tasks Correctly



Feedback from AI Controller

Response Time Change to Solve Tasks Correctly

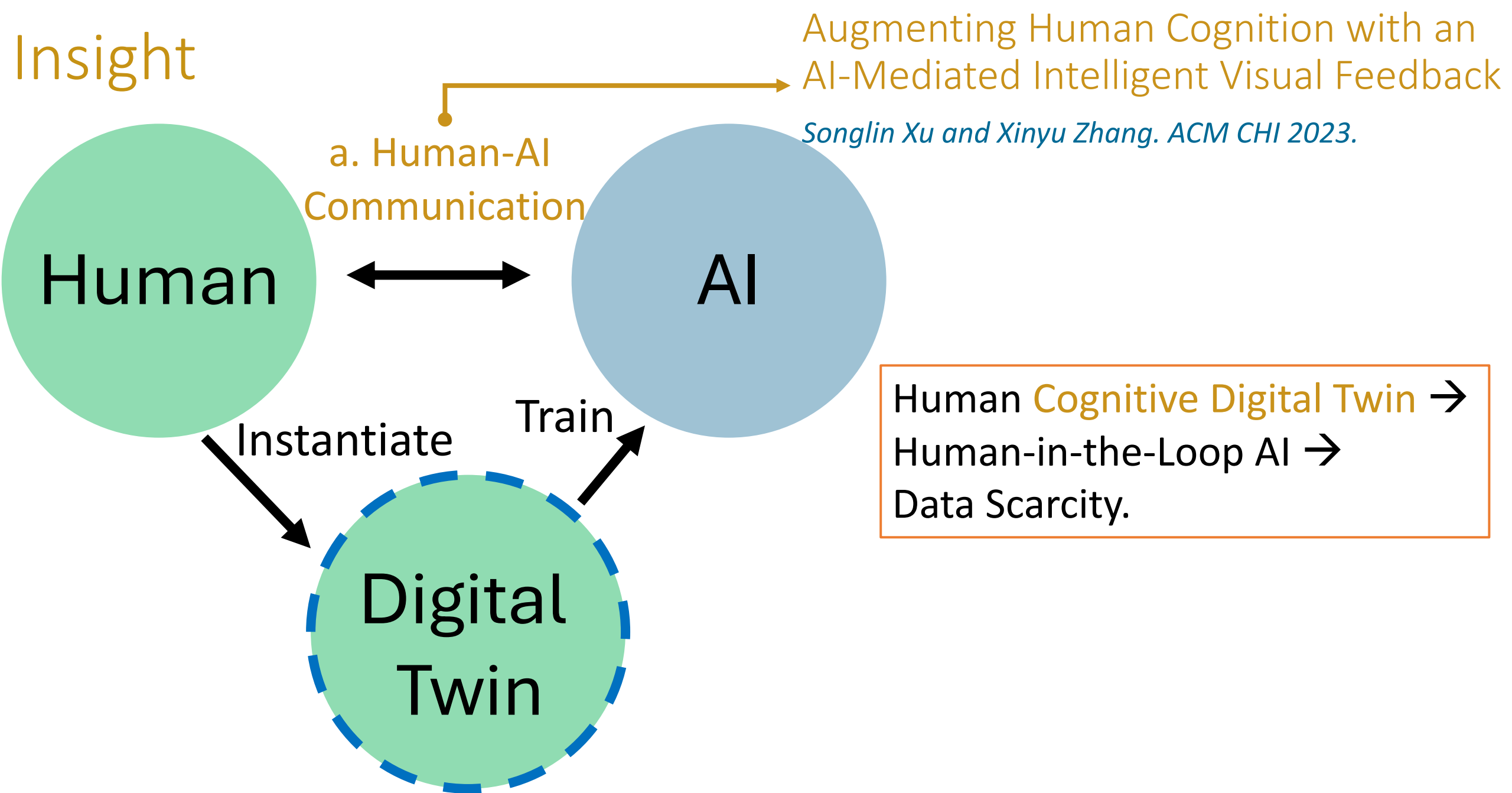


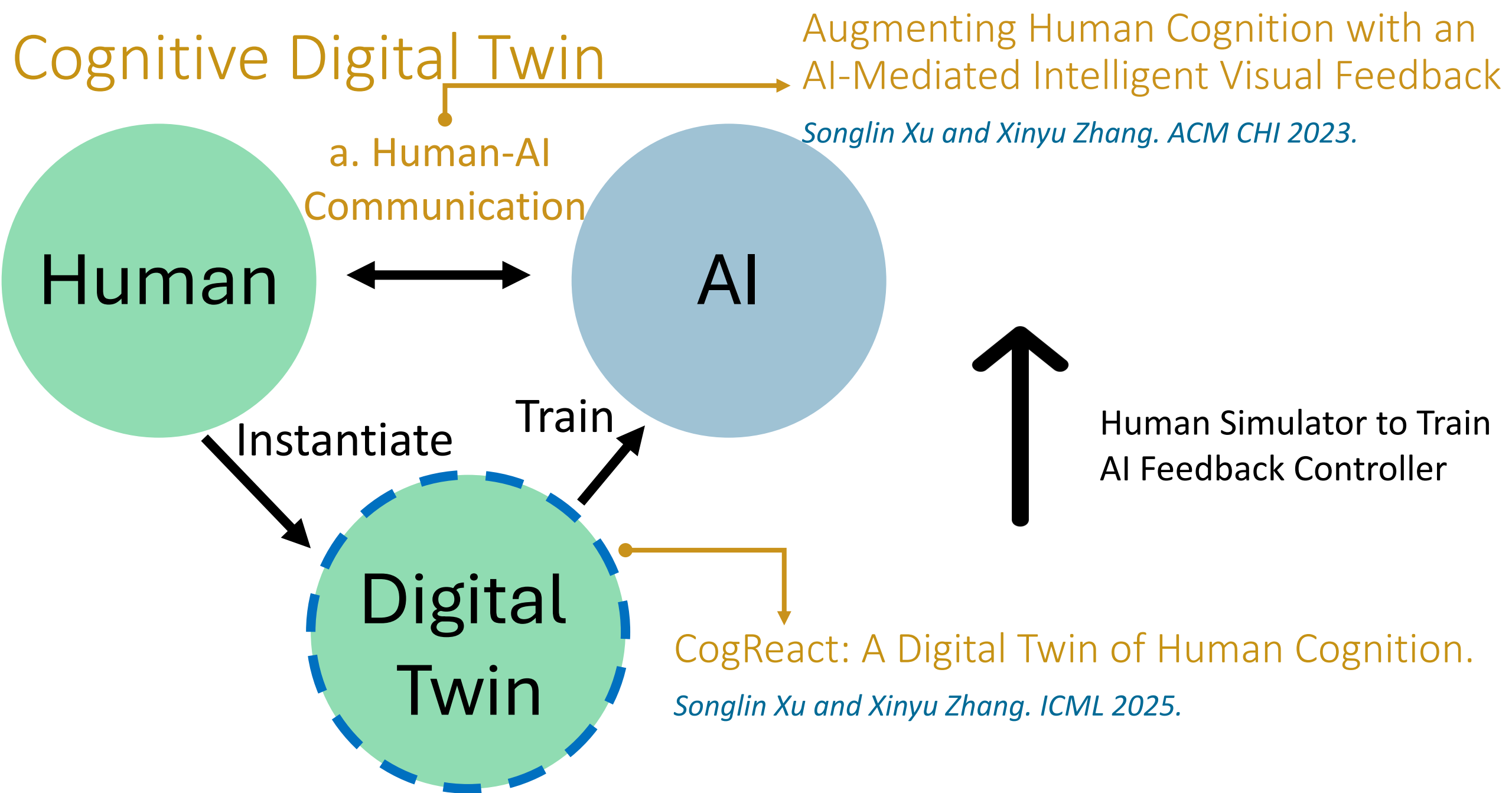
Feedback from Baseline

Participant ID

Participant ID

Insight





CogReact: A Digital Twin of Human Cognition

Songlin Xu and Xinyu Zhang. ICML 2025.

Decades of existing research efforts on modeling human cognition.

nature human behaviour

Article

<https://doi.org/10.1038/s41562-022-01510-8>

Modelling human behaviour in cognitive tasks with latent dynamical systems

Jaffe et al. 2023

CogReact: A Digital Twin of Human Cognition

Songlin Xu and Xinyu Zhang. ICML 2025.

Decades of existing research efforts on modeling human cognition.

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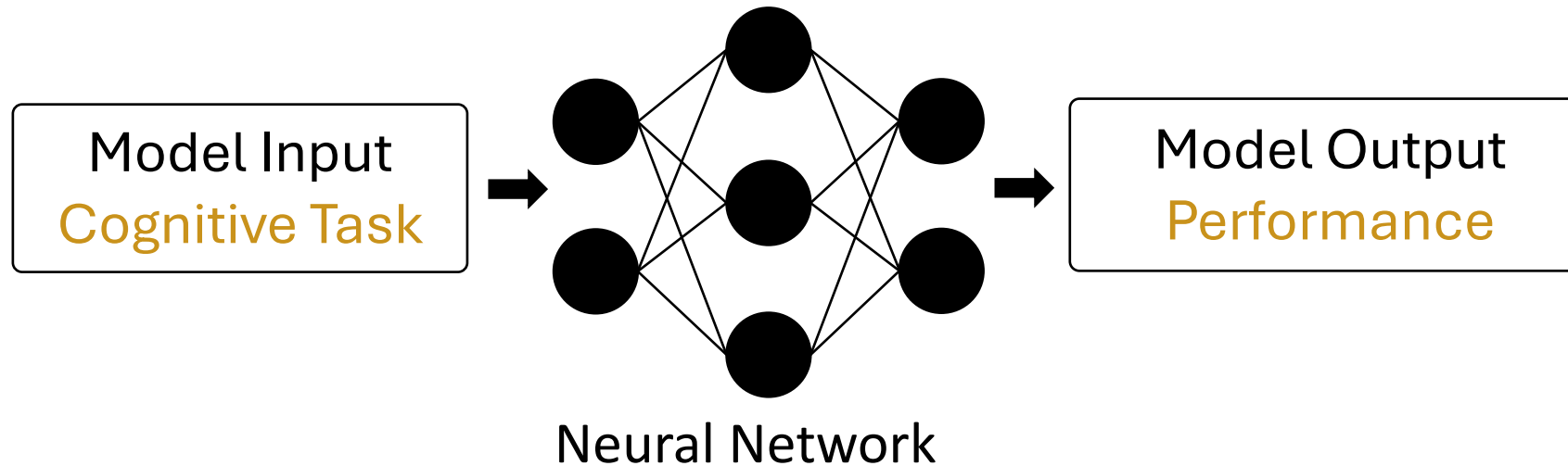
A foundation model to predict and capture human cognition

Binz et al. 2025

CogReact: A Digital Twin of Human Cognition

Songlin Xu and Xinyu Zhang. ICML 2025.

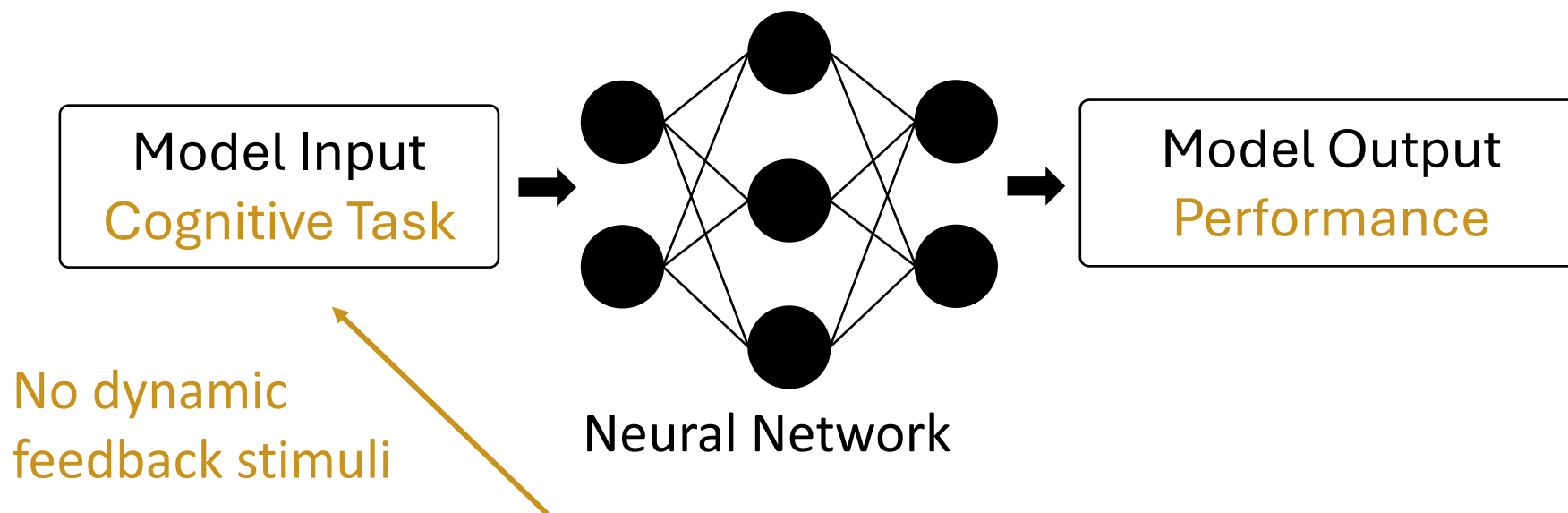
Decades of existing research efforts on modeling human cognition.



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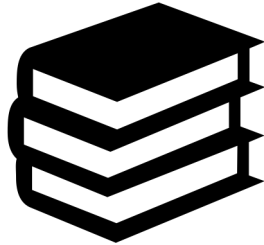


→ Assume idealized/static conditions.

Dynamic Feedback Stimuli Are Ubiquitous

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Regulating users' attention and decision making on products.



\$ 5.99

Dynamic Feedback Stimuli Are Ubiquitous

Regulating users' attention and decision making on products.



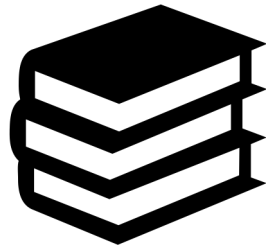
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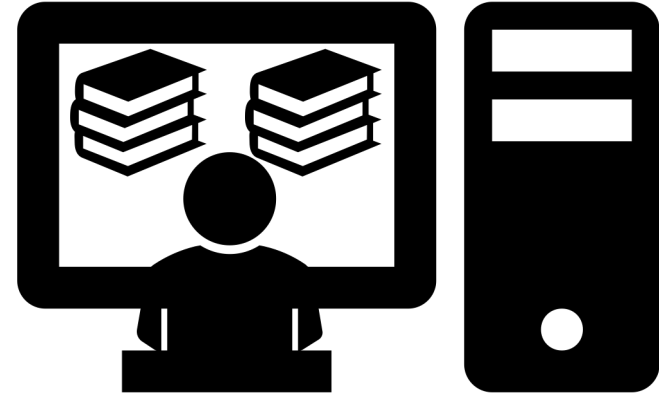


Dynamic Feedback Stimuli Are Ubiquitous

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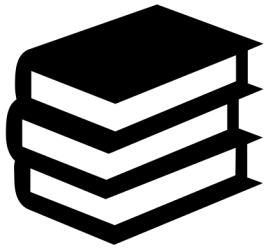
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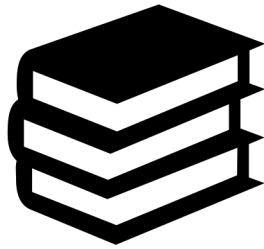
\$ 299.99

Dynamic Feedback Stimuli Are Ubiquitous

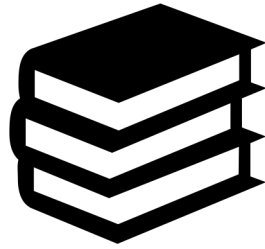
Regulating users' attention and decision making on products.



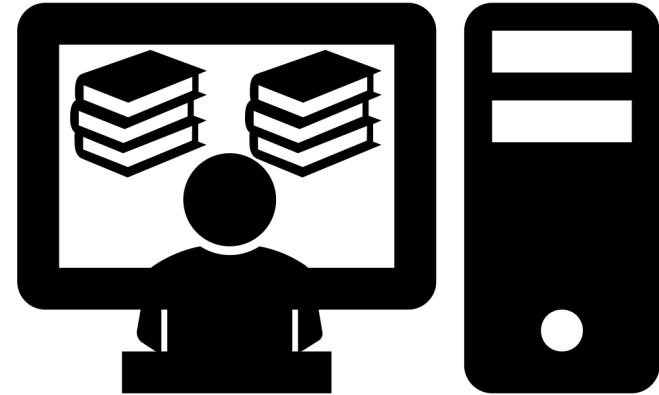
\$ 0.99



\$ 5.99



\$ 59.99



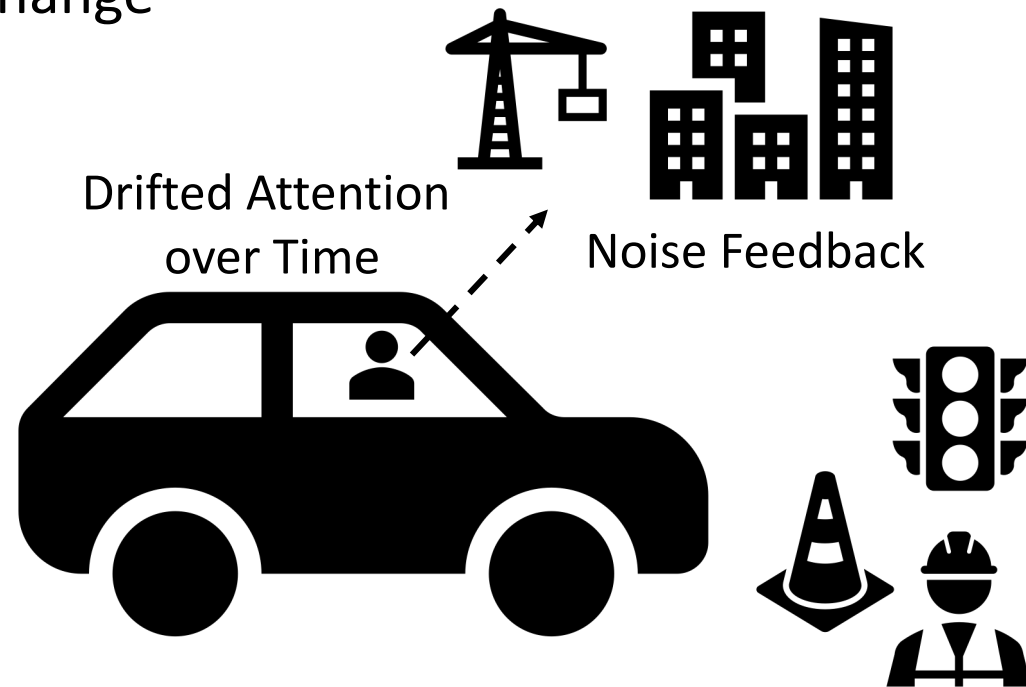
\$ 299.99

Dynamic Feedback Stimuli Are Ubiquitous



Dynamic Feedback Stimuli Are Ubiquitous

Failure to capture this change
is dangerous!

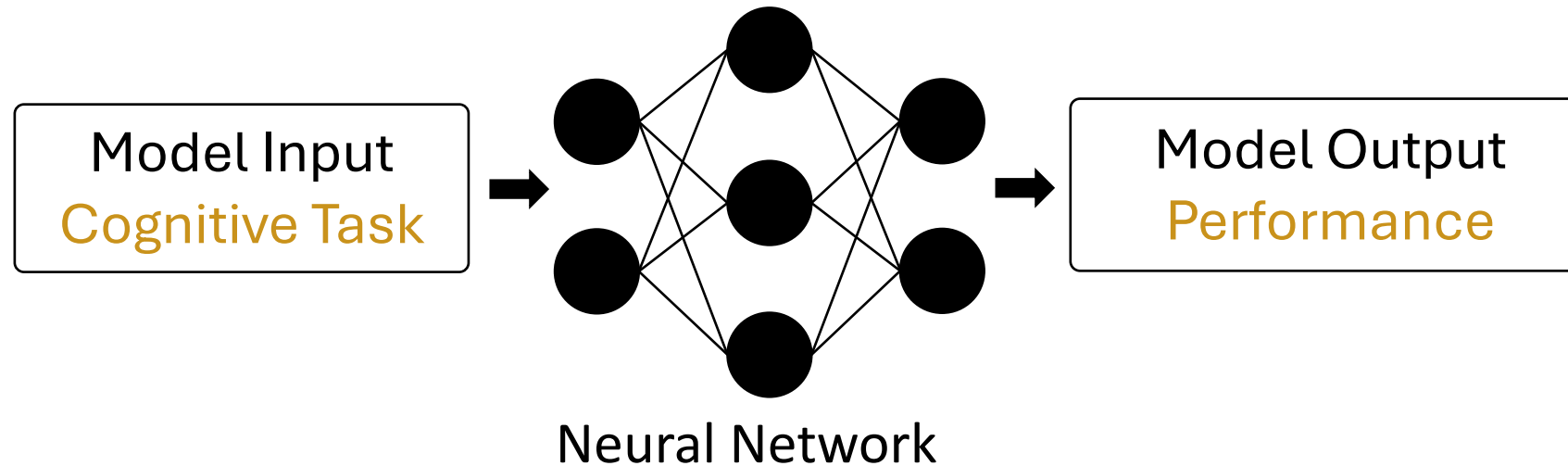


Modeling how **cognitive processes** are modulated
by dynamic feedback stimuli is essential.

Model Cognitive Process Modulated by Dynamic Feedback Stimuli

Model Cognitive Process Modulated by Dynamic Feedback Stimuli

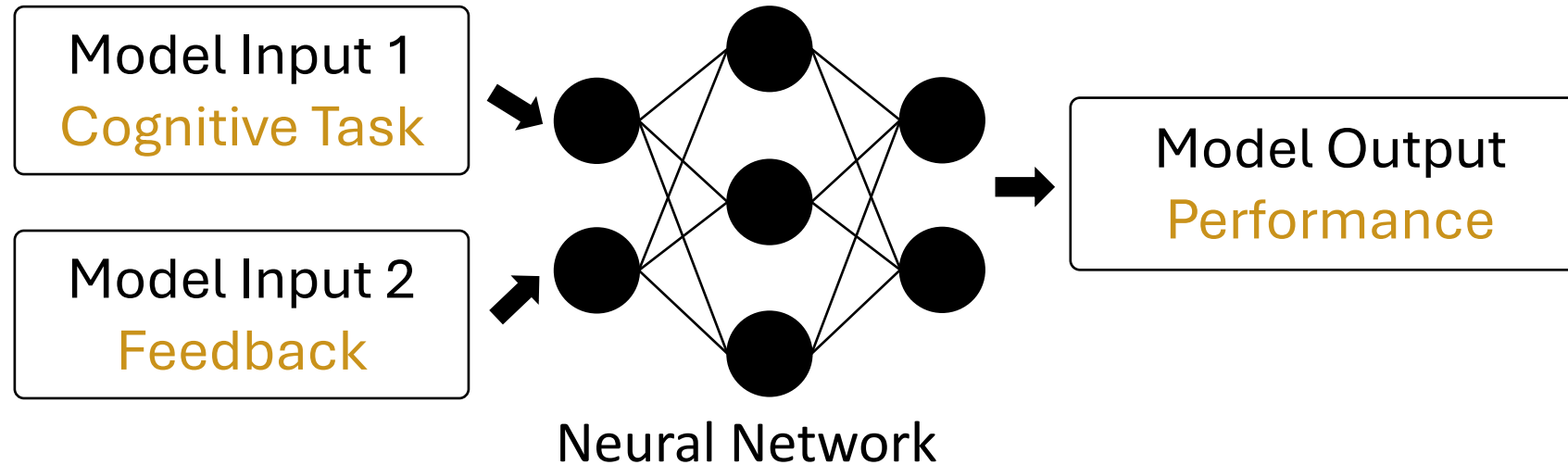
Existing approaches.



→ Assume **idealized/static** conditions.

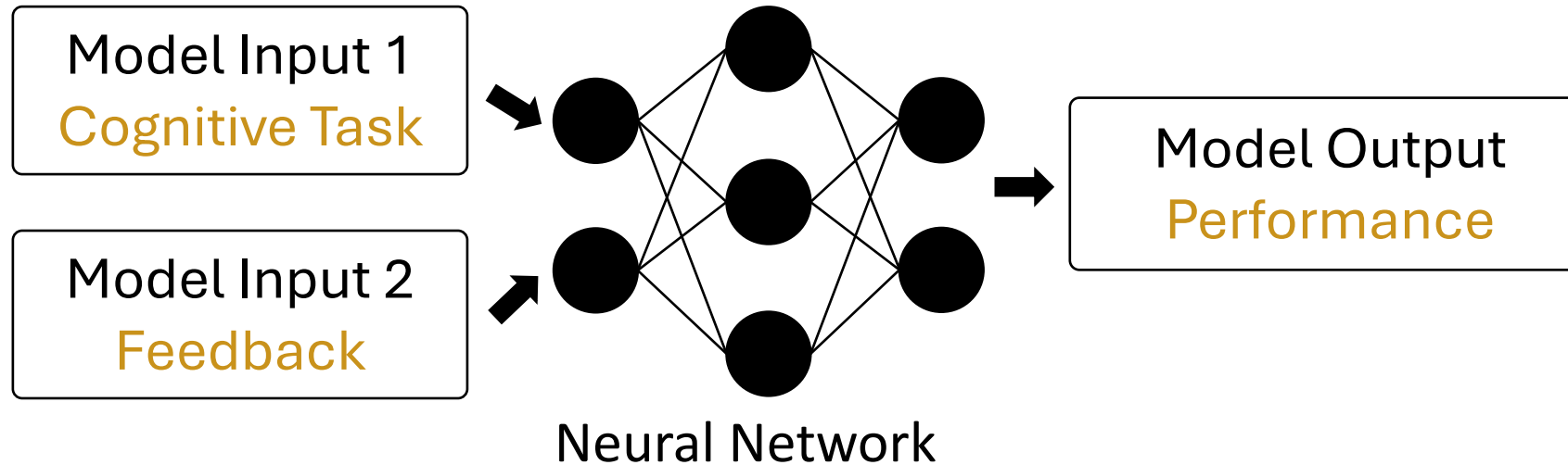
Model Cognitive Process Modulated by Dynamic Feedback Stimuli

Modifying existing approaches.



Model Cognitive Process Modulated by Dynamic Feedback Stimuli

Modifying existing approaches.



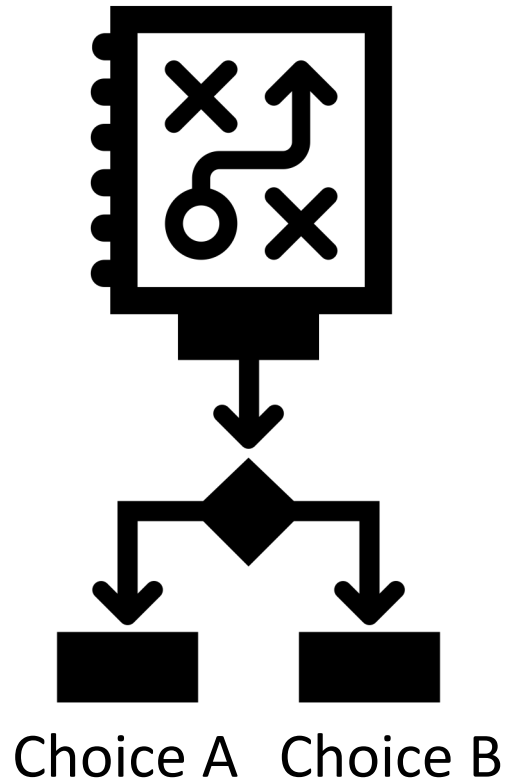
Predict Cognitive Output / Performance



Model Cognitive Process over Time by Feedback

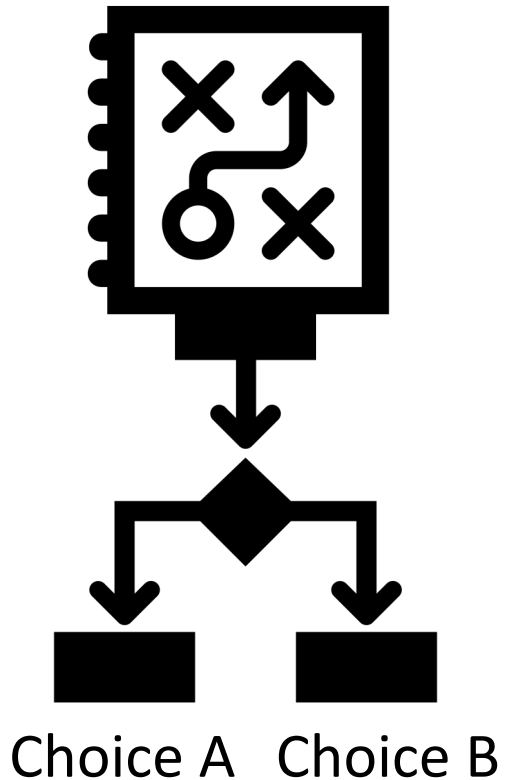
Our Approach: Insights from Cognitive Science

Logical Reasoning Task

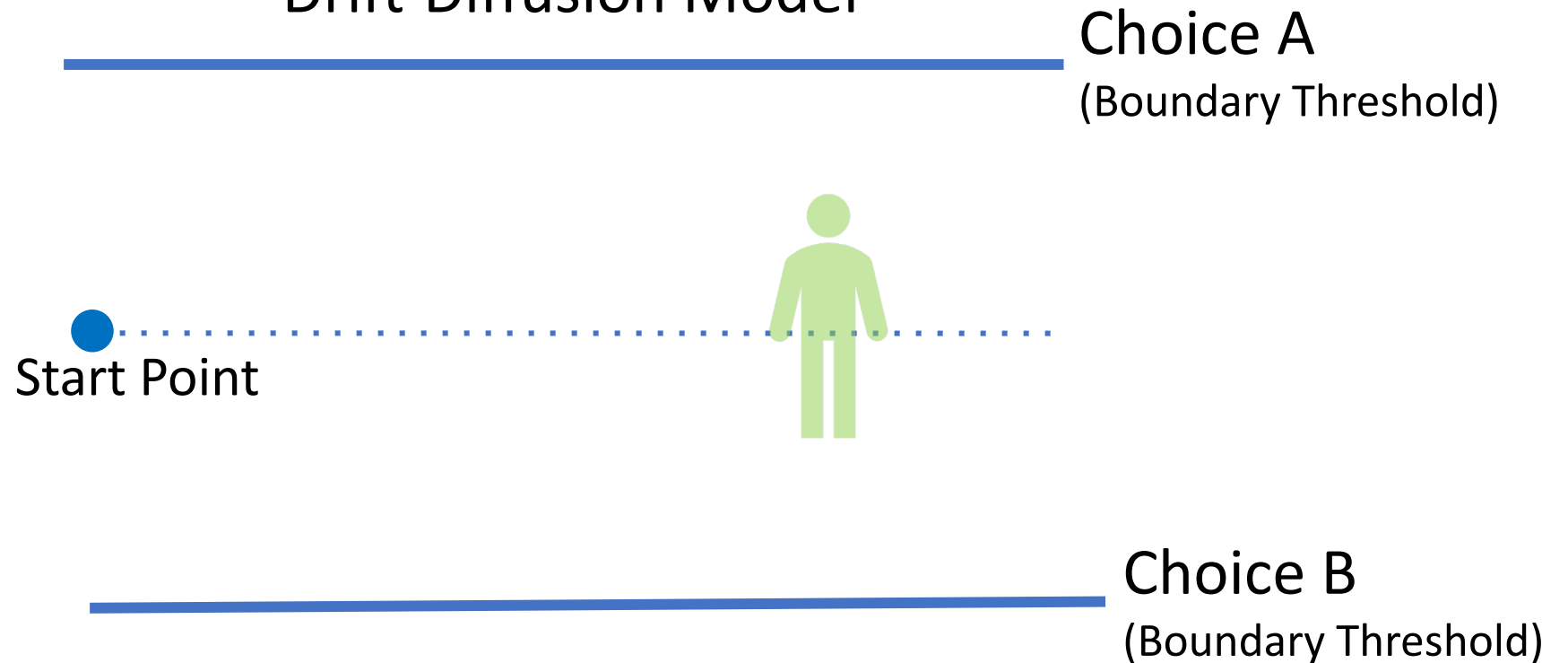


Our Approach: Insights from Cognitive Science

Logical Reasoning Task



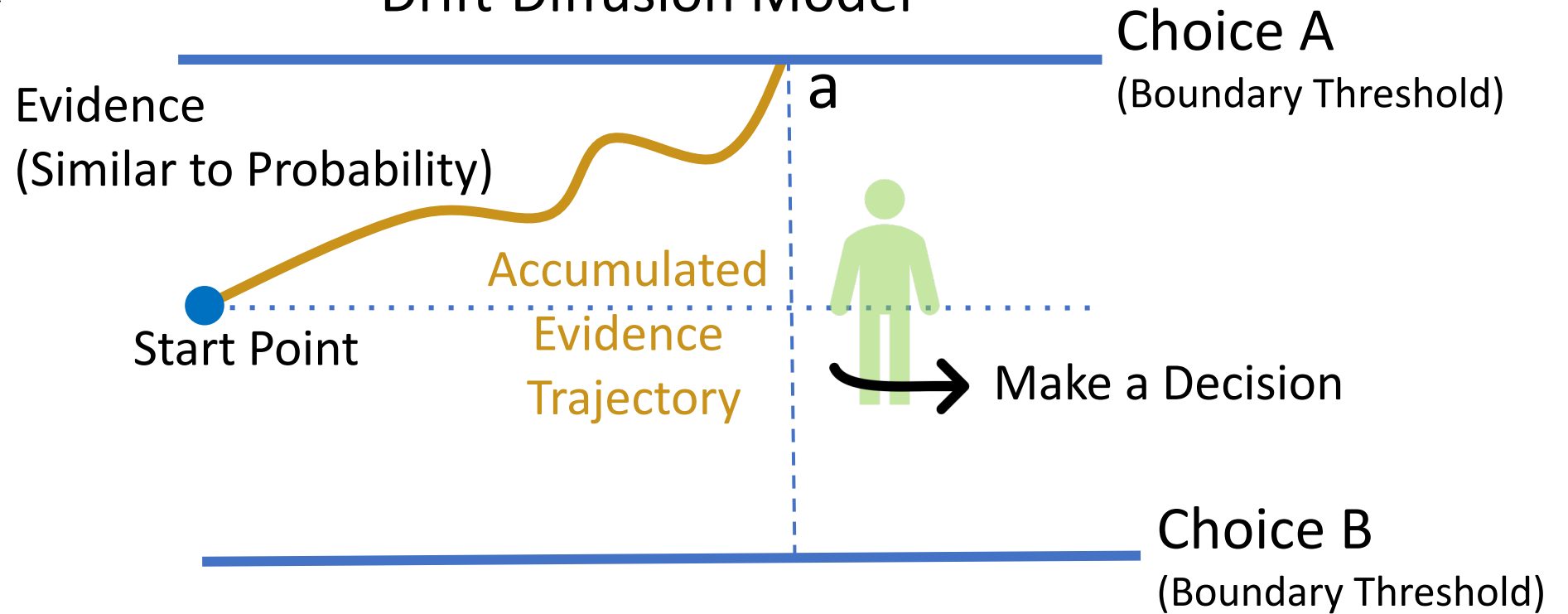
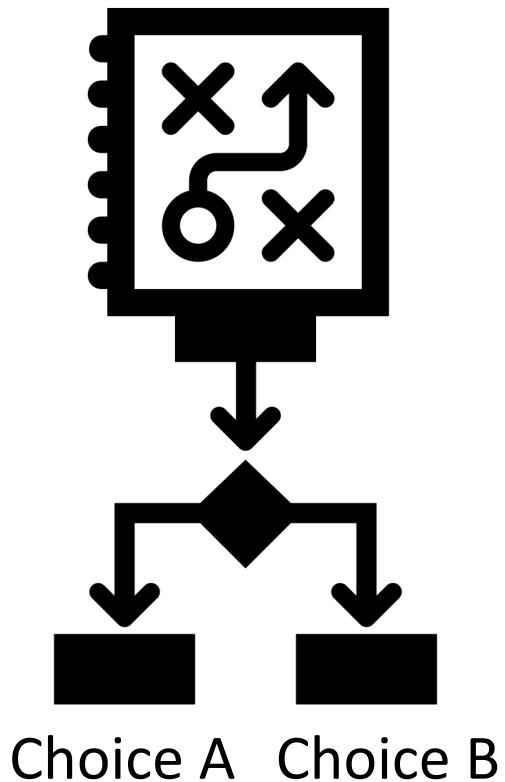
Drift-Diffusion Model



Our Approach: Insights from Cognitive Science

Logical Reasoning Task

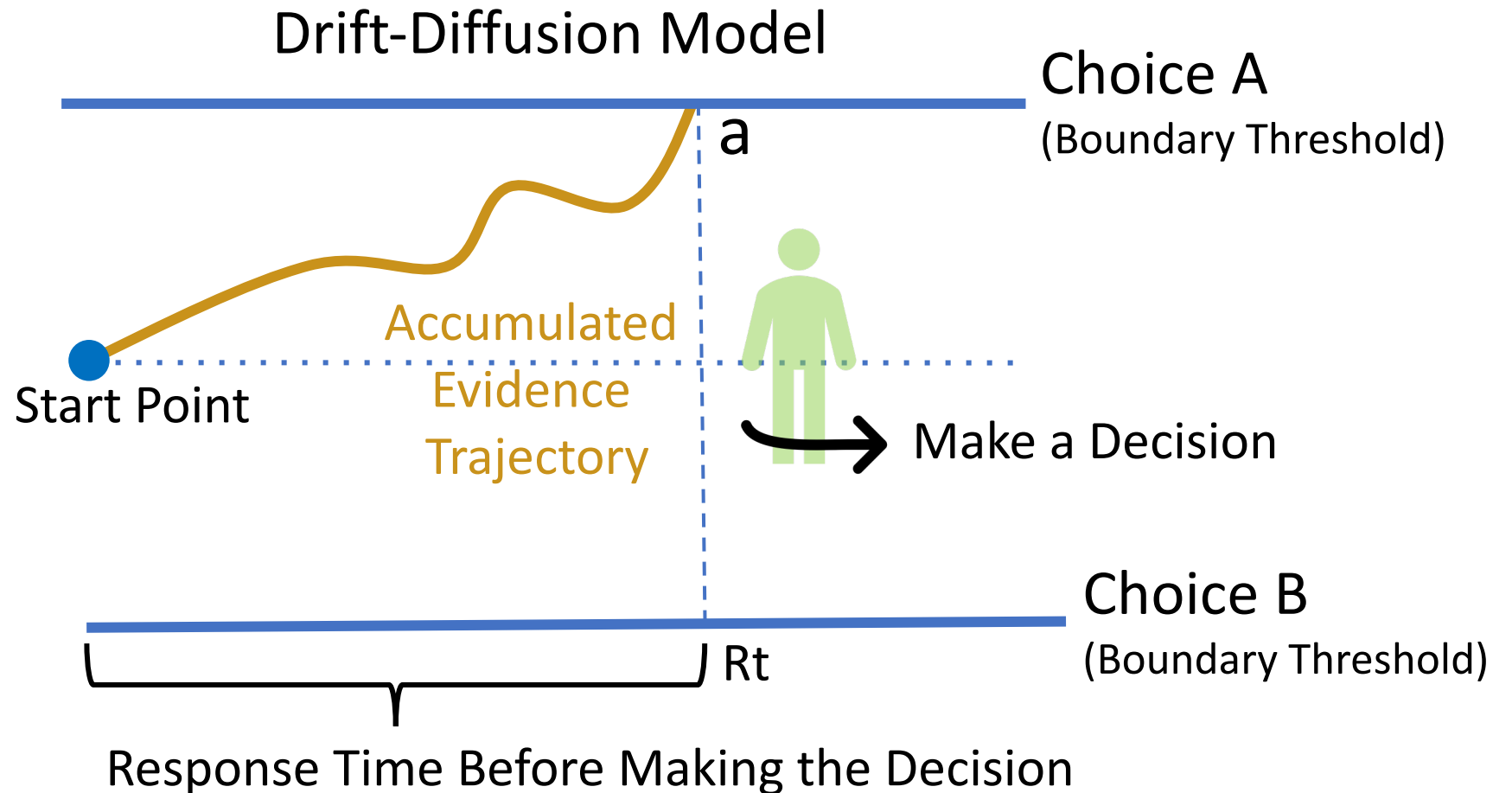
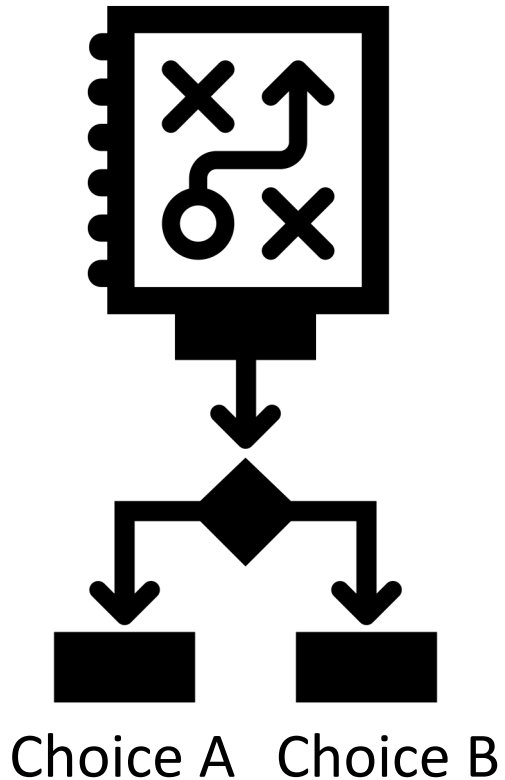
Drift-Diffusion Model



Evidence: how much the current information favors one choice over the other at each moment.

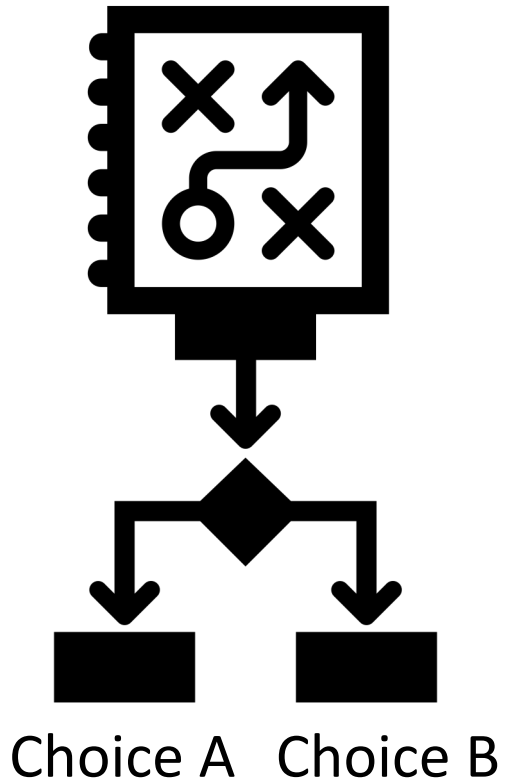
Our Approach: Insights from Cognitive Science

Logical Reasoning Task

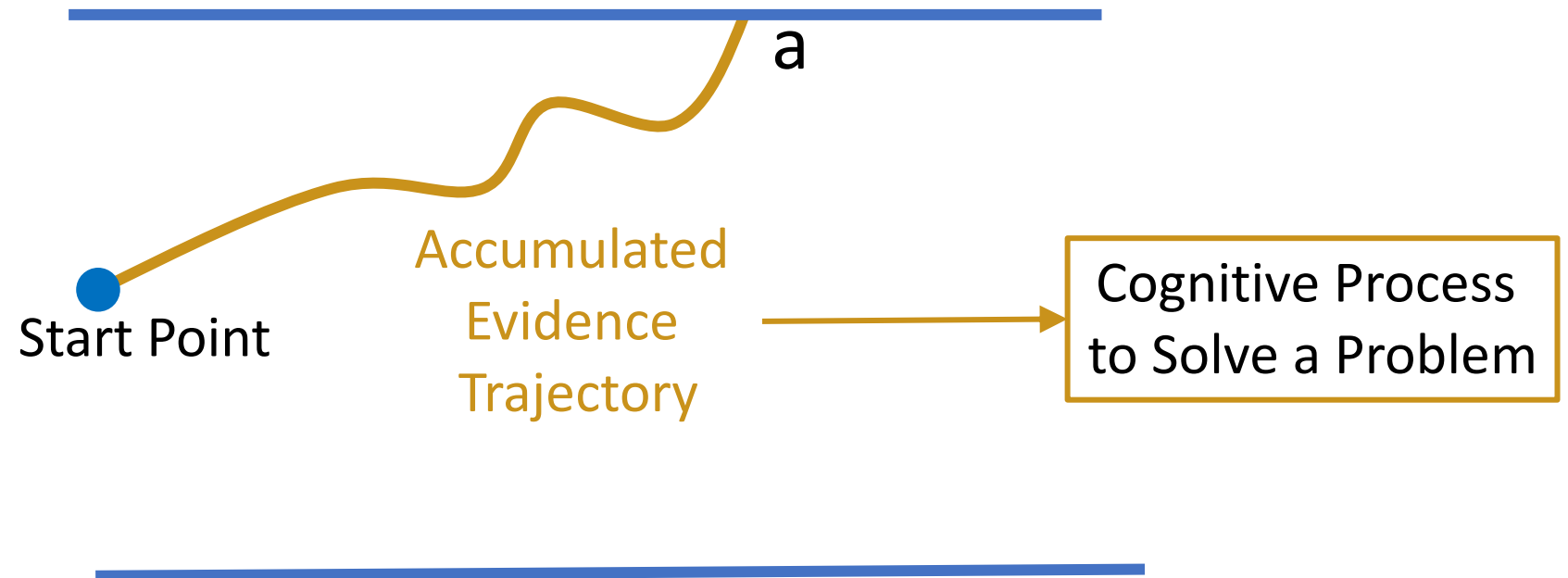


Our Approach: Insights from Cognitive Science

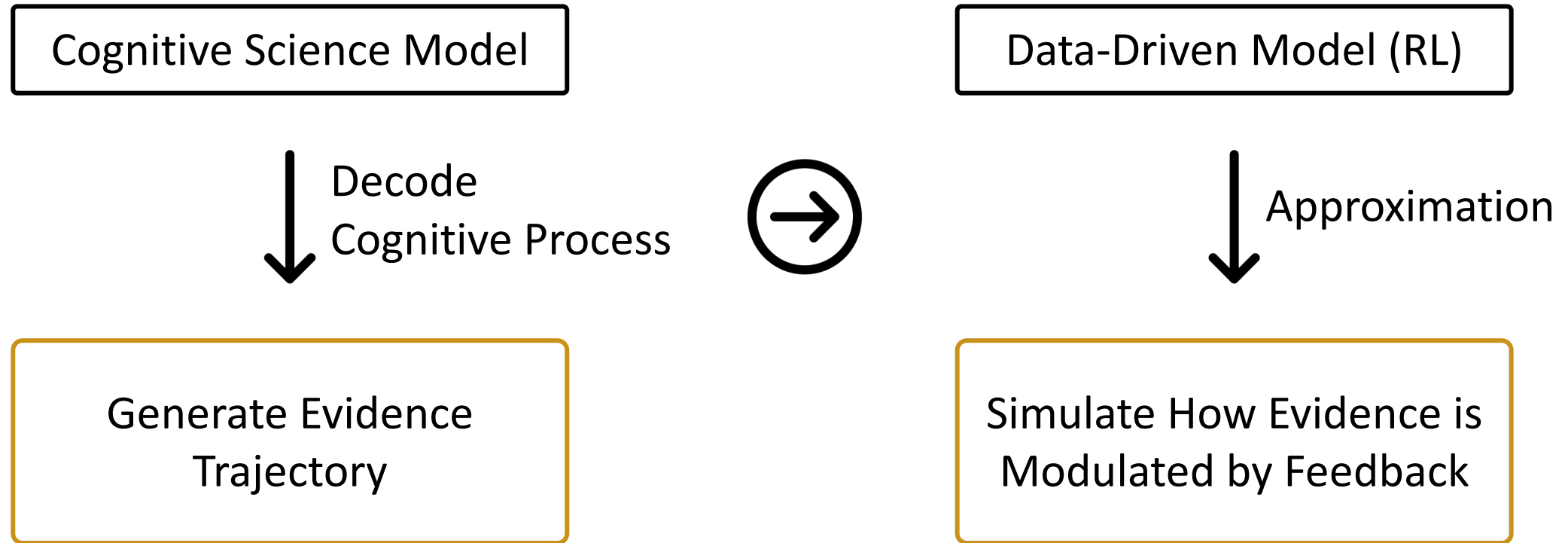
Logical Reasoning Task



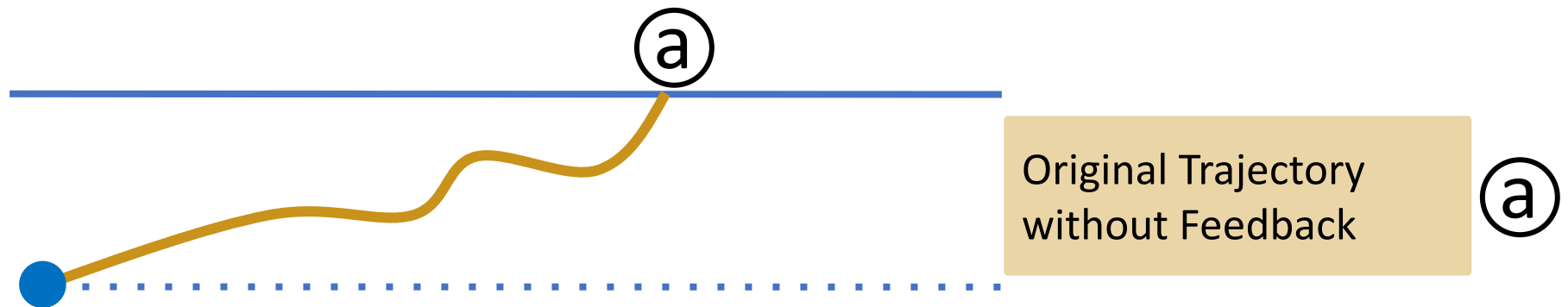
Drift-Diffusion Model



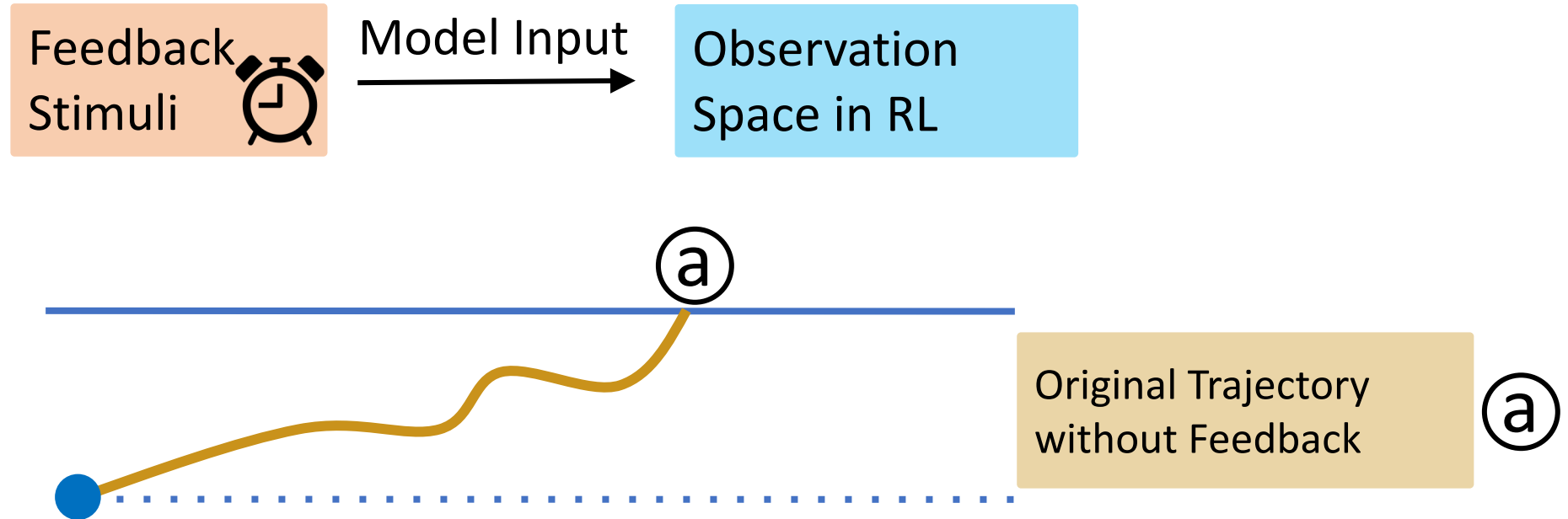
Encoding Feedback Stimuli Impact on Cognition



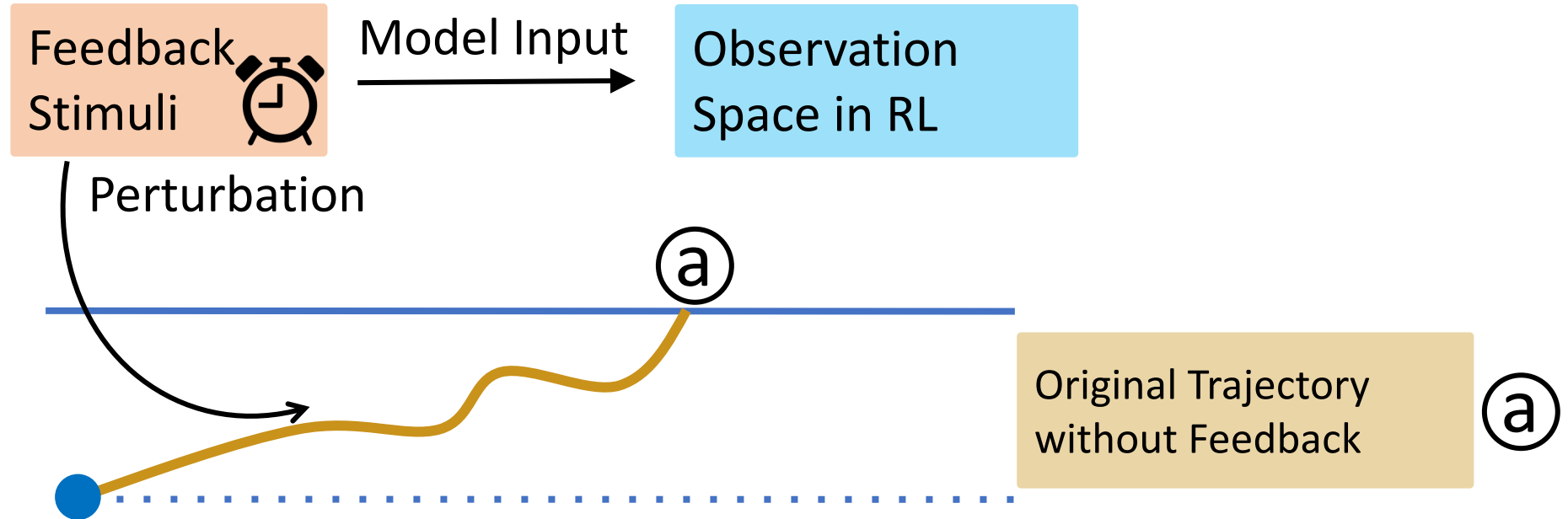
Encoding Feedback Stimuli Impact on Cognition



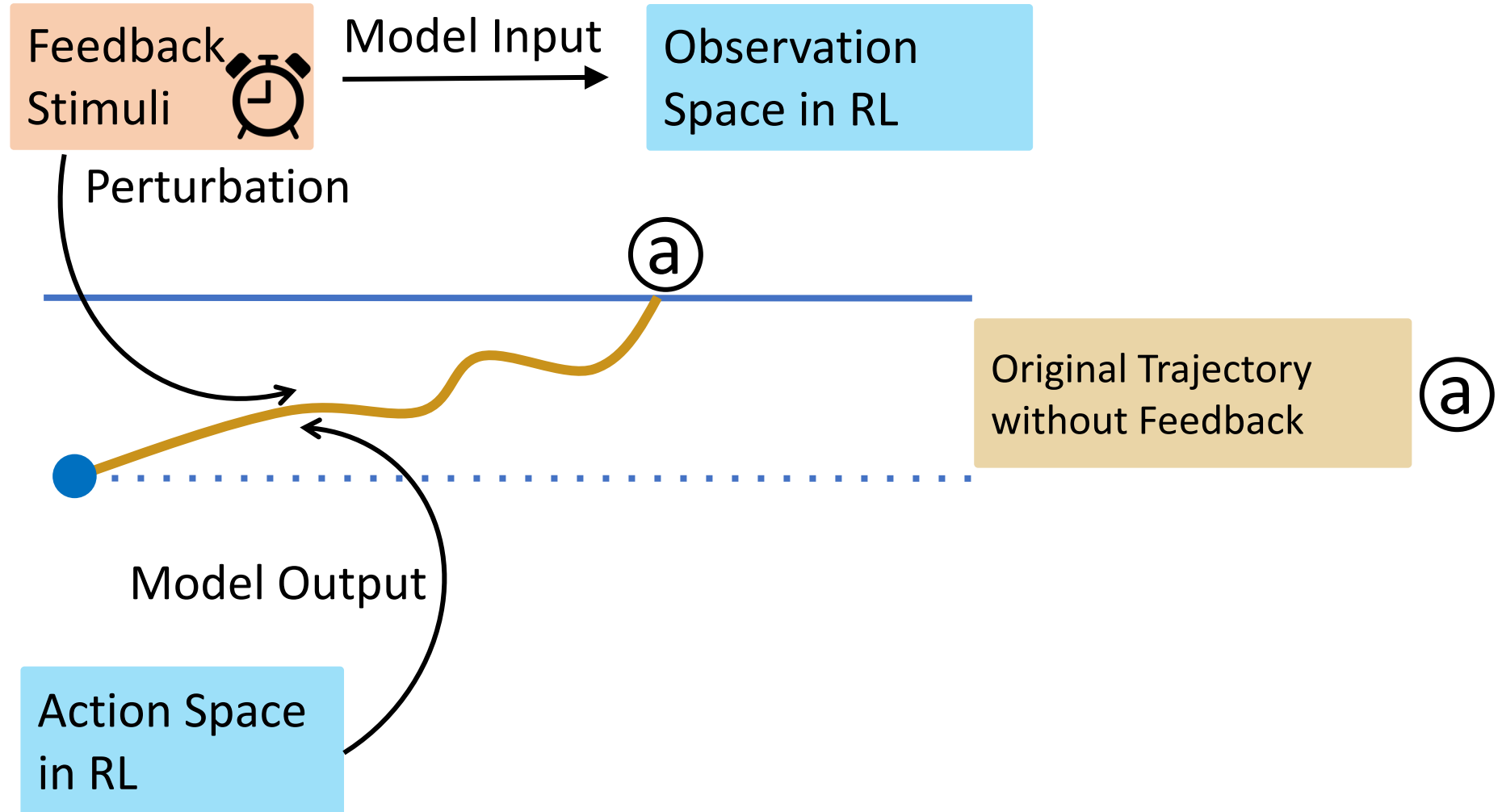
Encoding Feedback Stimuli Impact on Cognition



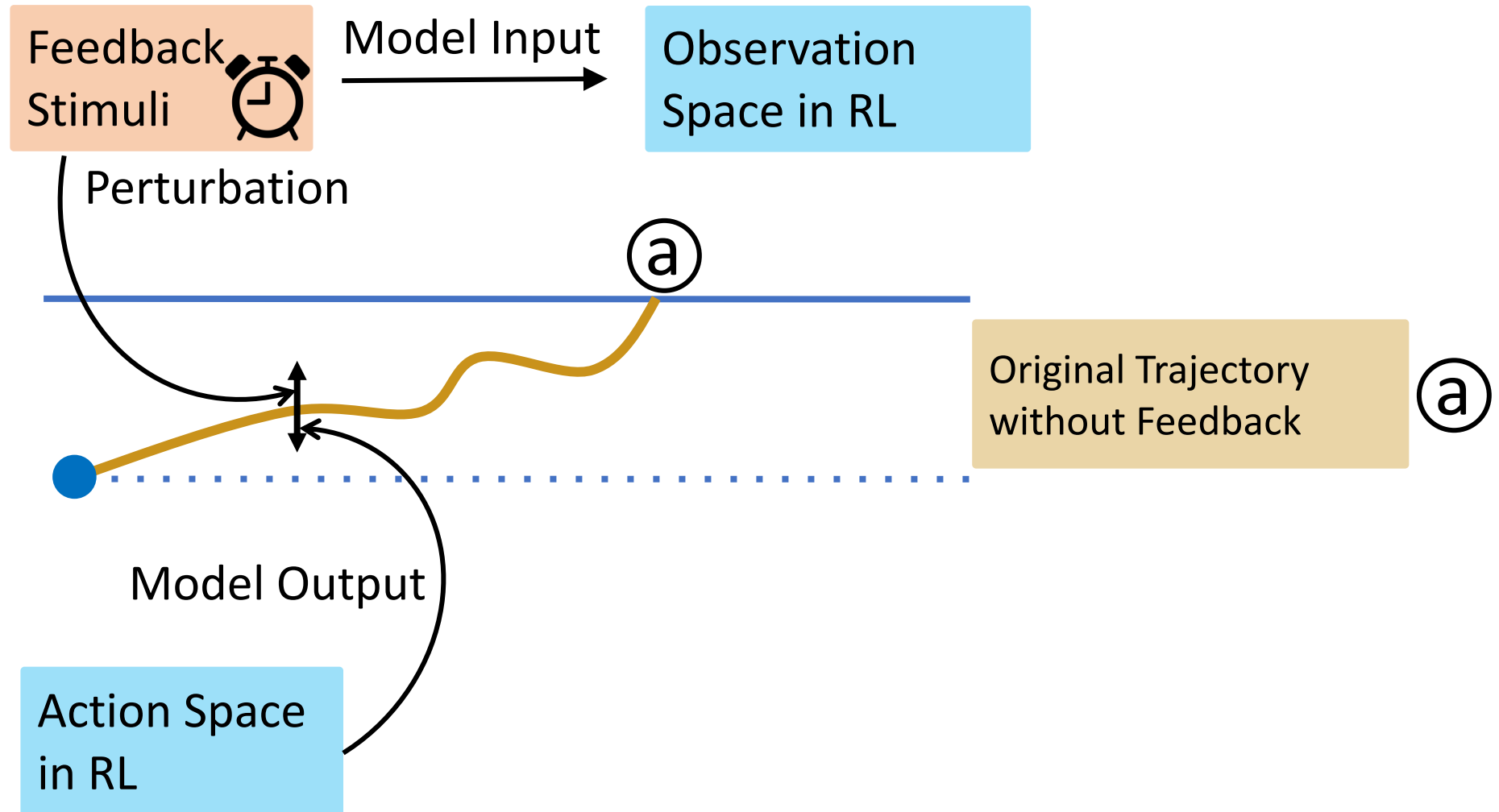
Encoding Feedback Stimuli Impact on Cognition



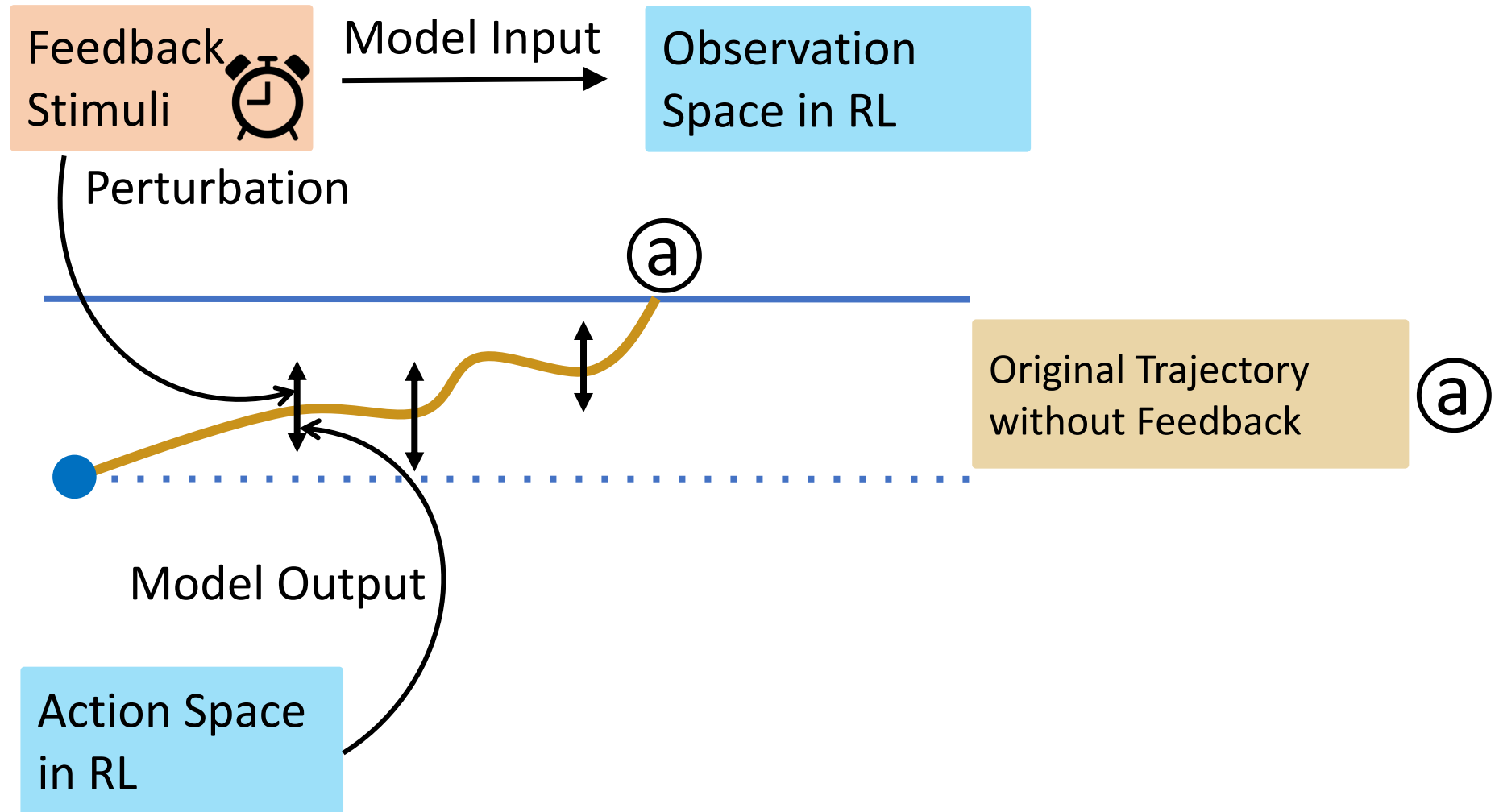
Encoding Feedback Stimuli Impact on Cognition



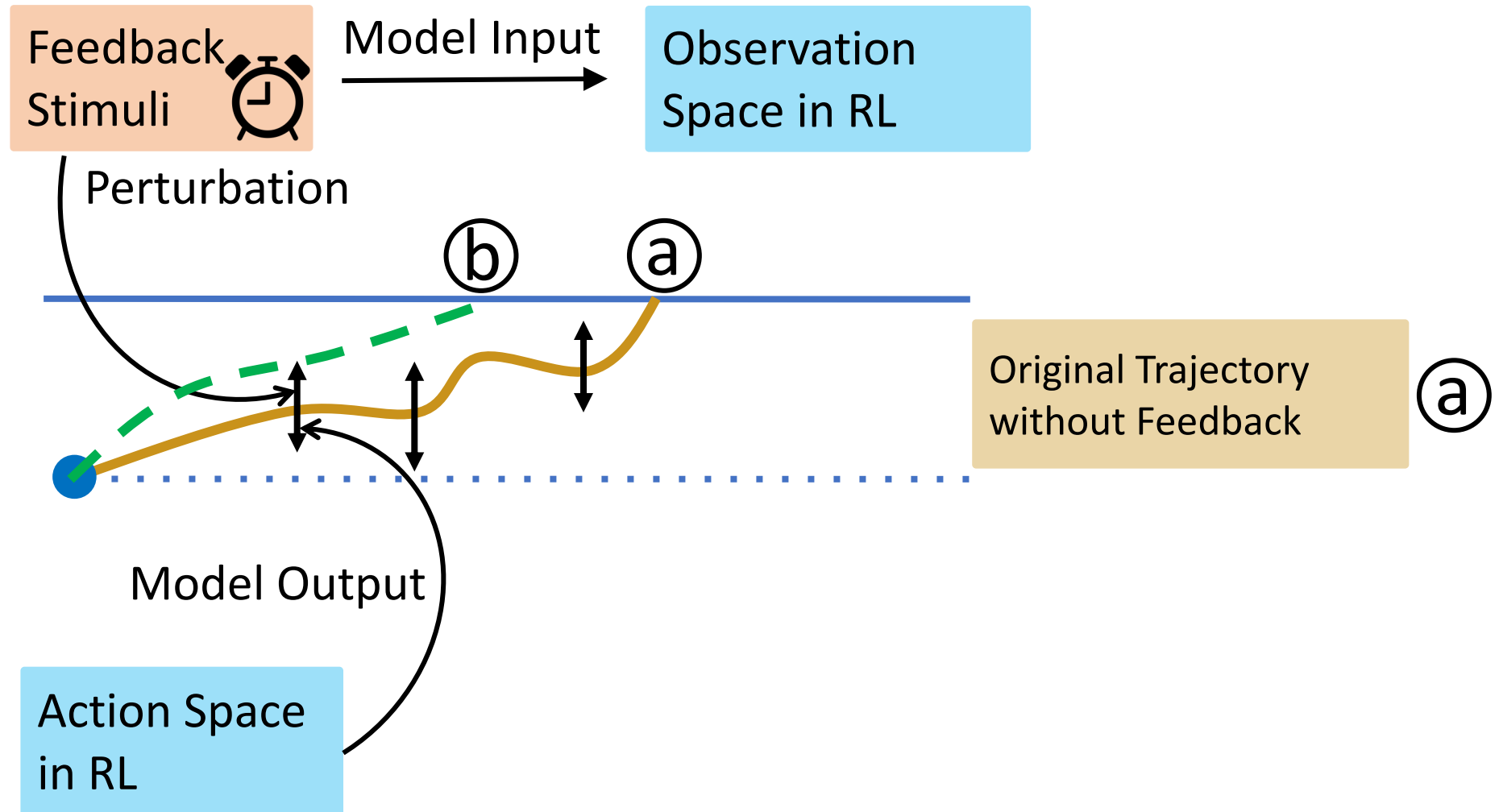
Encoding Feedback Stimuli Impact on Cognition



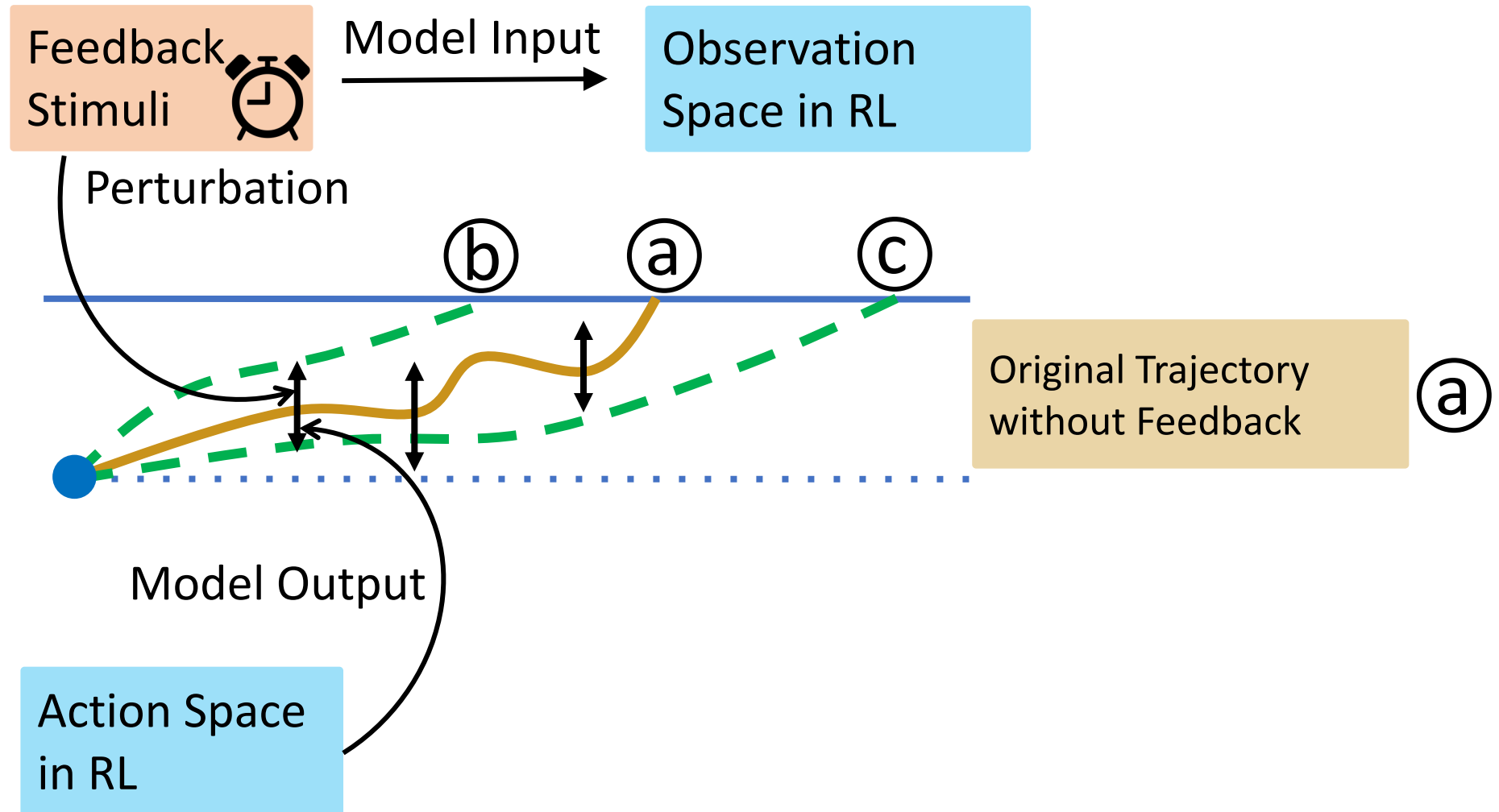
Encoding Feedback Stimuli Impact on Cognition



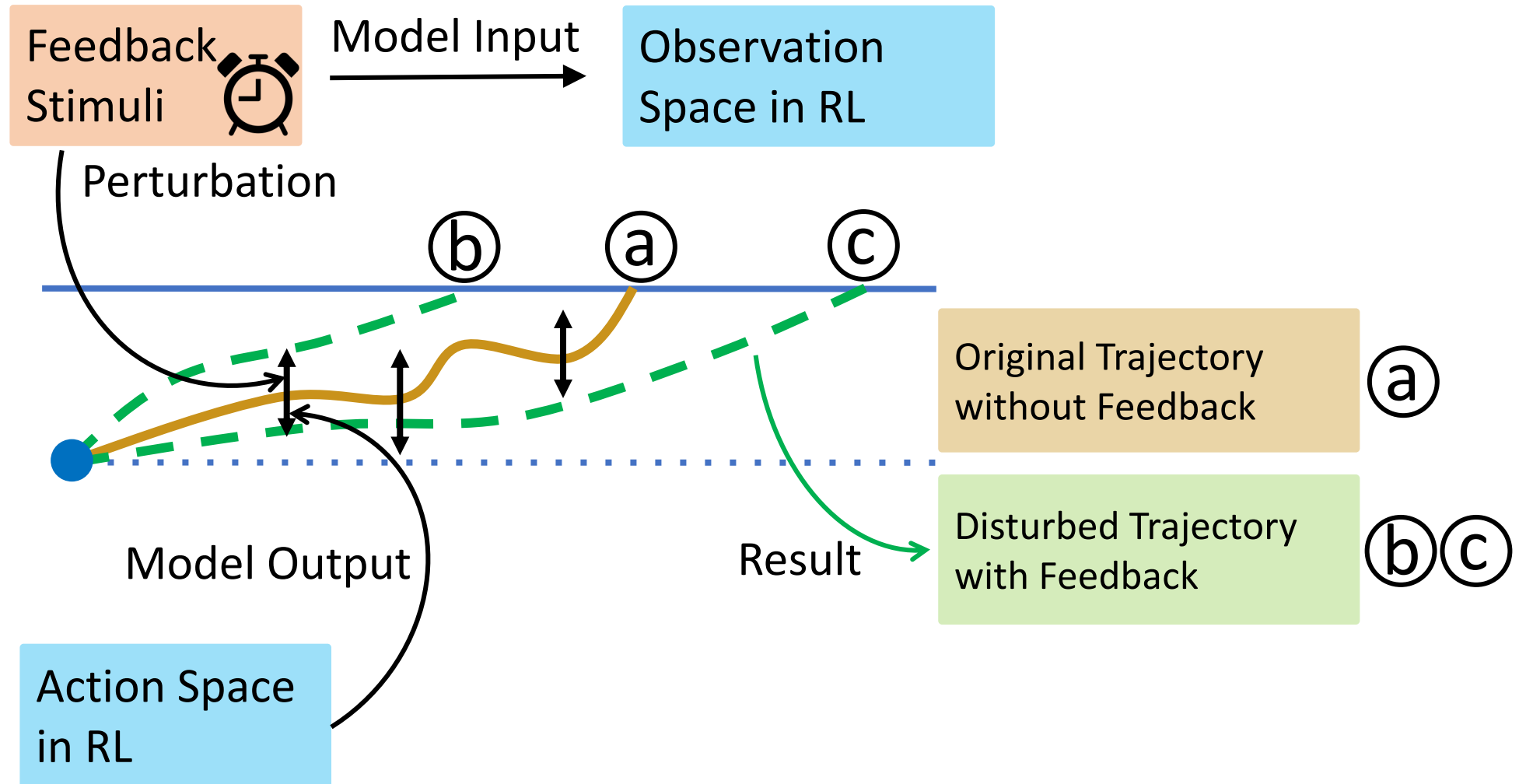
Encoding Feedback Stimuli Impact on Cognition



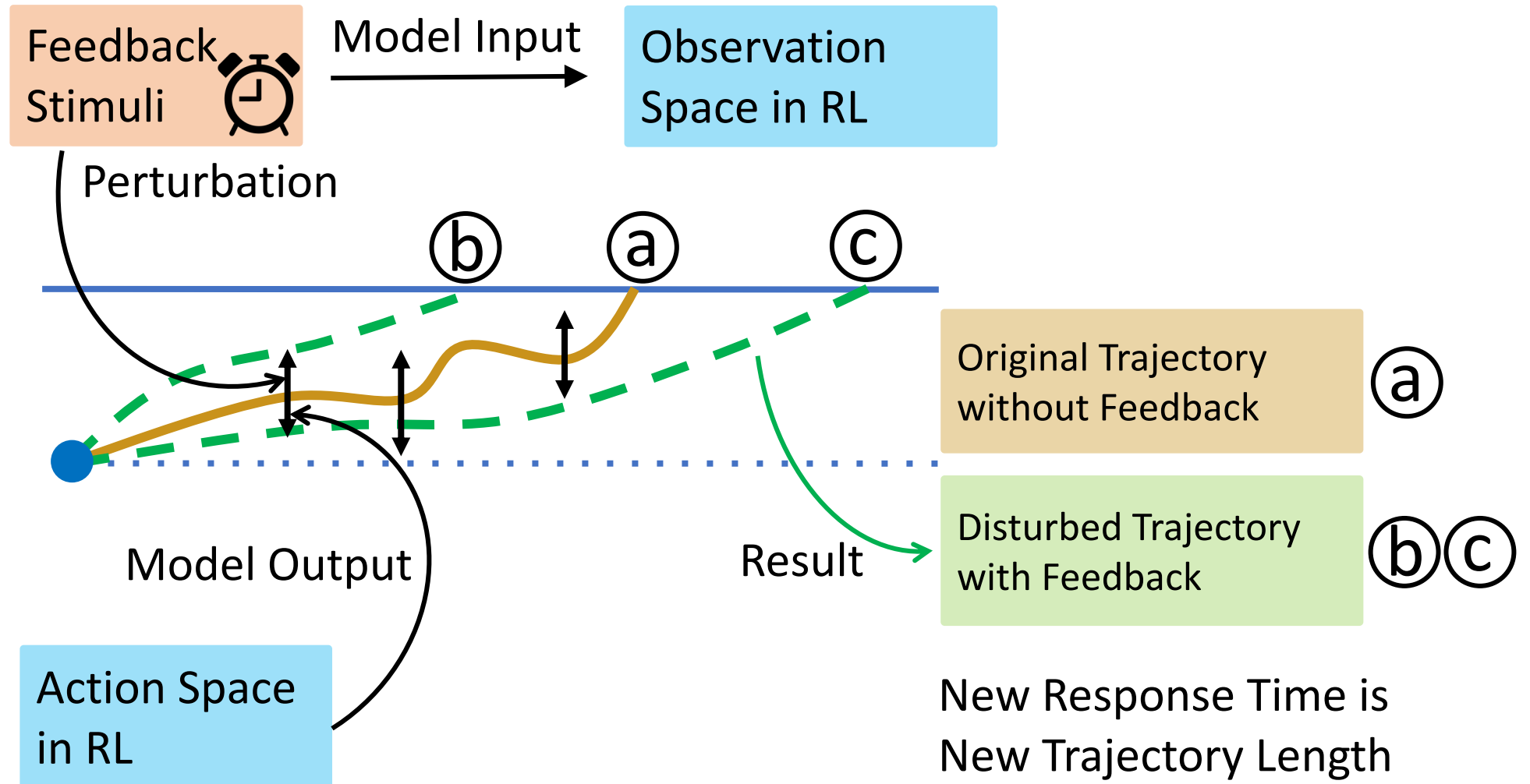
Encoding Feedback Stimuli Impact on Cognition



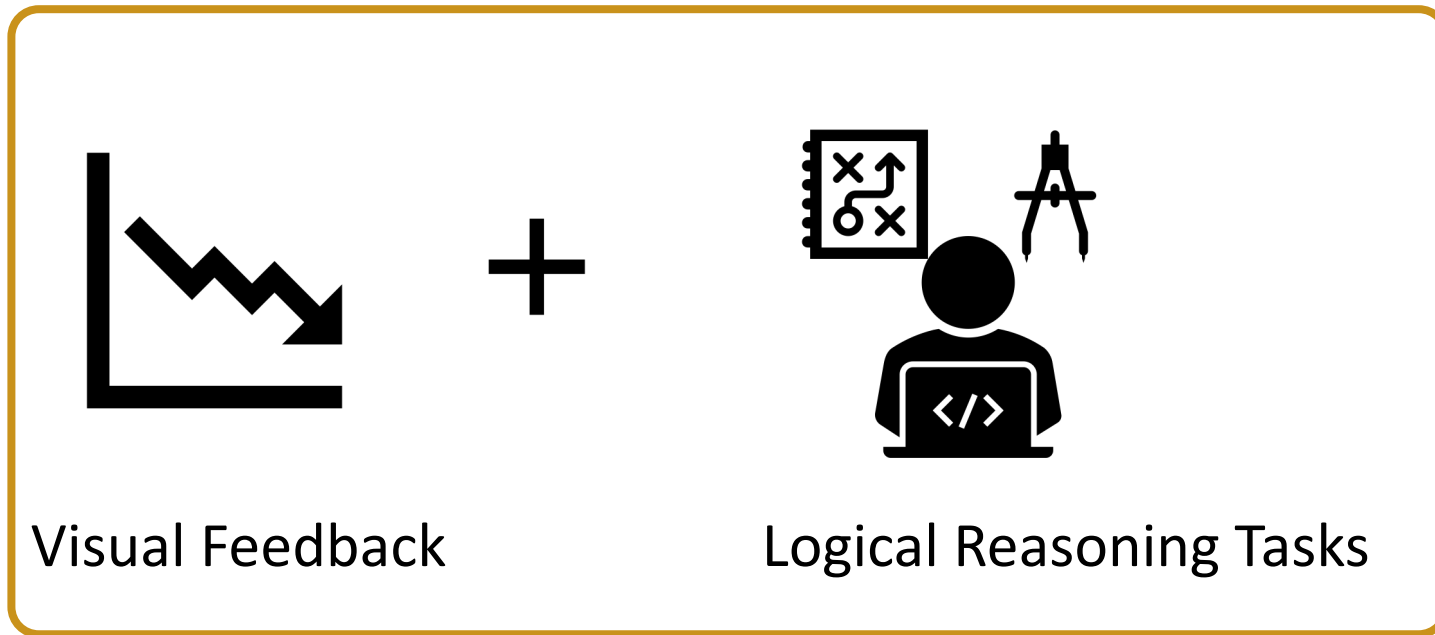
Encoding Feedback Stimuli Impact on Cognition



Encoding Feedback Stimuli Impact on Cognition



Evaluation: Data Collection

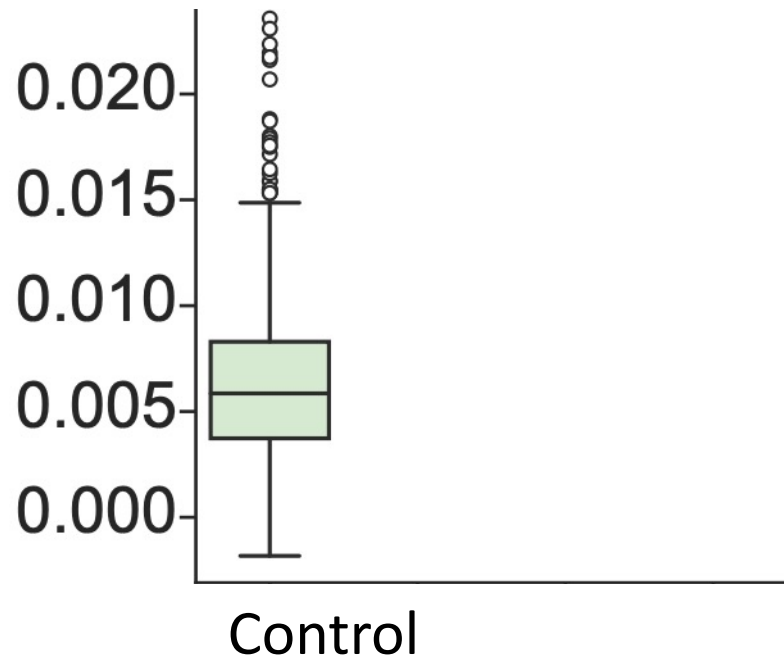


21,157 Responses

Interpretability

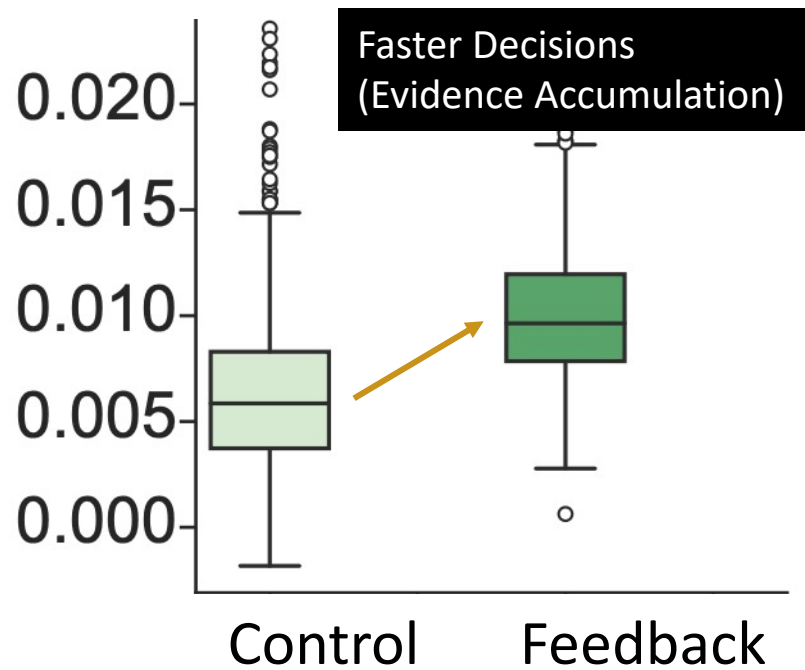
Interpretability

Slope of Simulated
Evidence Accumulation Trajectory



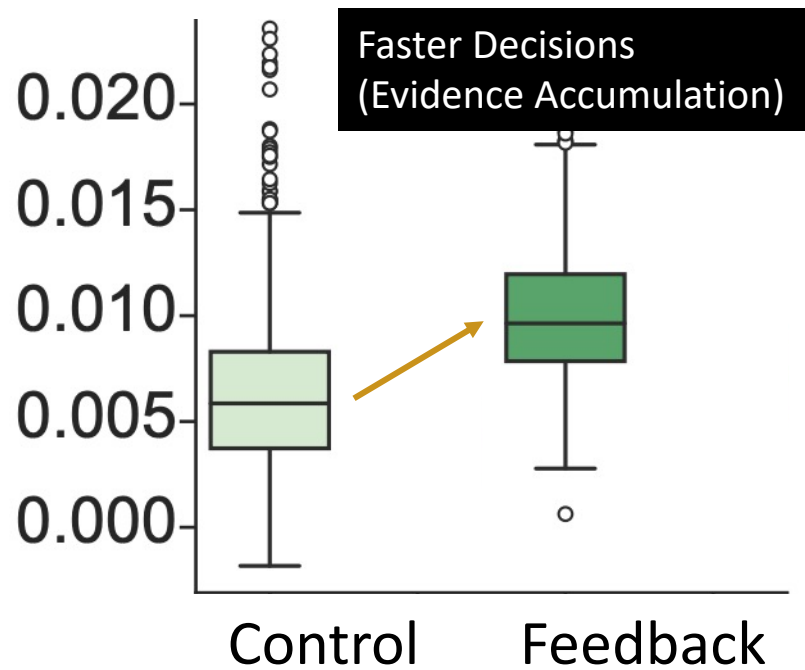
Interpretability

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Evidence Accumulation Trajectory

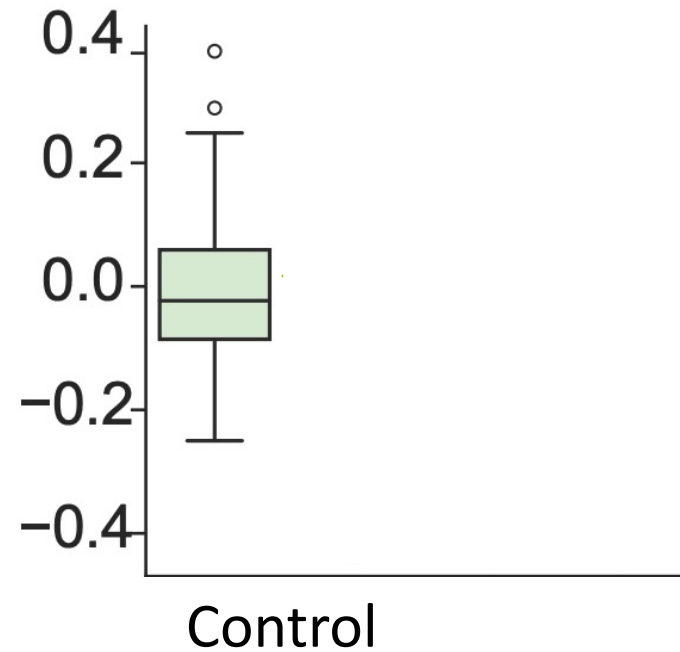


Interpretability

Slope of Simulated
Evidence Accumulation Trajectory

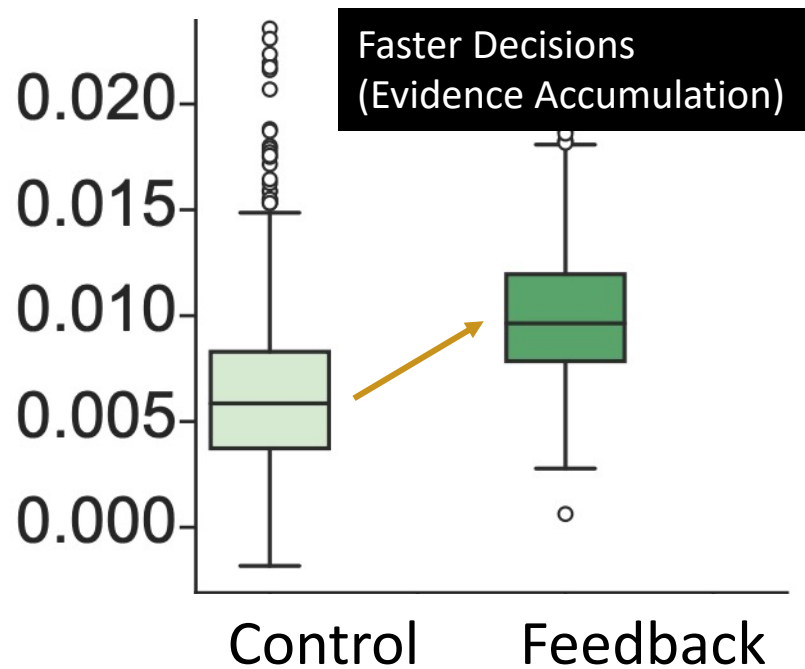


Real Human (Relative)
Response Time (Ground Truth)

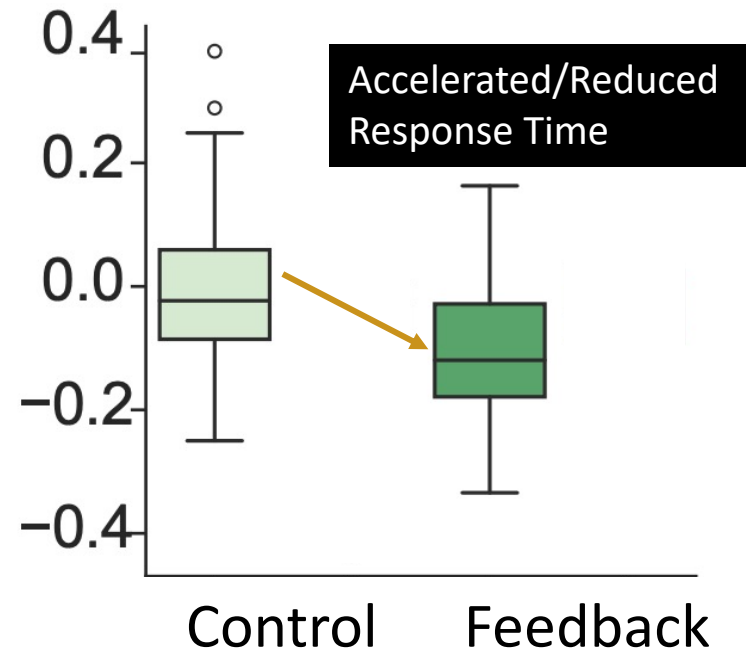


Interpretability

Slope of Simulated
Evidence Accumulation Trajectory



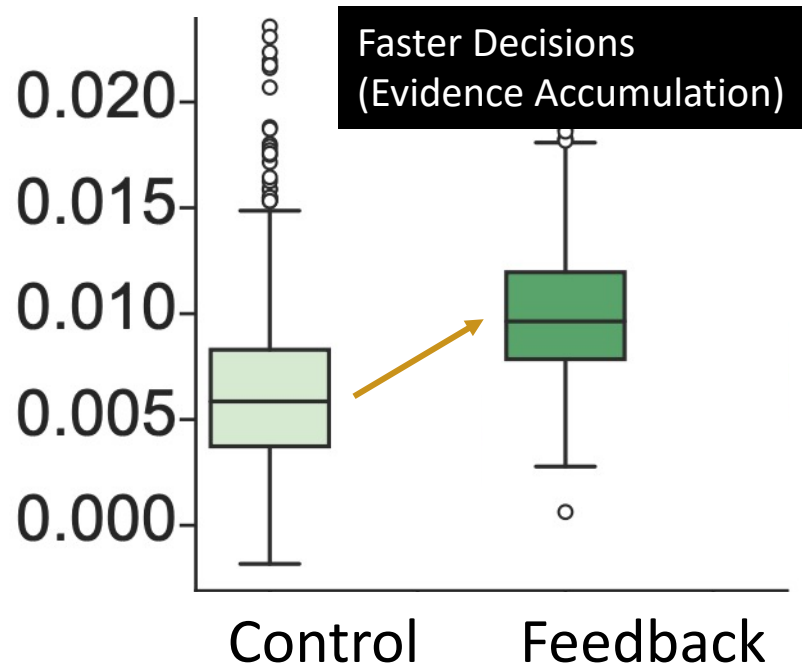
Real Human (Relative)
Response Time (Ground Truth)



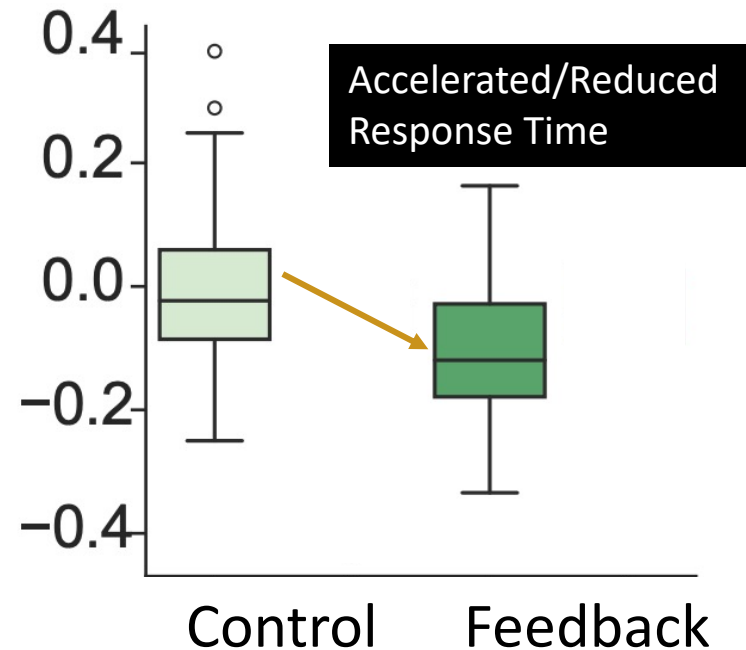
Interpretability

Simulated **cognitive process signatures** align with real humans' performance under feedback stimuli.

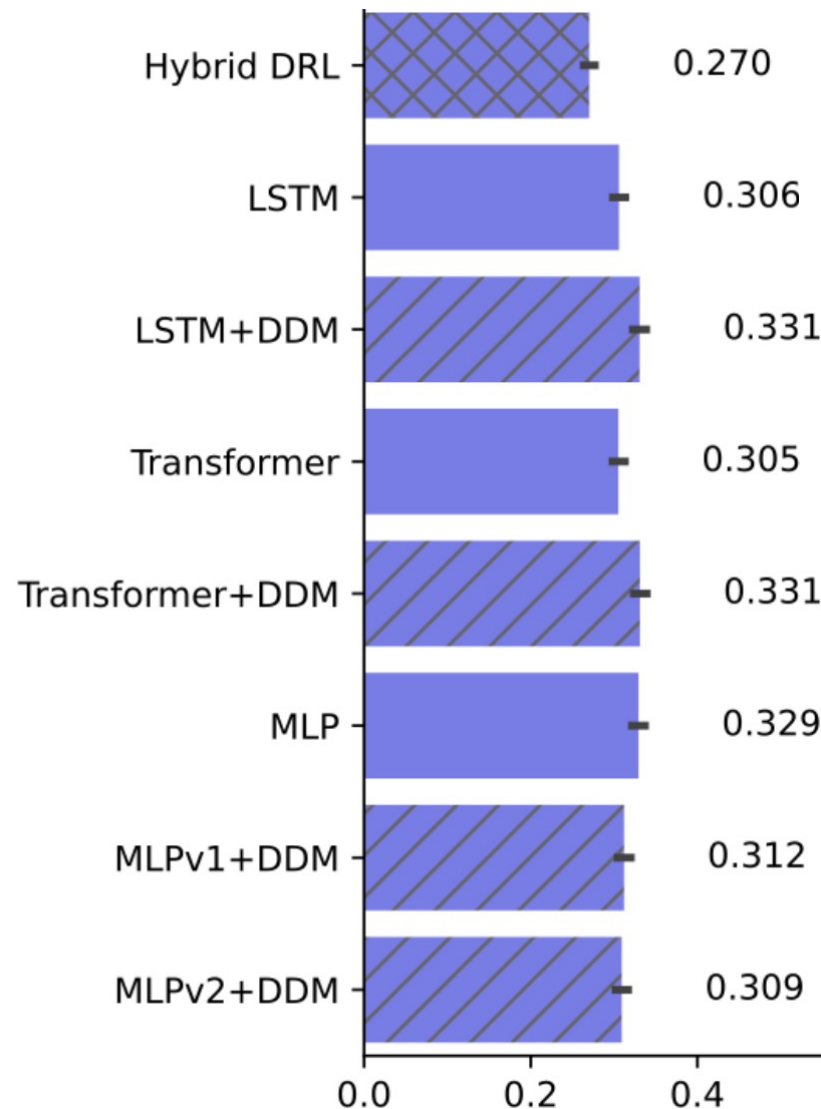
Slope of Simulated
Evidence Accumulation Trajectory



Real Human (Relative)
Response Time (Ground Truth)



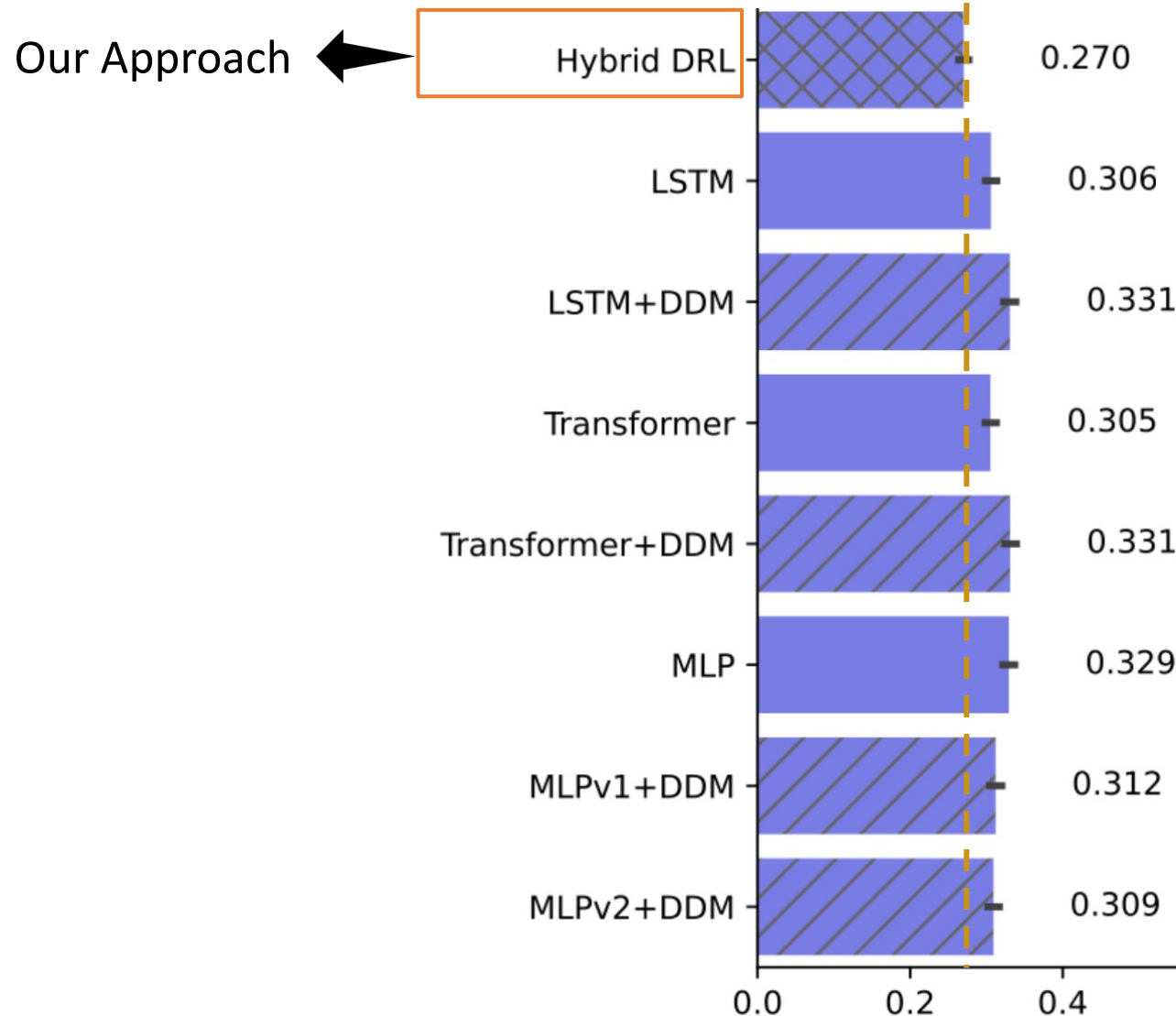
Simulation Error



Lower error than
27 baselines.

Mean Absolute Percentage Error

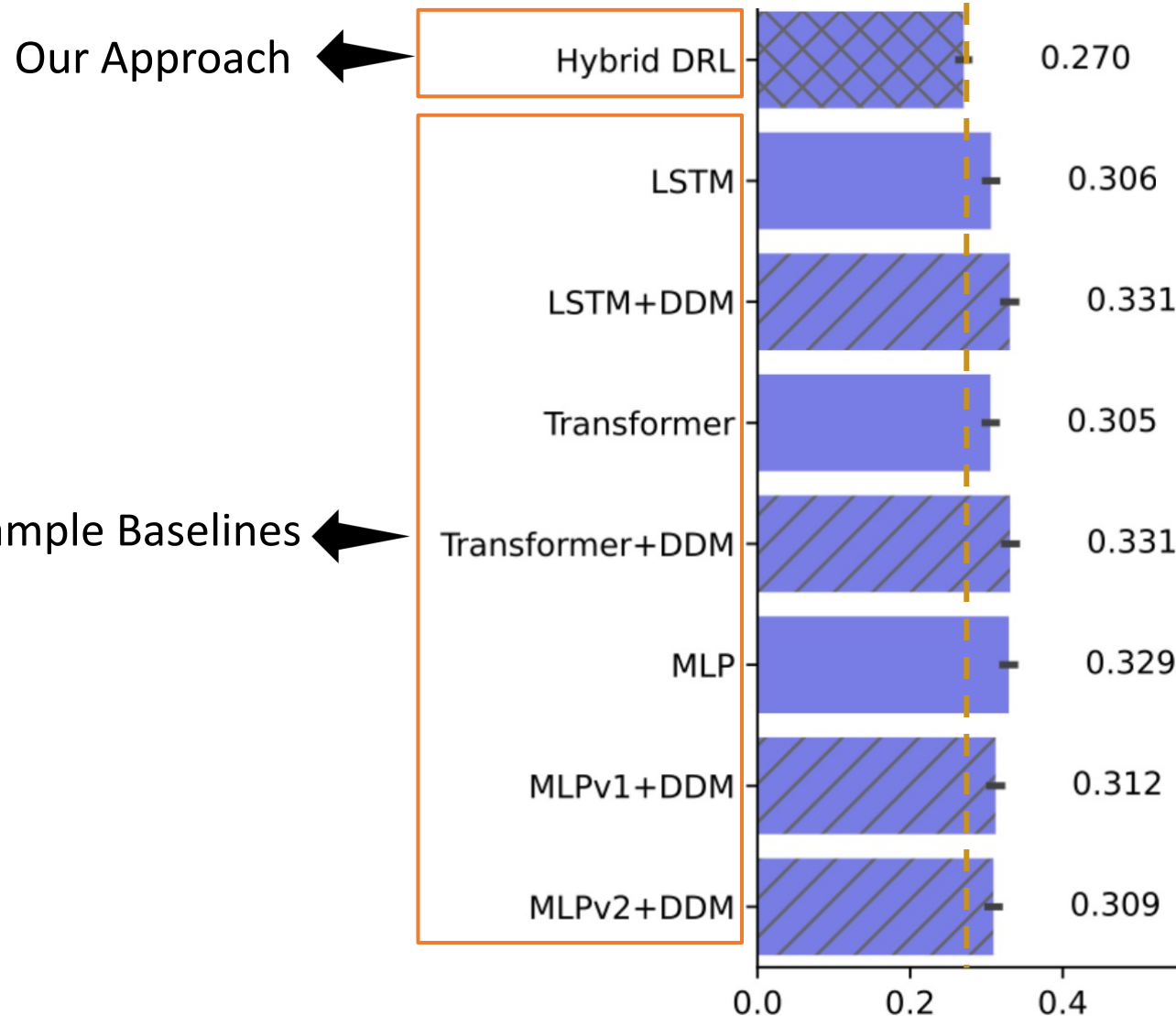
Simulation Error



Lower error than
27 baselines.

Mean Absolute Percentage Error

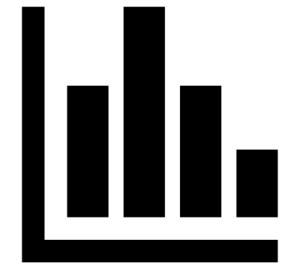
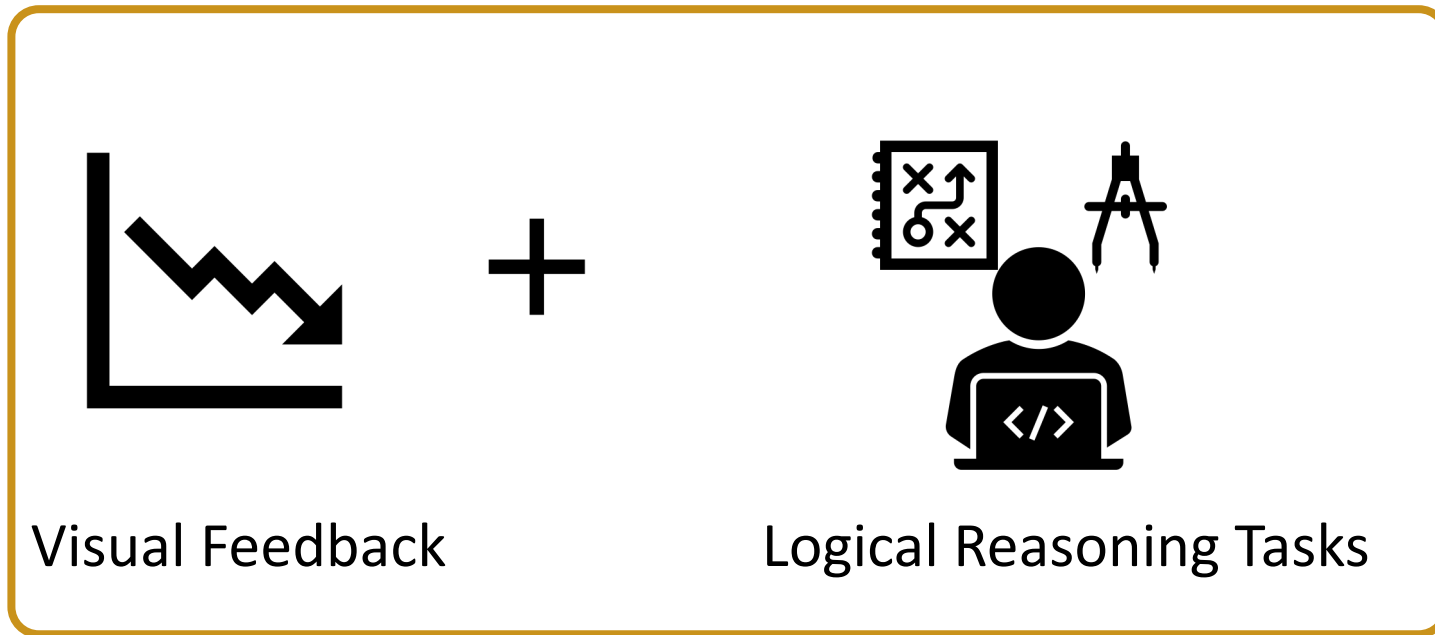
Simulation Error



Lower error than
27 baselines.

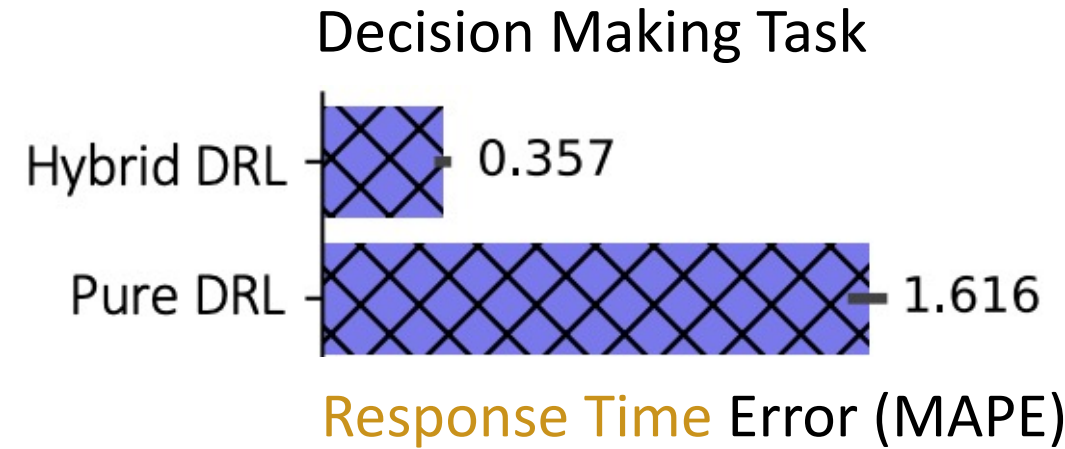
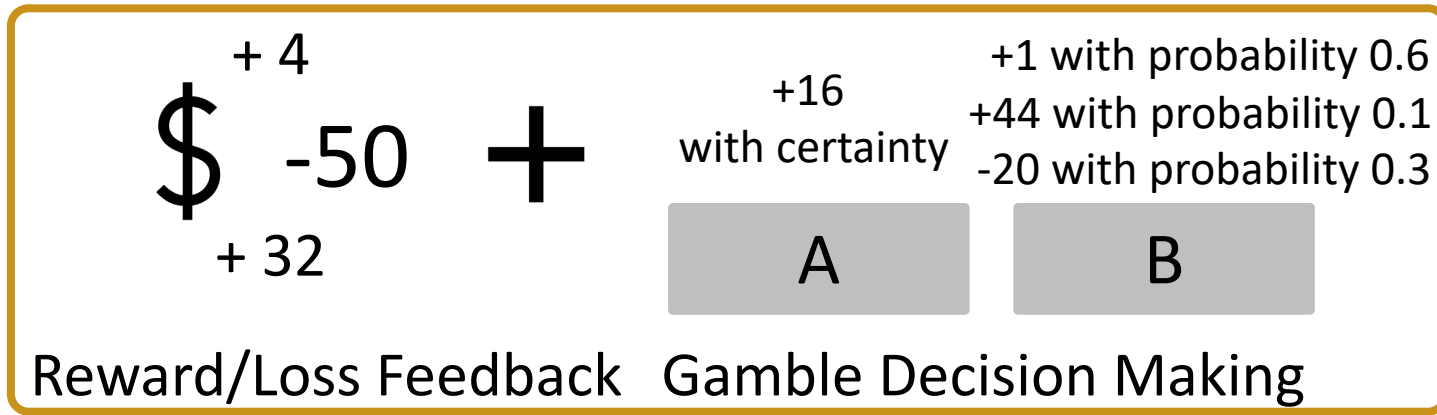
Mean Absolute Percentage Error

Generalization across Cognitive Task Domains

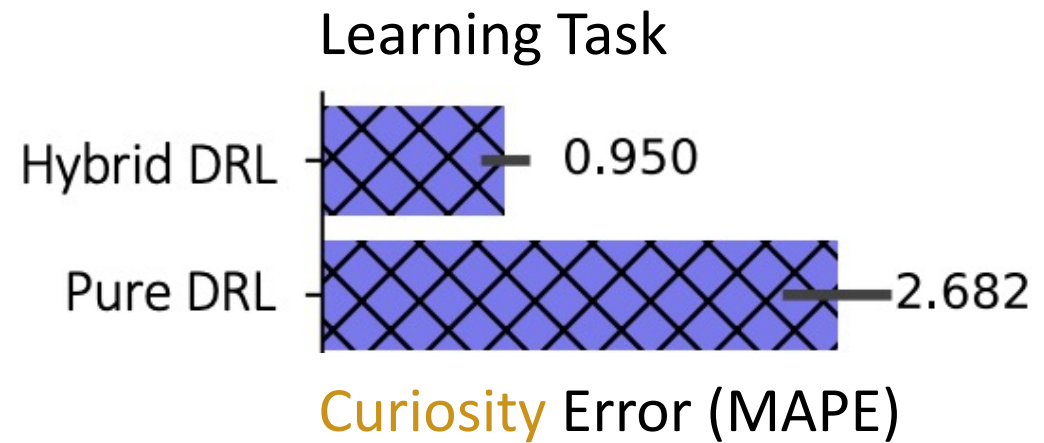
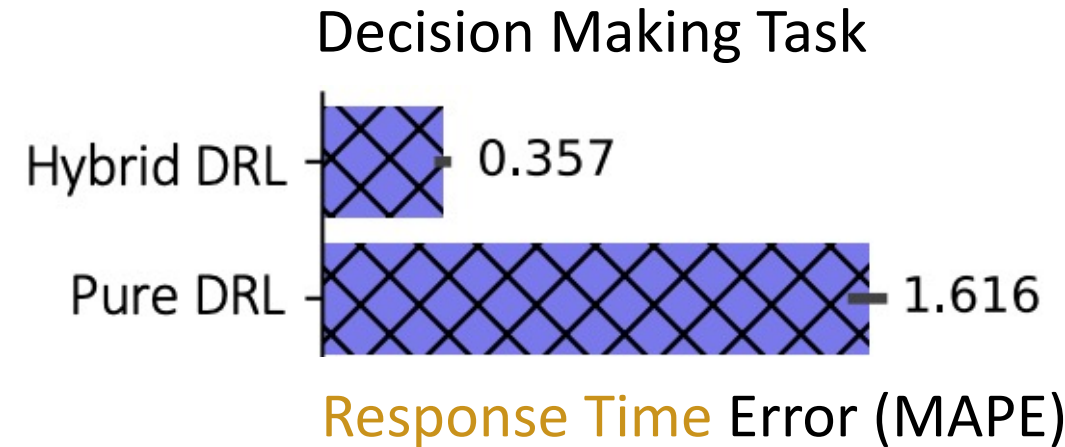
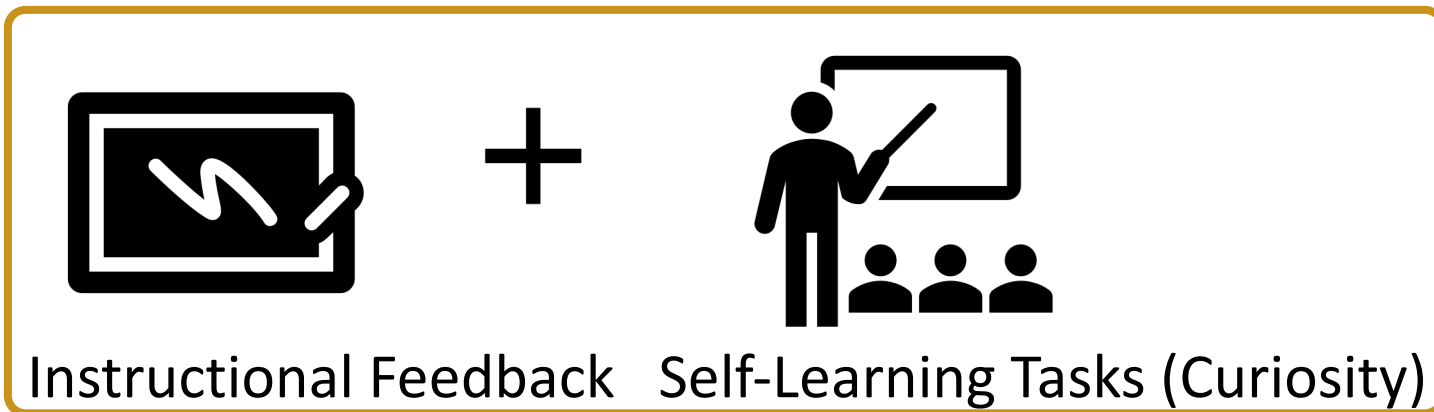
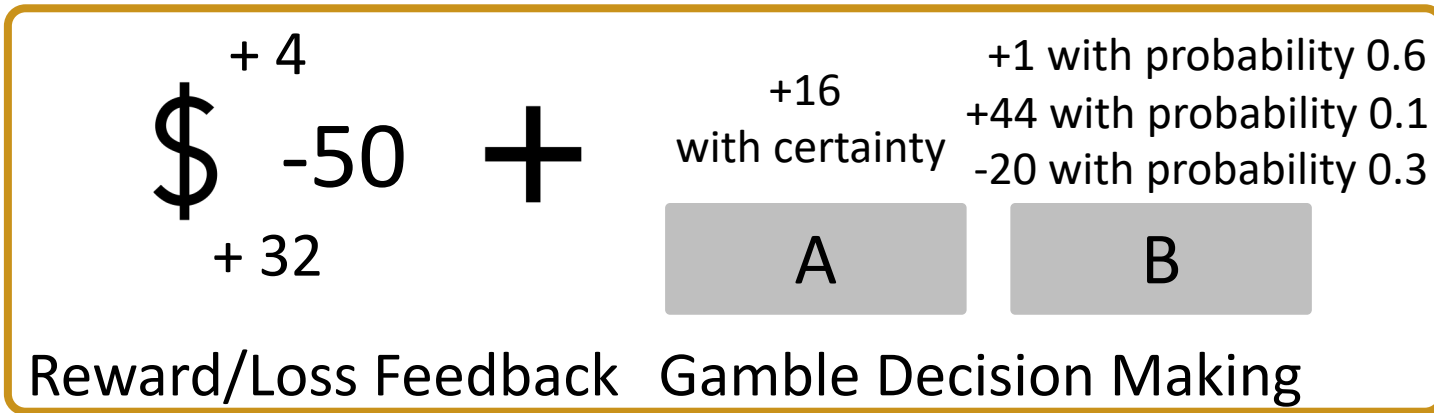


21,157 Responses

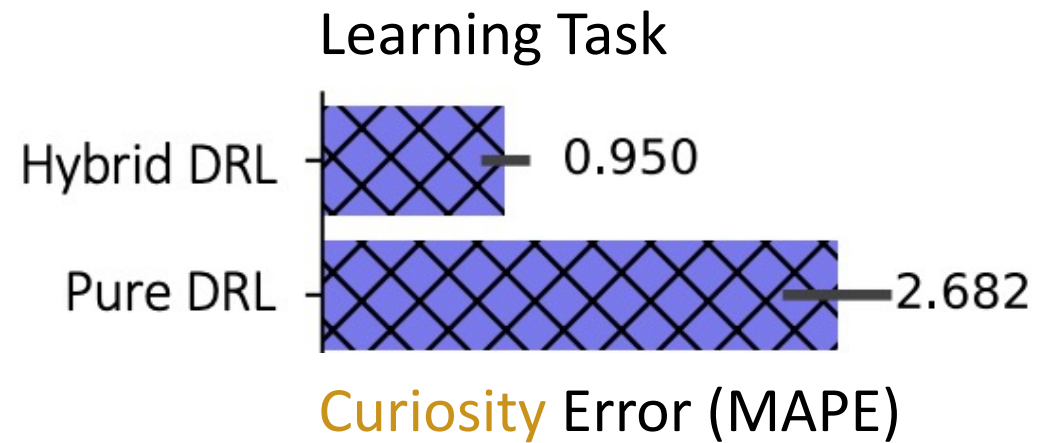
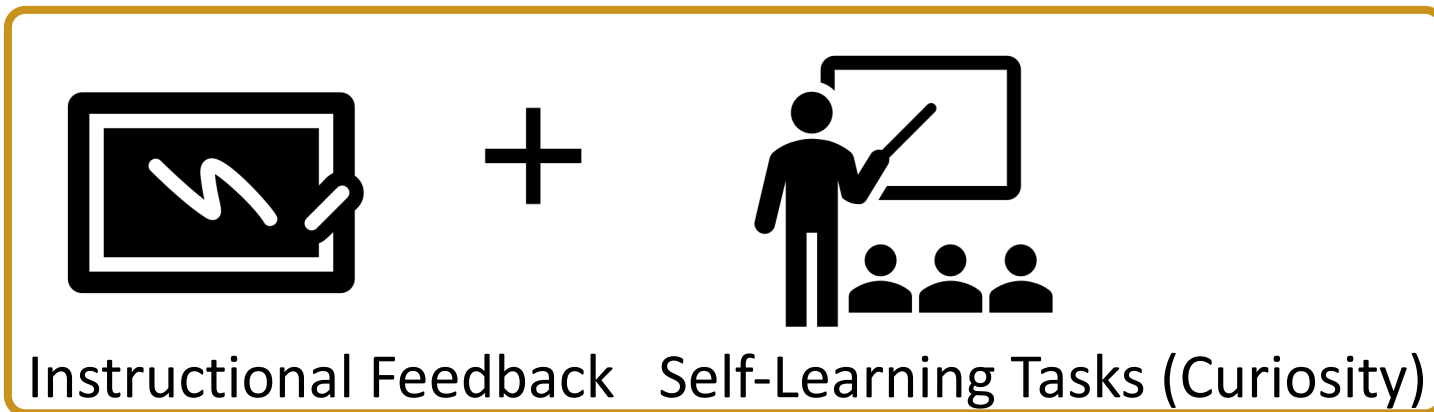
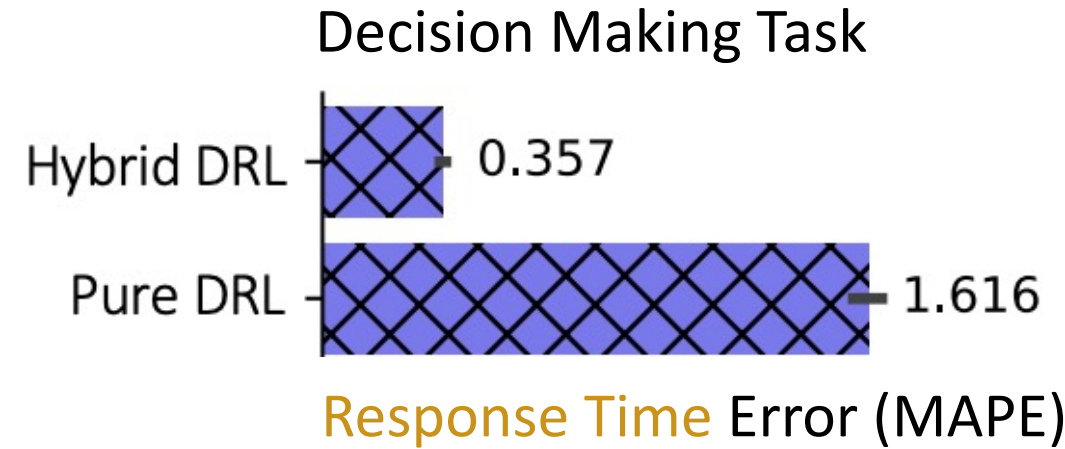
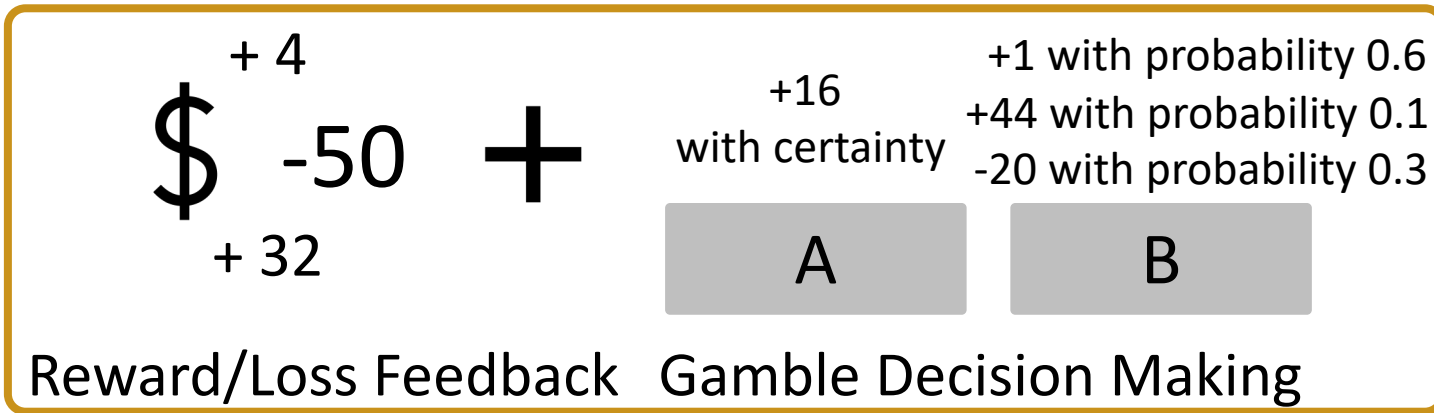
Generalization across Cognitive Task Domains



Generalization across Cognitive Task Domains



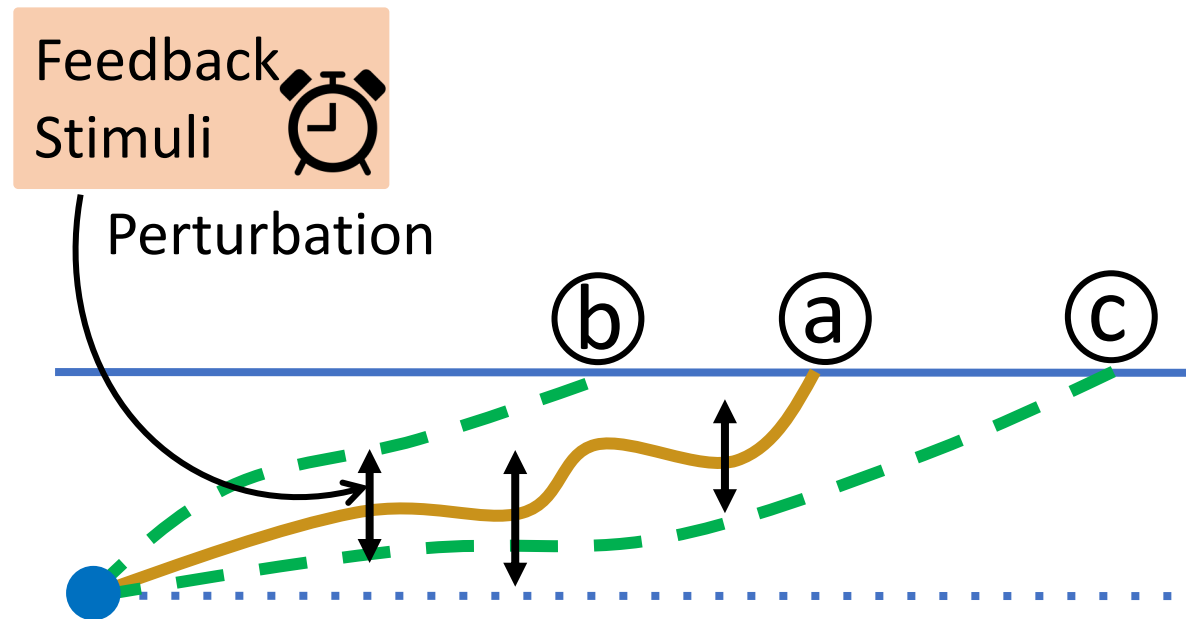
Generalization across Cognitive Task Domains



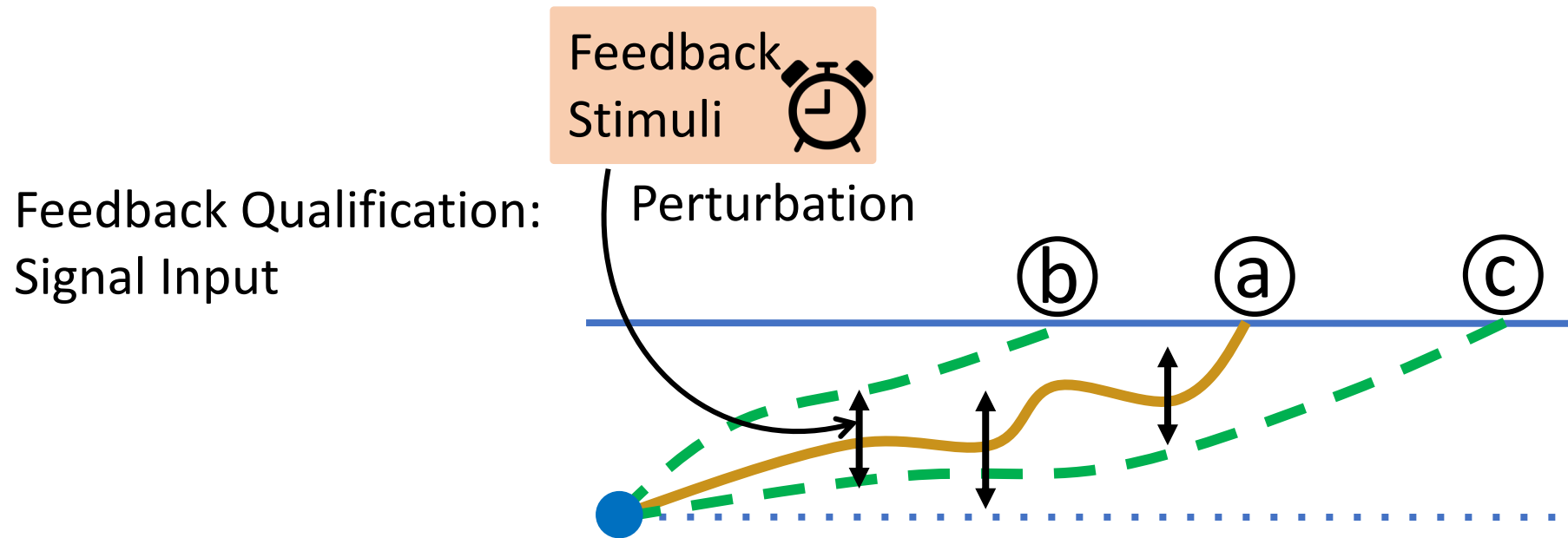
Difference in Feedback Modality, Cognitive Function.

Task/Feedback Qualification for Generalization

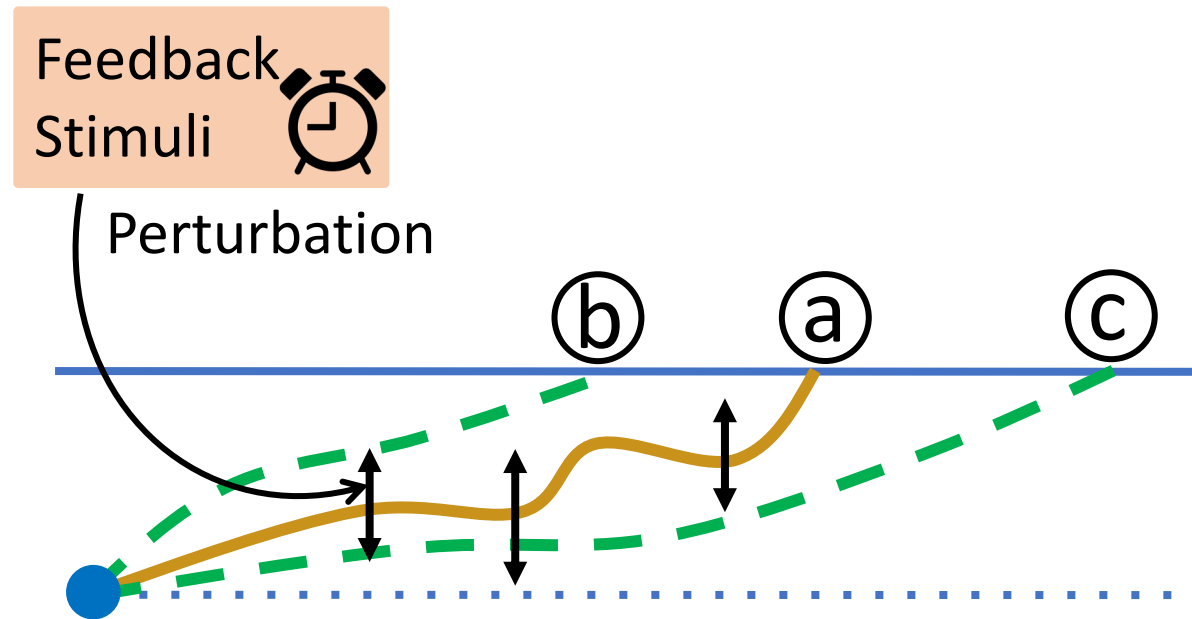
Task/Feedback Qualification for Generalization



Task/Feedback Qualification for Generalization

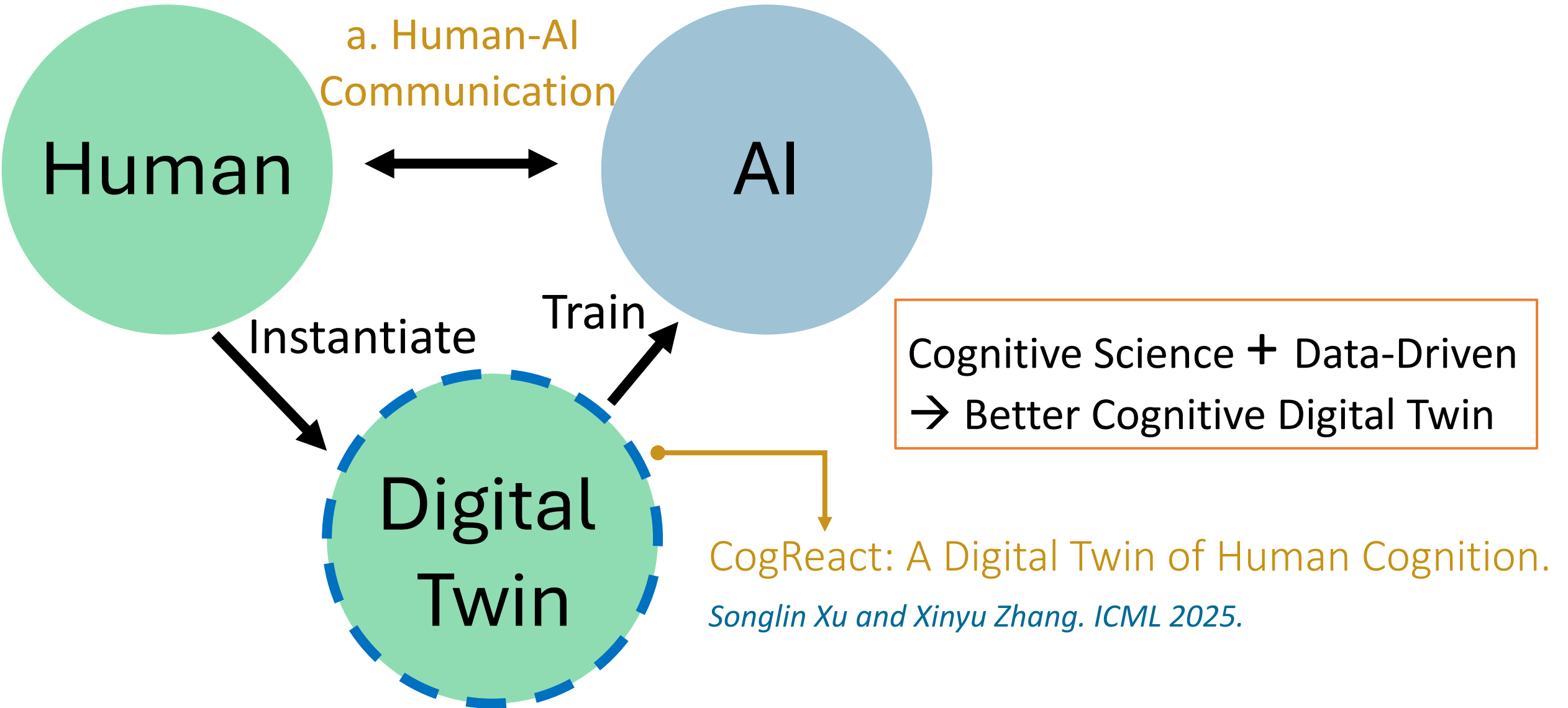


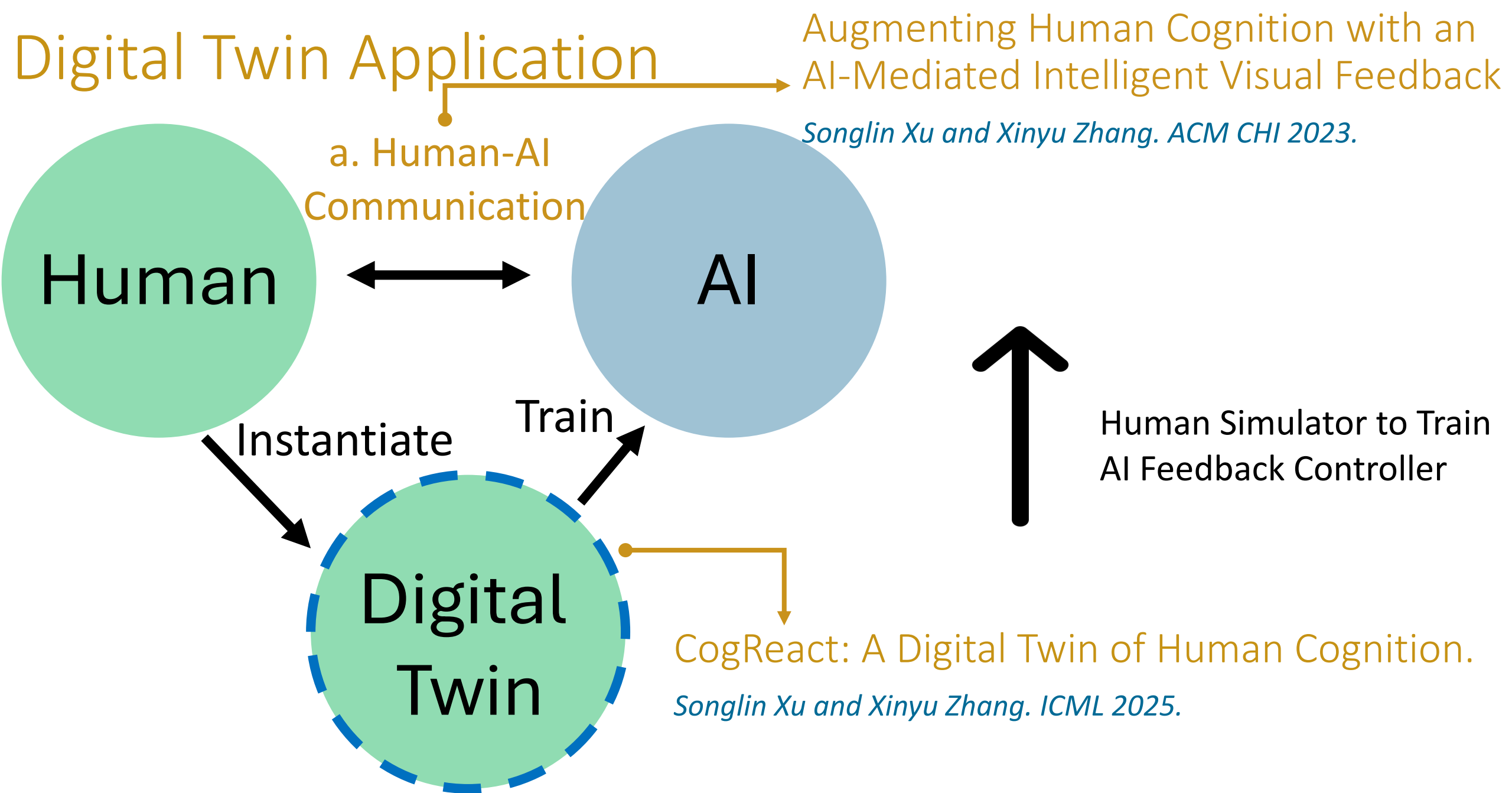
Task/Feedback Qualification for Generalization



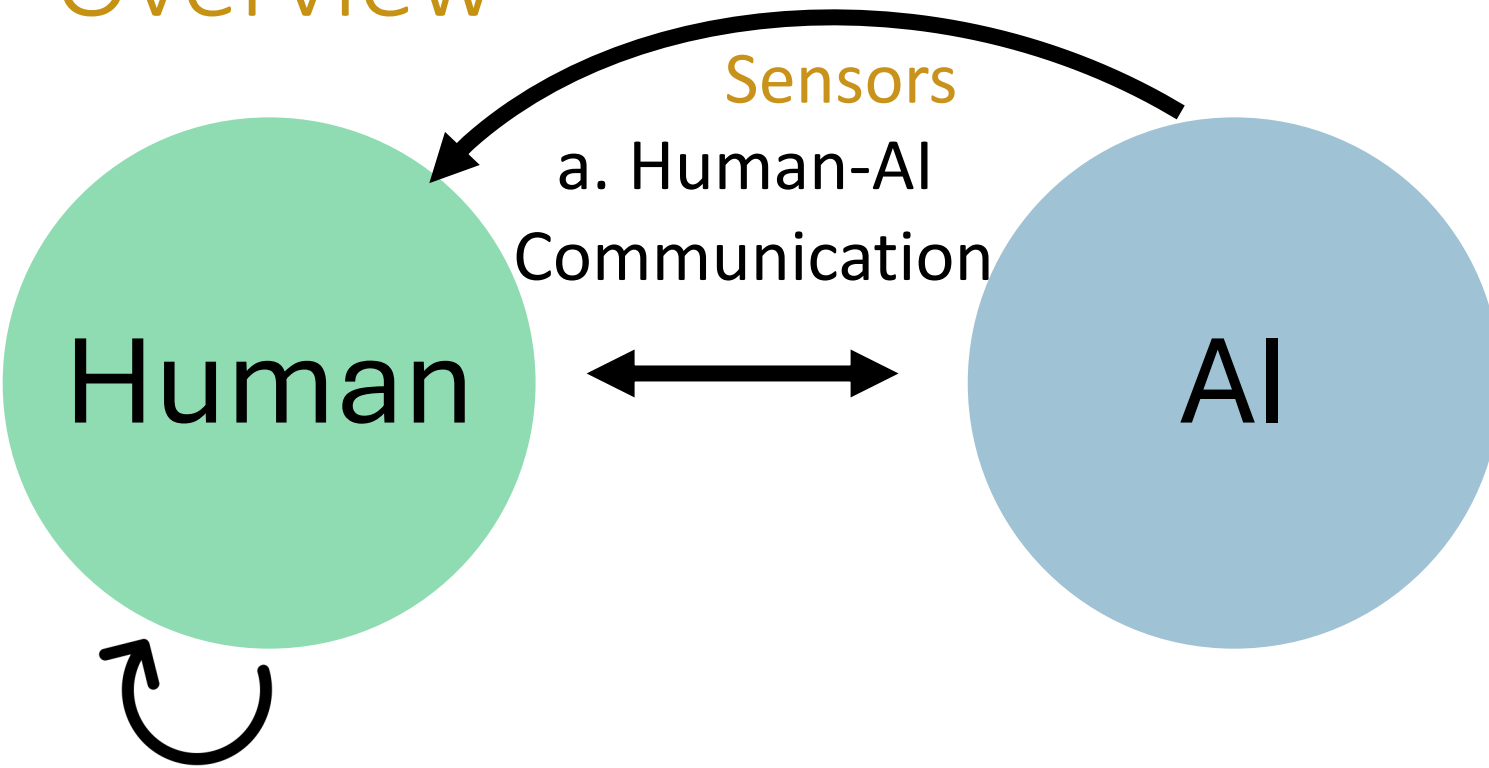
Task Qualification:
Mental Cognitive Process during Task Solving

Insight



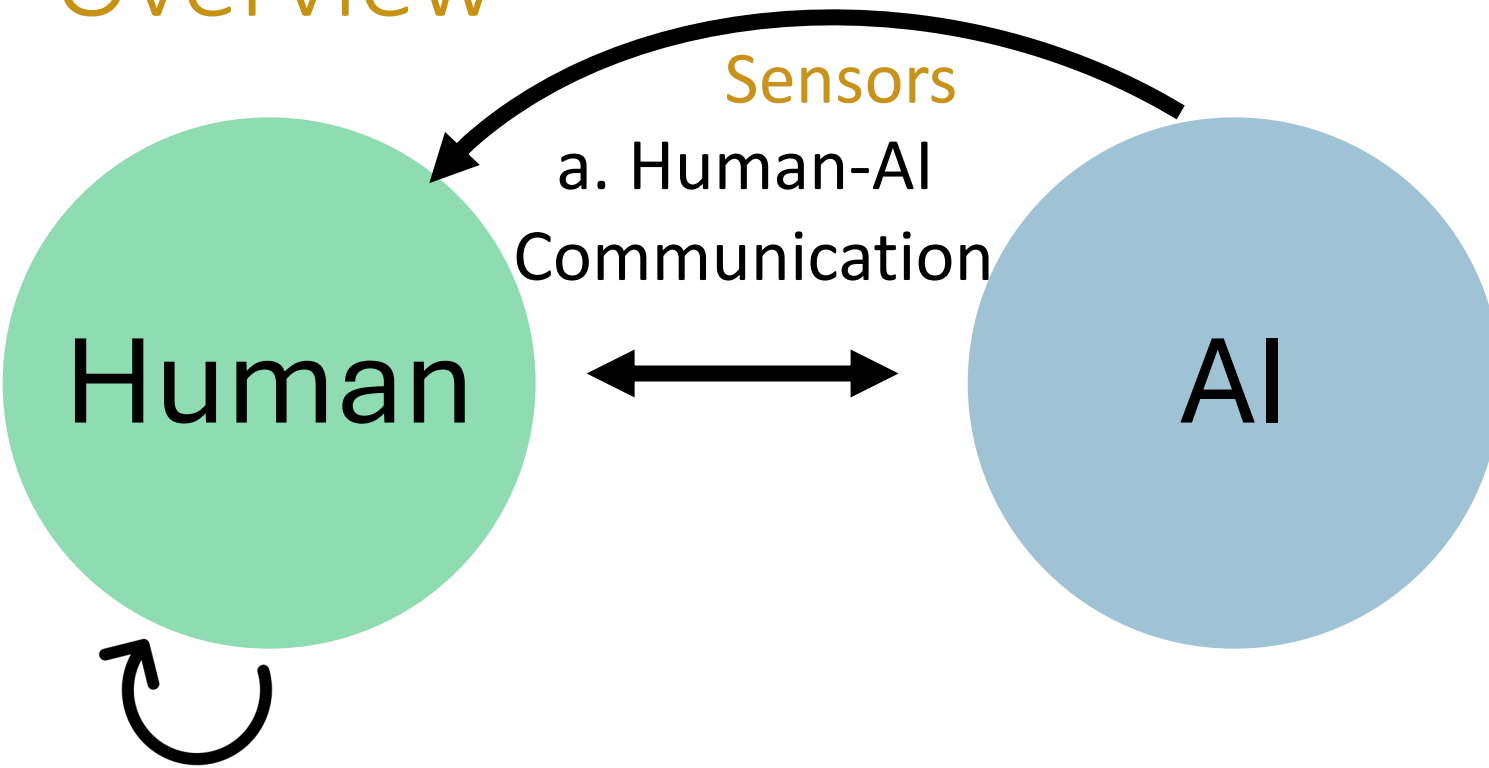


Overview



b. AI-Augmented
Human-Human
Communication

Overview



b. AI-Augmented
Human-Human
Communication

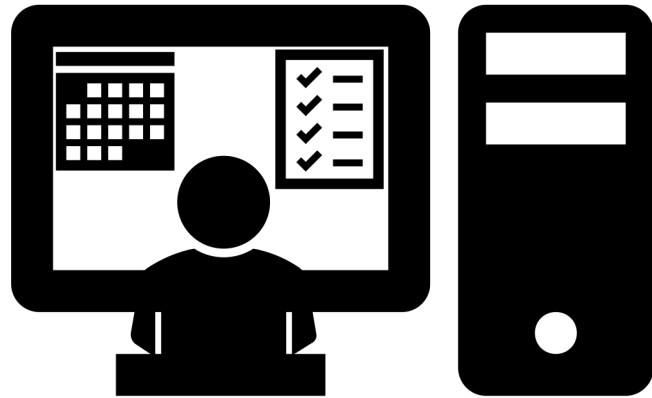
Social Influence



PeerEdu: Understanding Student Learning
Behaviors Augmented from Peer Feedback

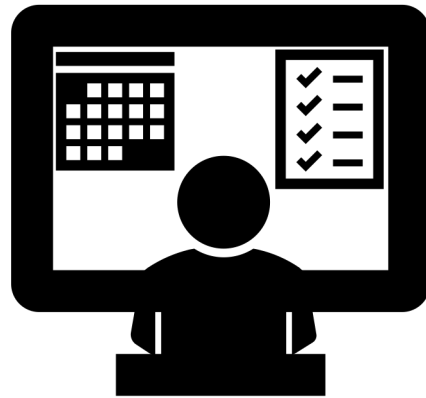
Xu et al. ACM CHI 2025.

Peer Feedback Is Ubiquitous

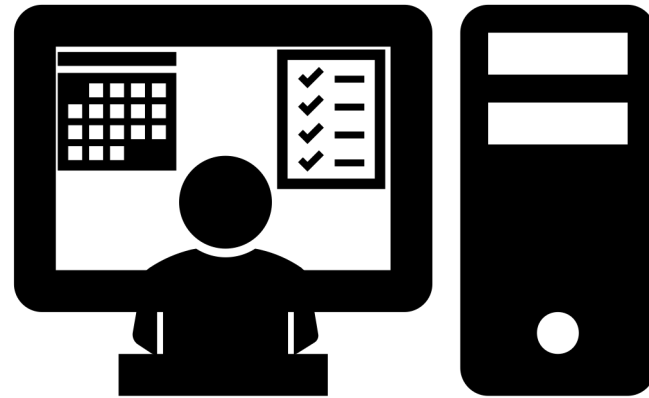


Working over Time

Peer Feedback Is Ubiquitous



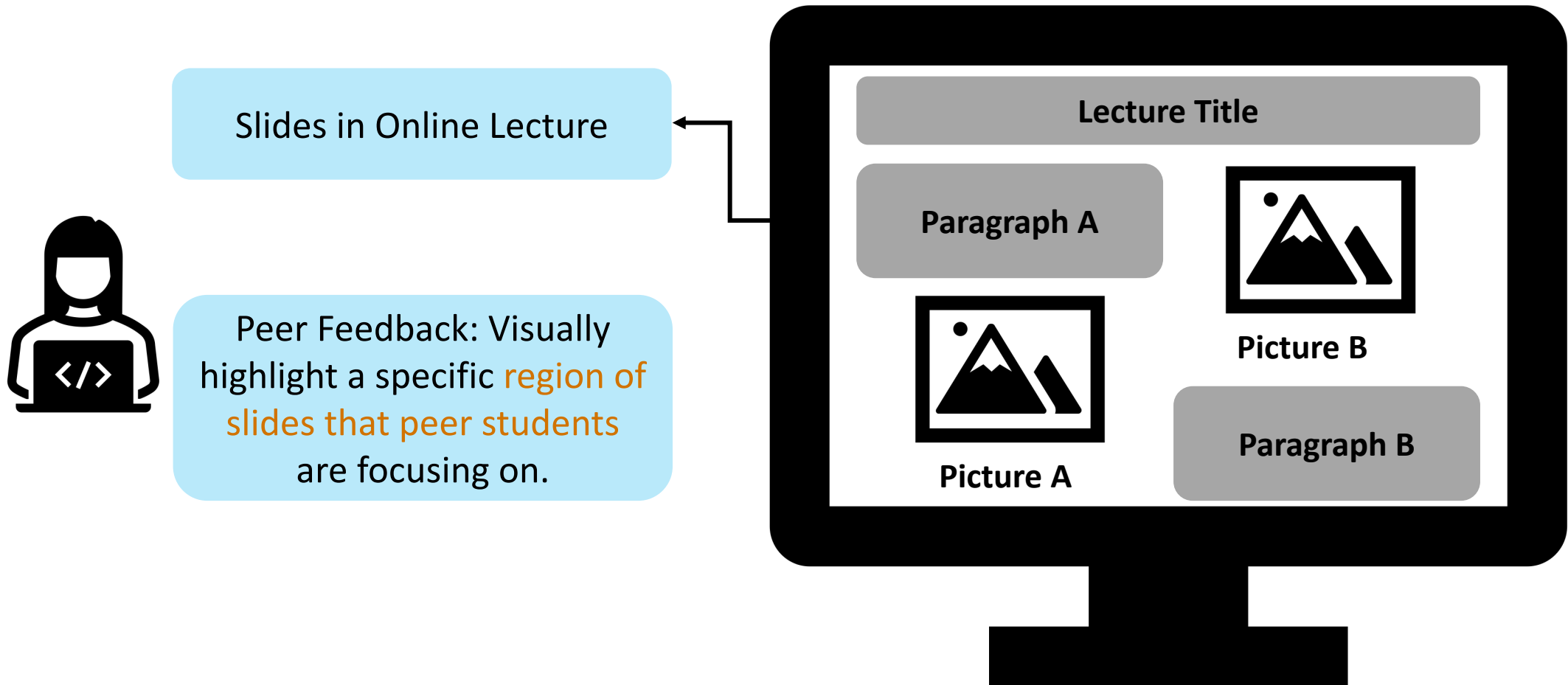
Working over Time



Peer Feedback

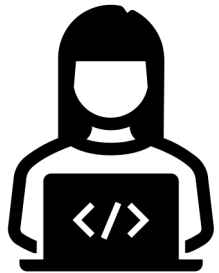
Peer Feedback (Visual Focus from Peer Gaze)

Students are learning online course videos by themselves.

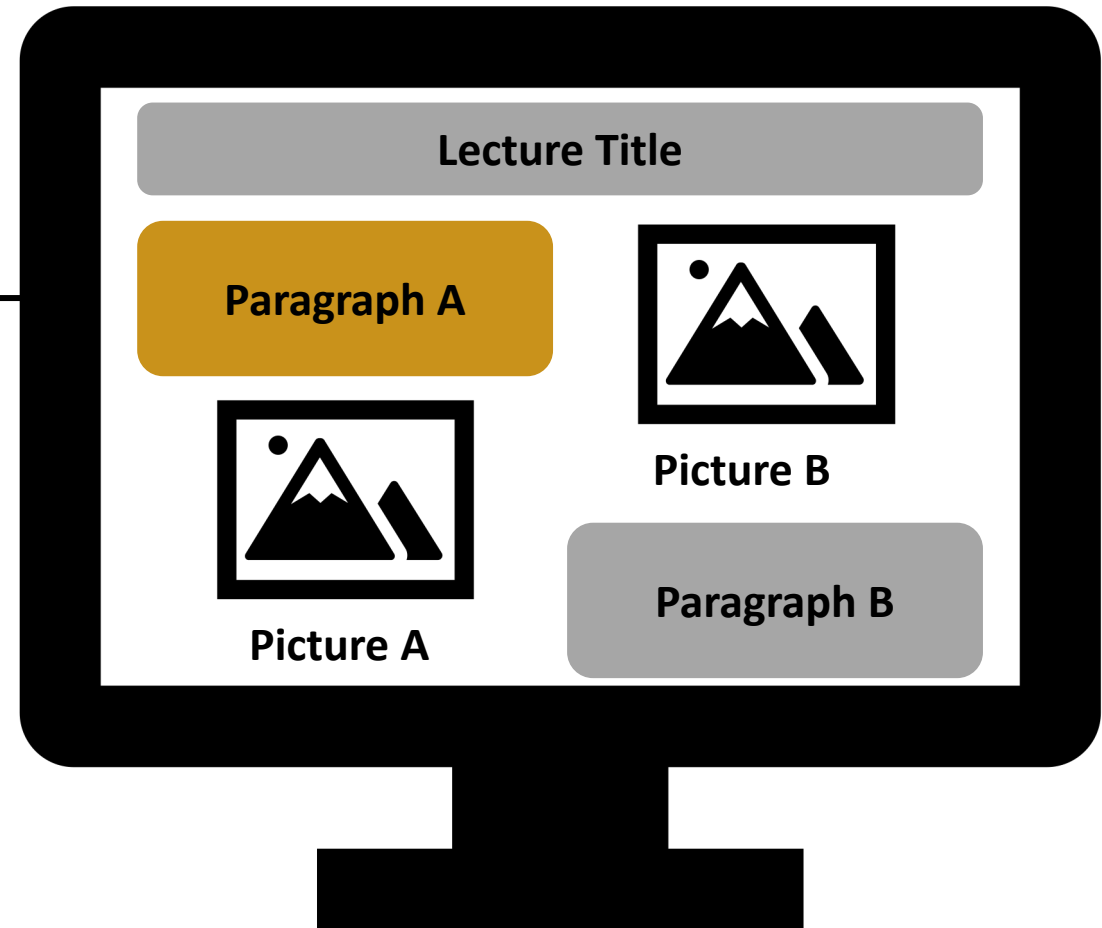


Peer Feedback (Visual Focus from Peer Gaze)

Students are learning online course videos by themselves.

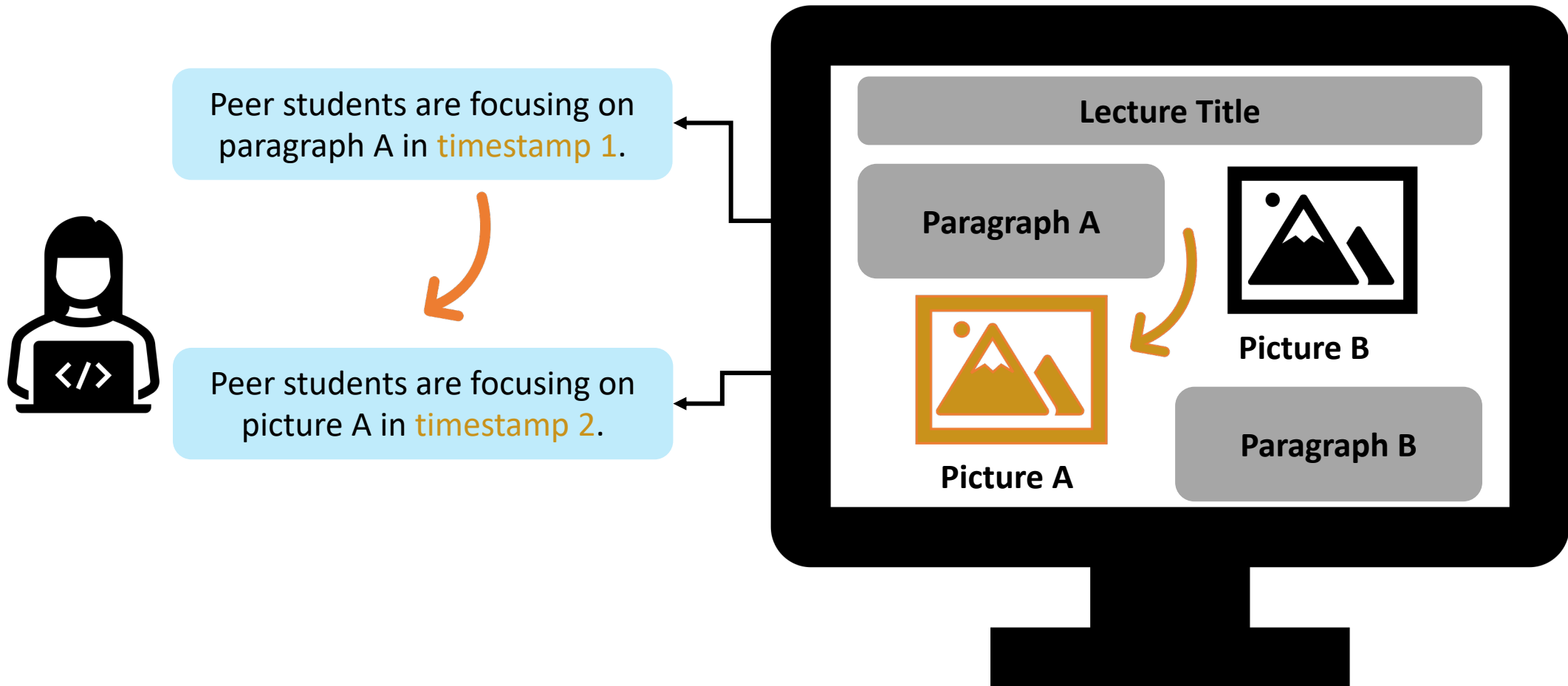


Peer students are focusing on paragraph A in **timestamp 1**.



Peer Feedback (Visual Focus from Peer Gaze)

Students are learning online course videos by themselves.



Peer Feedback Research in Social Influence

The social influence of peer feedback has been studied for decades.

PNAS

RESEARCH ARTICLE

PSYCHOLOGICAL AND COGNITIVE SCIENCES

Predicting the attention of others

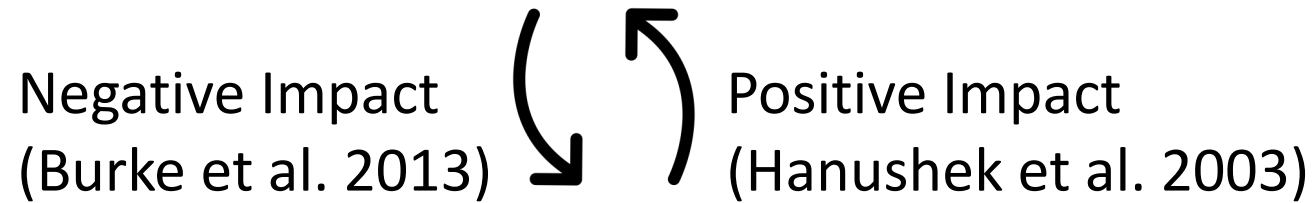
Ziman et al. 2023

Long-Standing Debate in Learning Science

How does peer feedback modulate student learning behaviors?

Long-Standing Debate in Learning Science

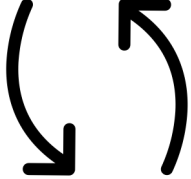
How does peer feedback modulate student learning behaviors?



Long-Standing Debate in Learning Science

How does peer feedback modulate student learning behaviors?

Negative Impact
(Burke et al. 2013)



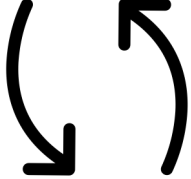
Positive Impact
(Hanushek et al. 2003)

Lack of Fine-Grained Behavioral Evidence in Large-Scale Population.

Long-Standing Debate in Learning Science

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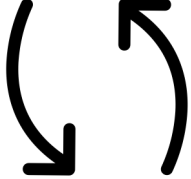
Sensors

Deployable at Scale (Hundreds of Users)

Long-Standing Debate in Learning Science

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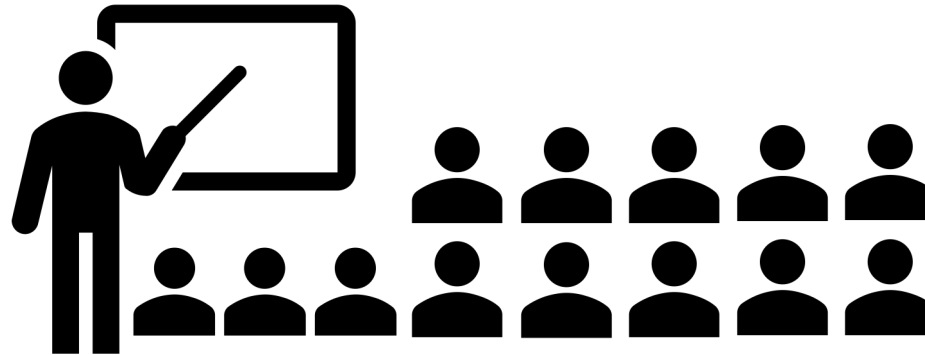
Positive Impact
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Lack of **Fine-Grained Behavioral Evidence** in **Large-Scale** Population.

Sensors

Deployable at Scale (Hundreds of Users)

Web Camera + Production-Ready Software System for Online Education



CogTeach Platform: Online Learning Demo

The screenshot displays the CogTeach Platform interface during an online learning session. The browser address bar shows the URL `.com/admin.html`. The interface includes a top navigation bar with "Info Hub" and "Zoom" tabs. The main content area shows a presentation slide titled "The details of V-learning".

The slide content includes:

- r is not consistent
- evaluation may depend on other unknown factors

Below the list, it states: "solution: pull V to r so that V approaches its average value."

Further down, it defines the new state and reward:

new state = s
reward = r

It then defines the value function of s :

$$V = V_{old} + 0.1 * (r - V_{old})$$

A note states: "note: V_{old} is the old value of V ".

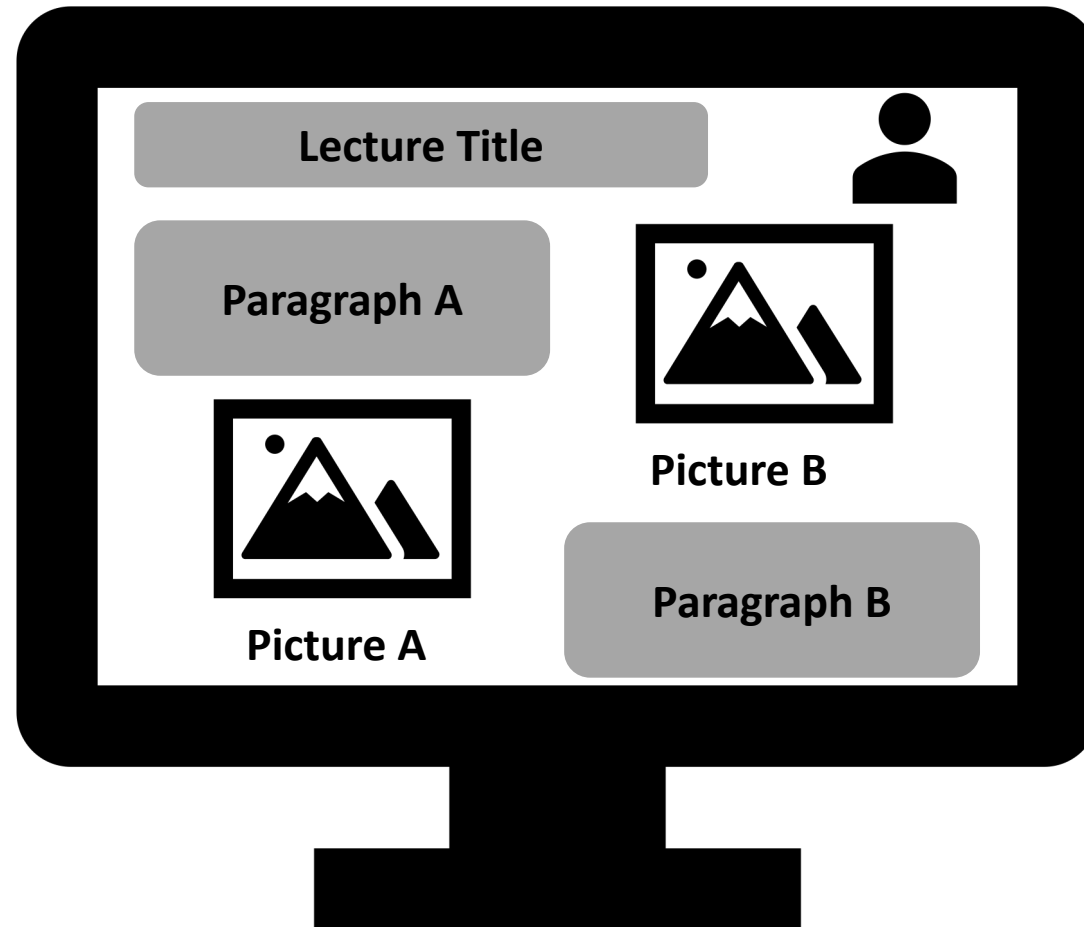
An arrow points from the text "pull V to r " to a box labeled "Image Demonstration".

The interface also features a sidebar on the right with a "Talking: One talking student" indicator and a user icon. At the bottom, there is a control bar with icons for "Join Audio", "Start Video", "Participants", "Share Screen", "Chat", and "More". A "Leave" button is also present.

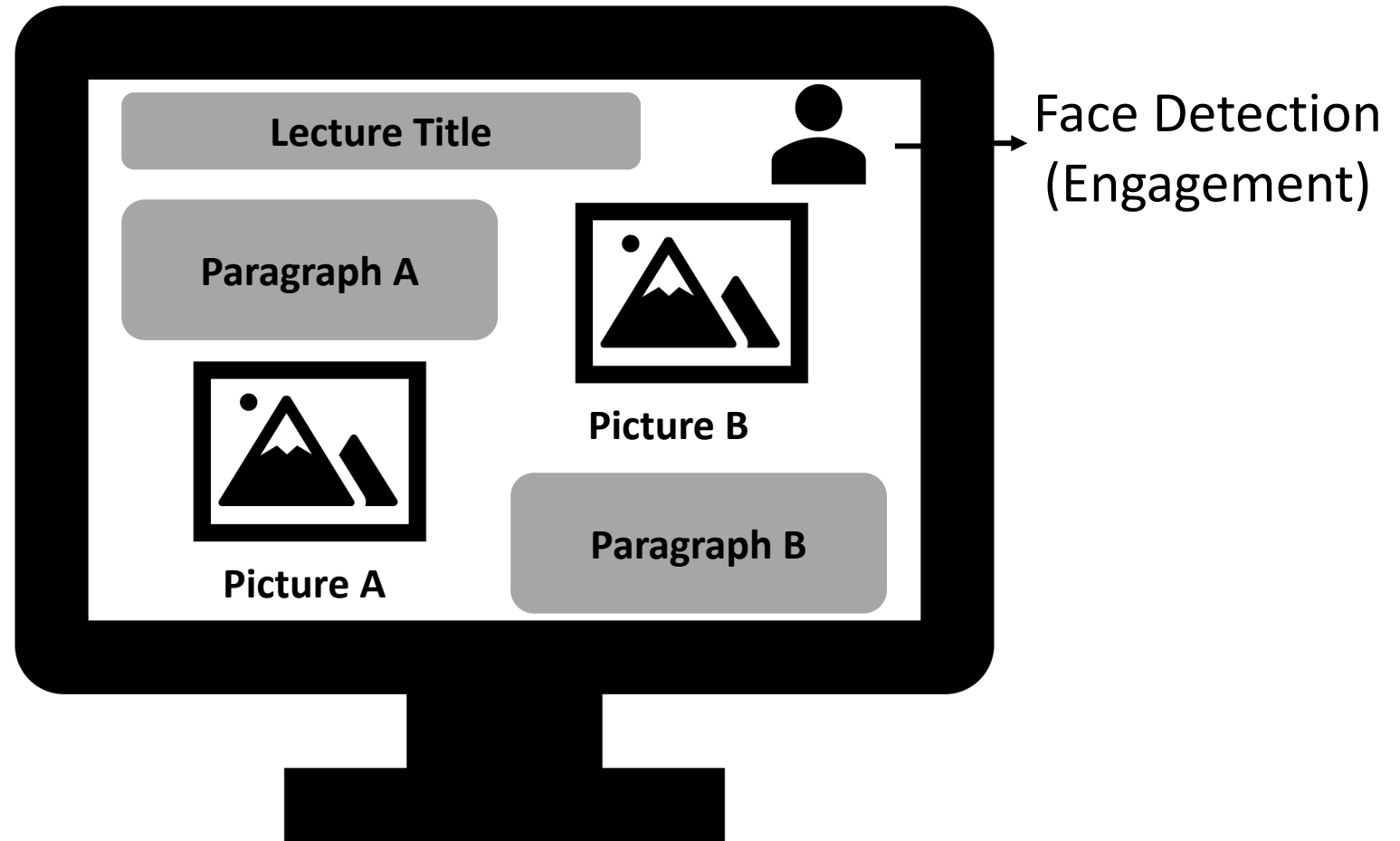
At the bottom left, a progress bar shows the following data:

Metric	Value
Knowledge	33.3%
Attention	100.0%

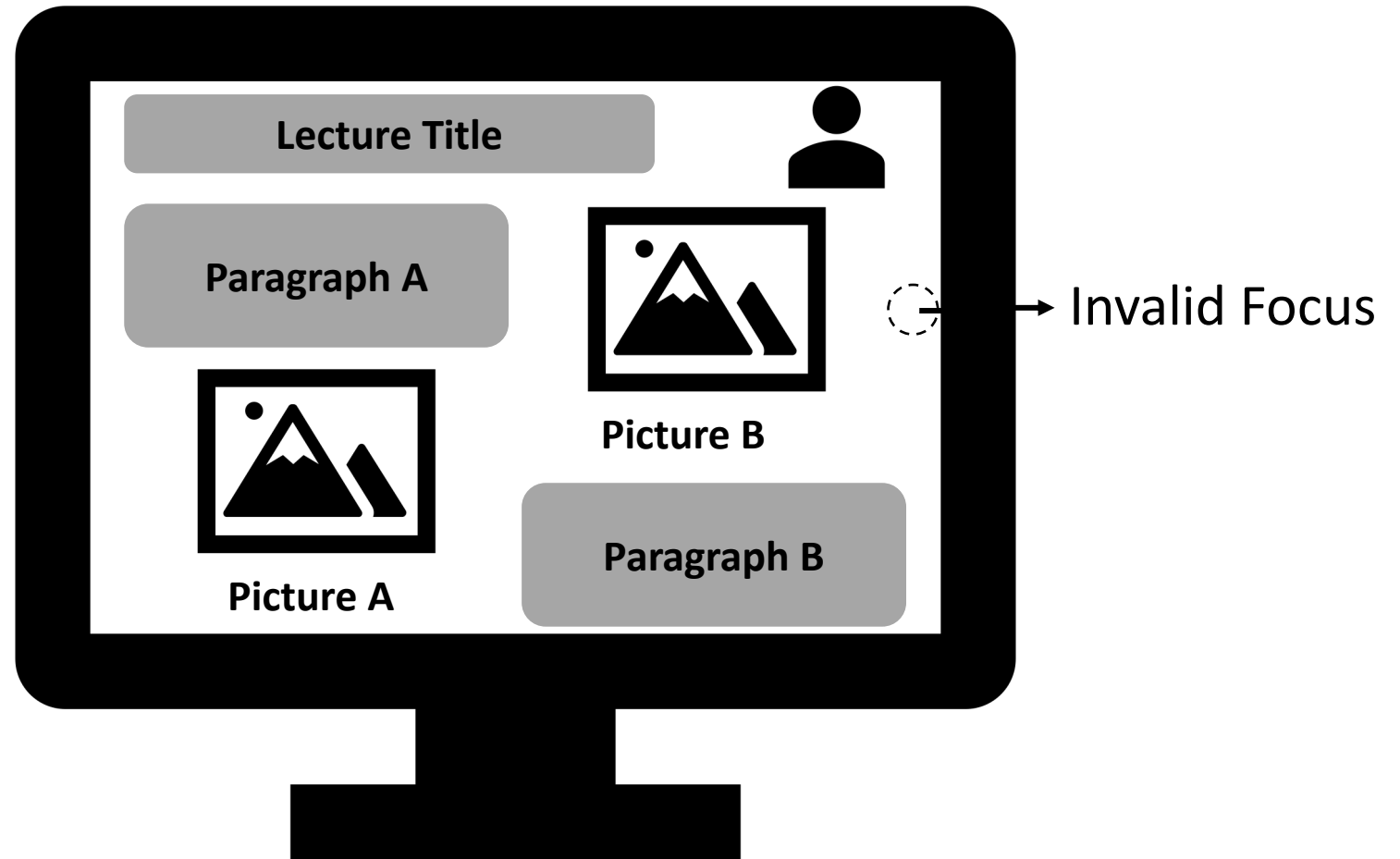
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



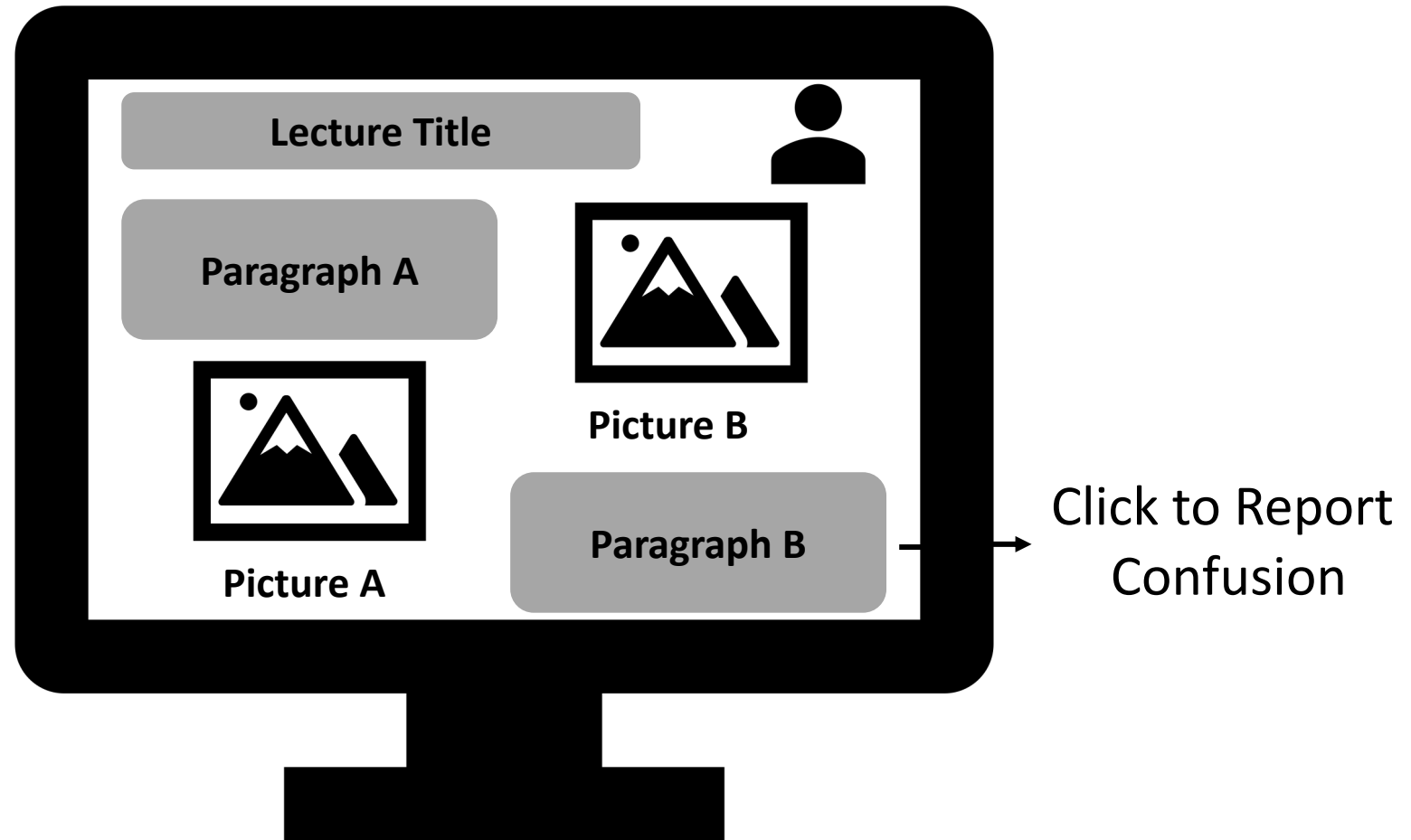
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



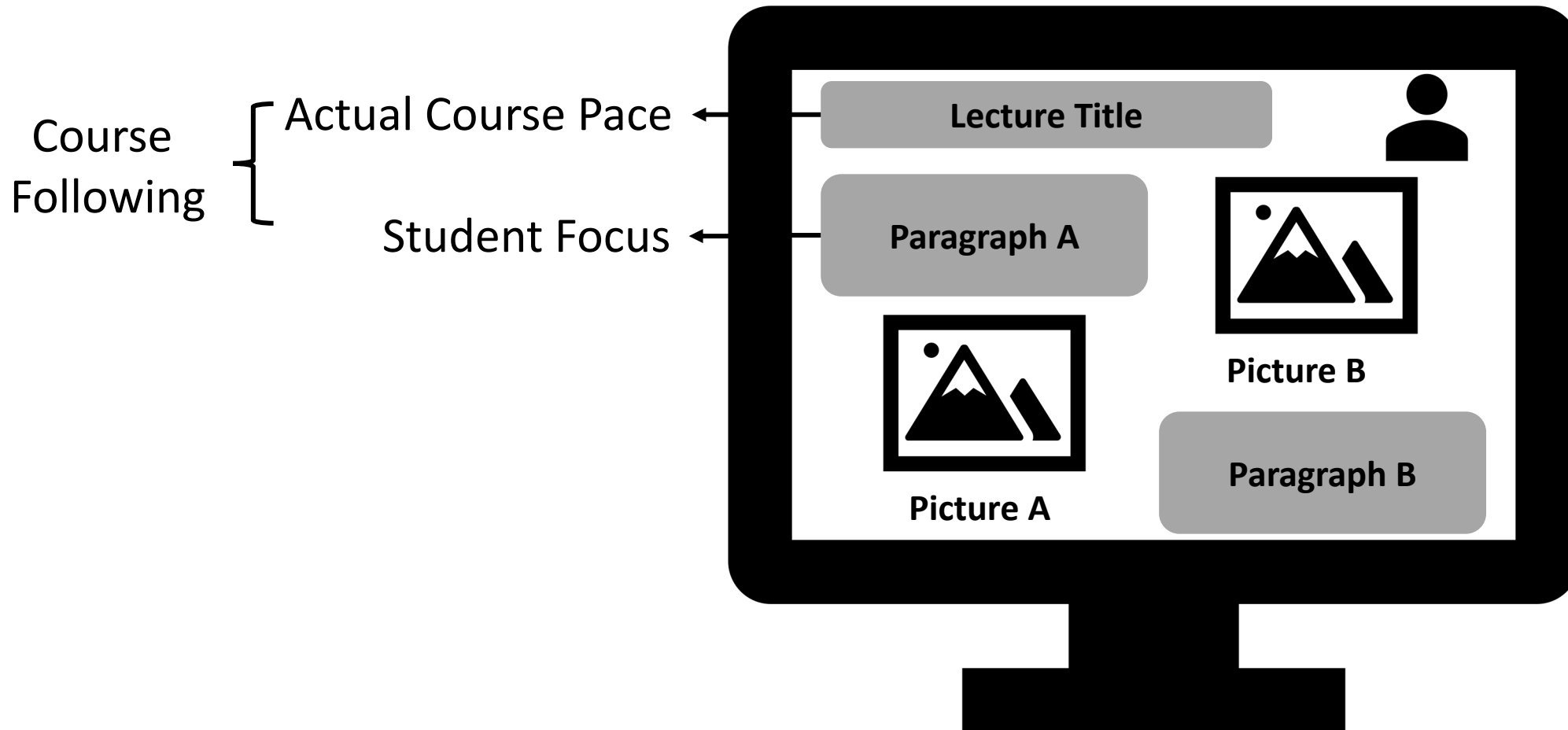
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



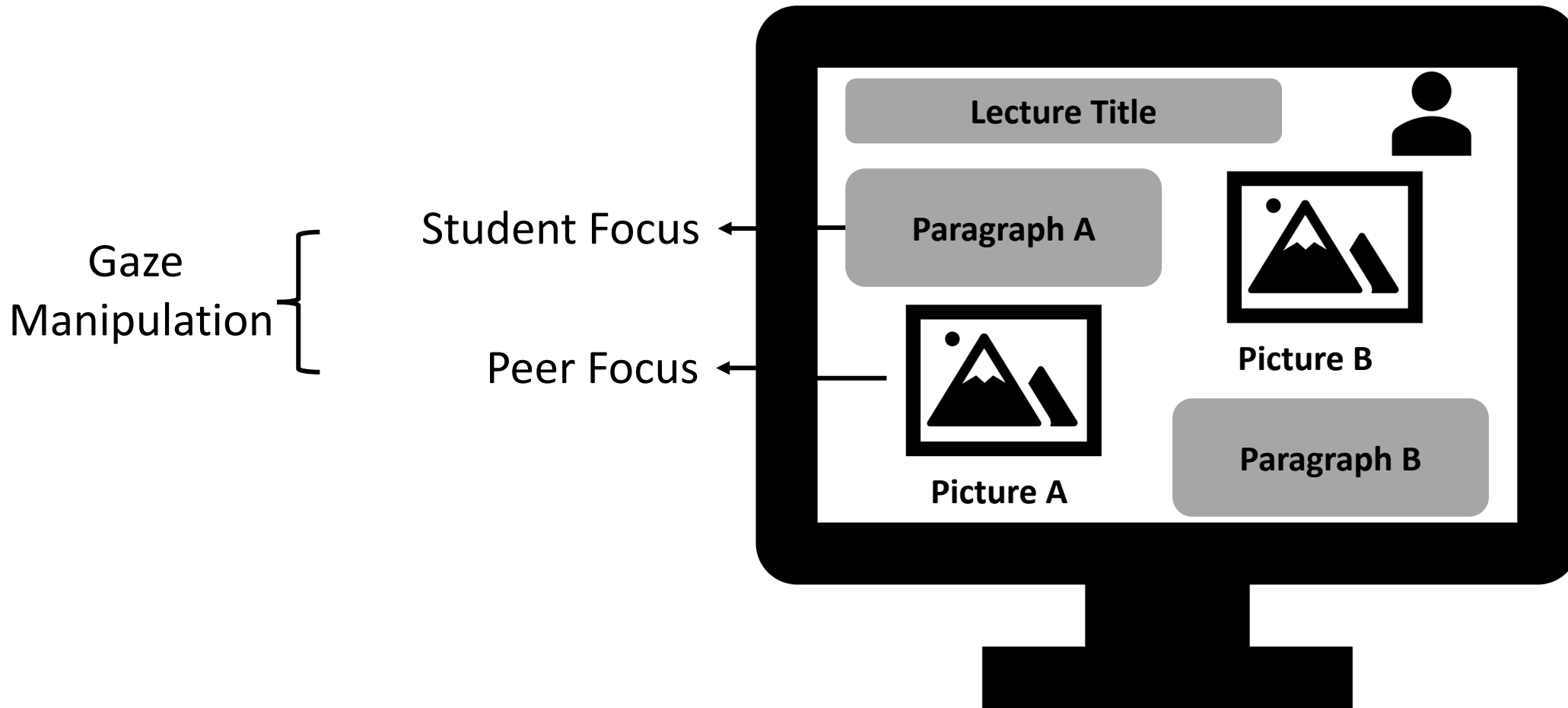
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



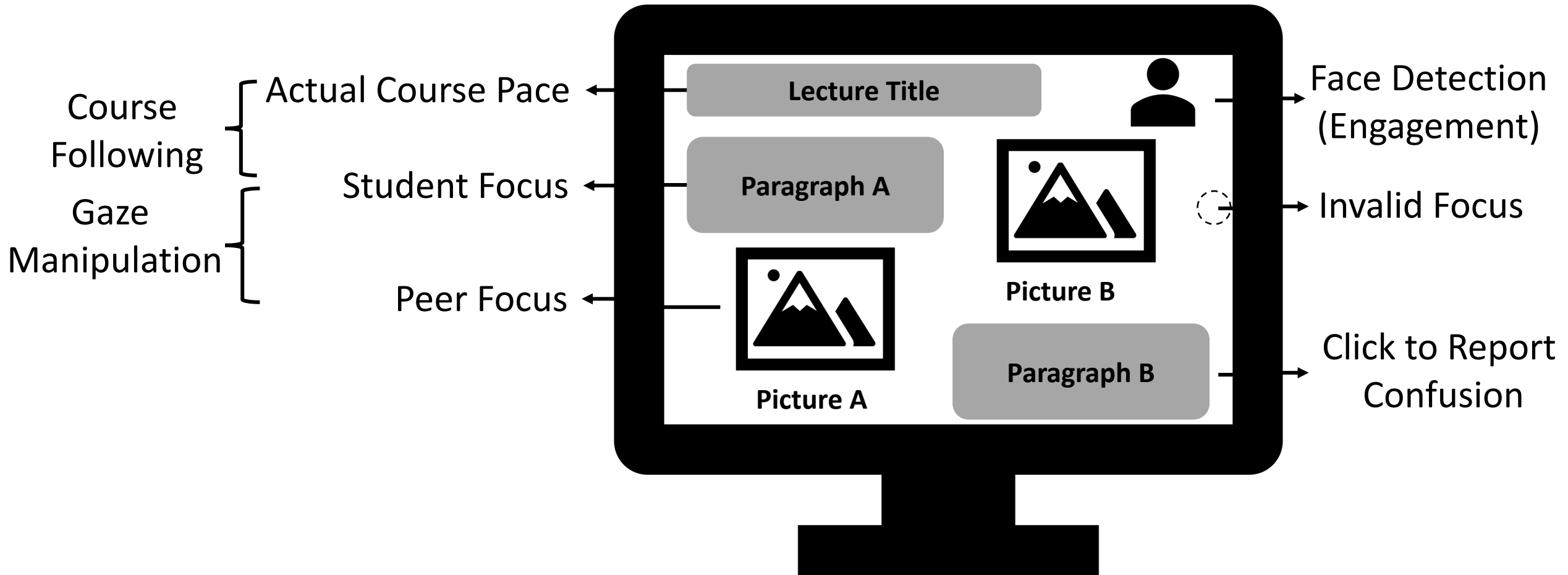
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



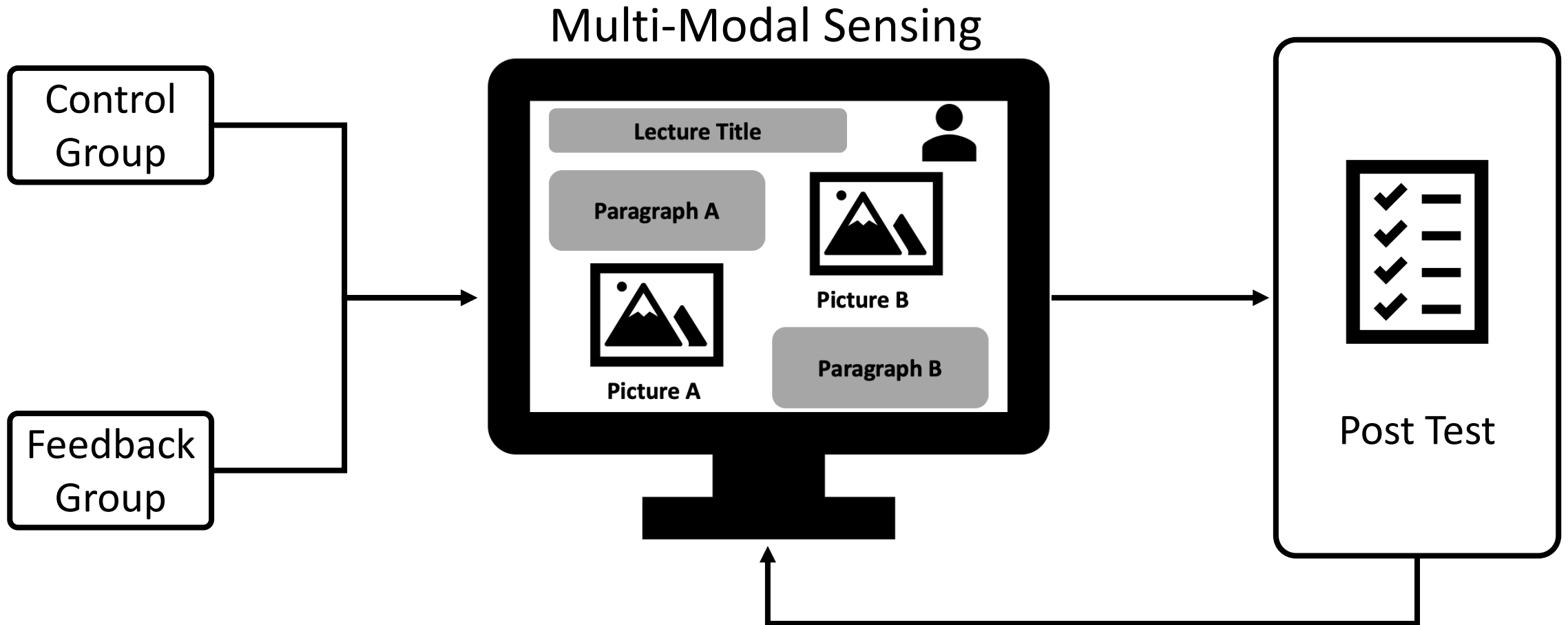
Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States



Multimodal Sensing to Capture Fine-Grained Behaviors and Cognitive States

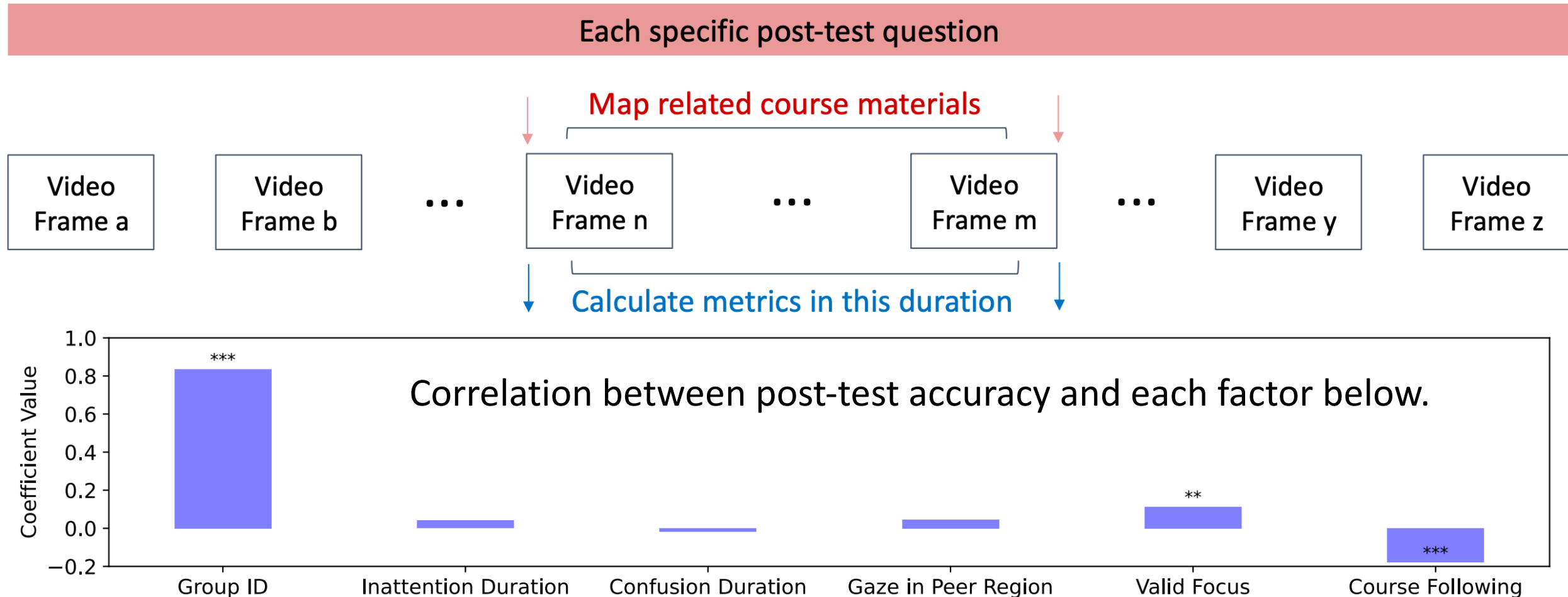


Evaluation from a Study of N = 311 Students' Online Learning

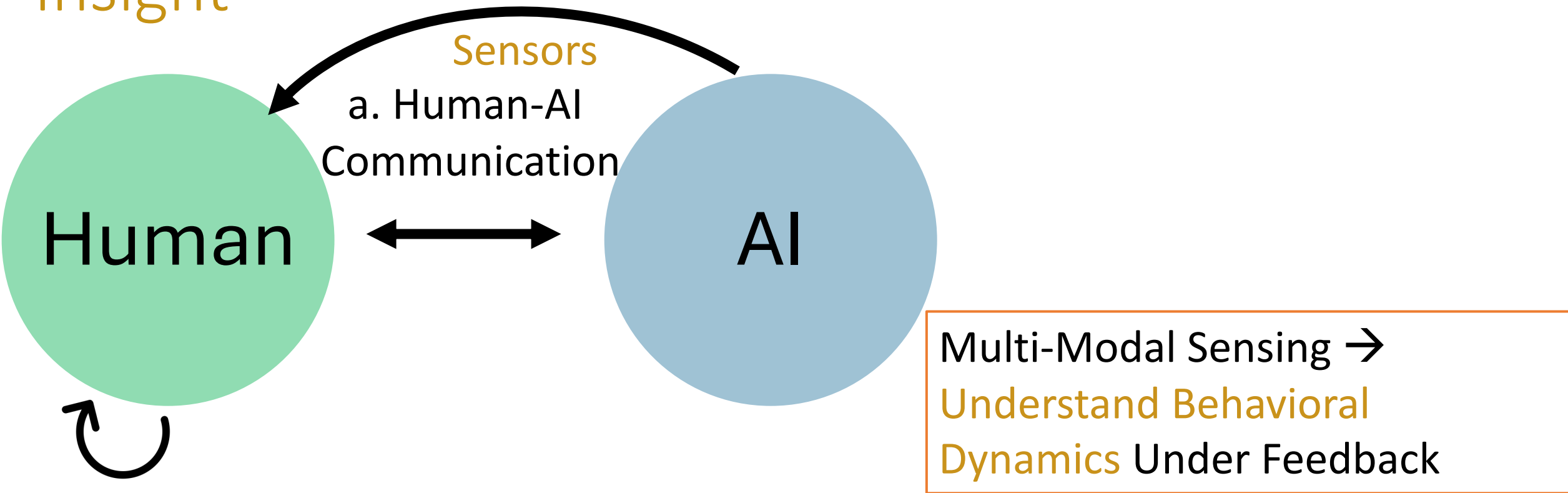


Map each question to the specific course duration.

Results: N = 311 Students' Online Learning



Insight

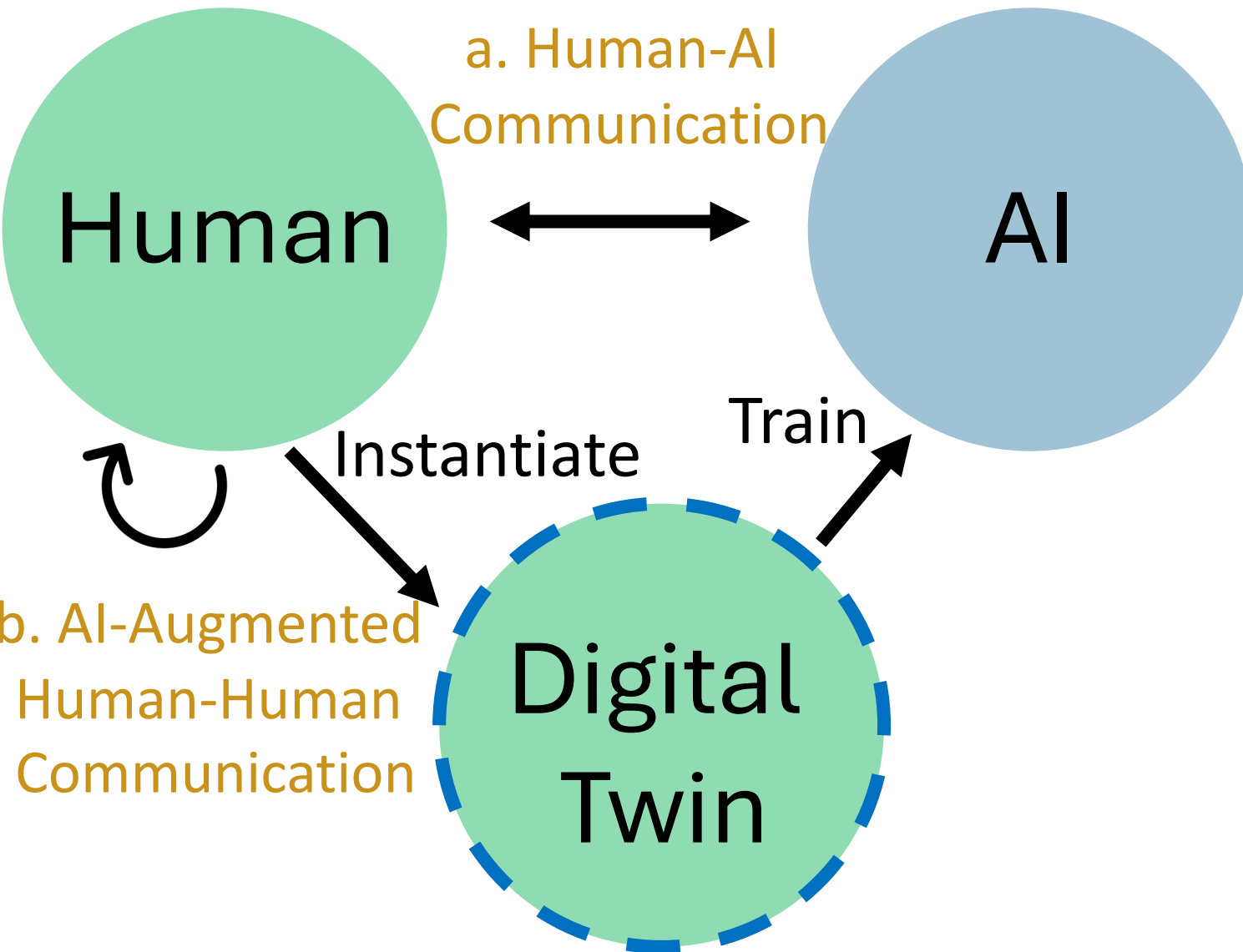


b. AI-Augmented Human-Human Communication

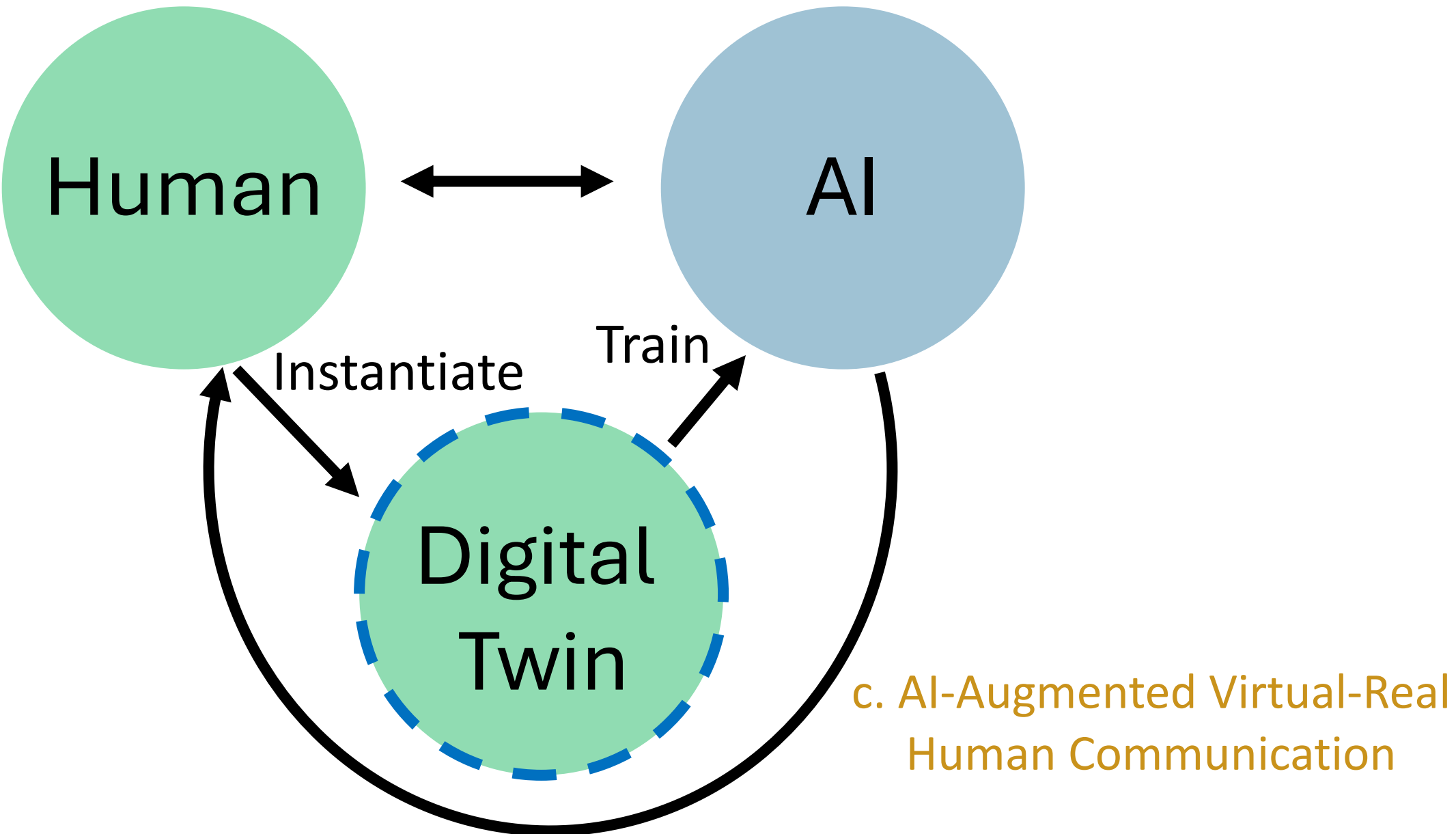
PeerEdu: Understanding Student Learning Behaviors Augmented from Peer Feedback

Xu et al. ACM CHI 2025.

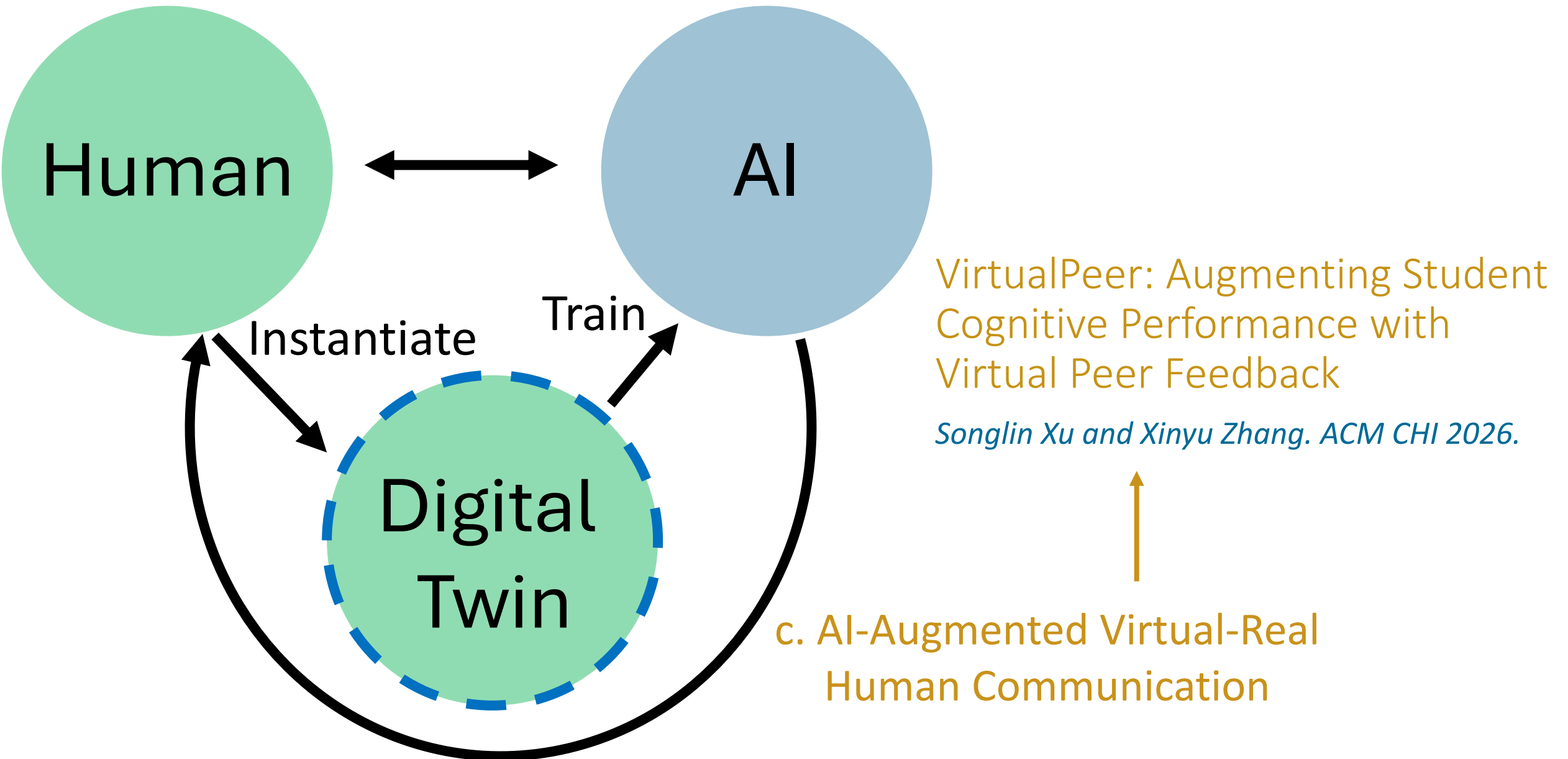
Overview



Overview

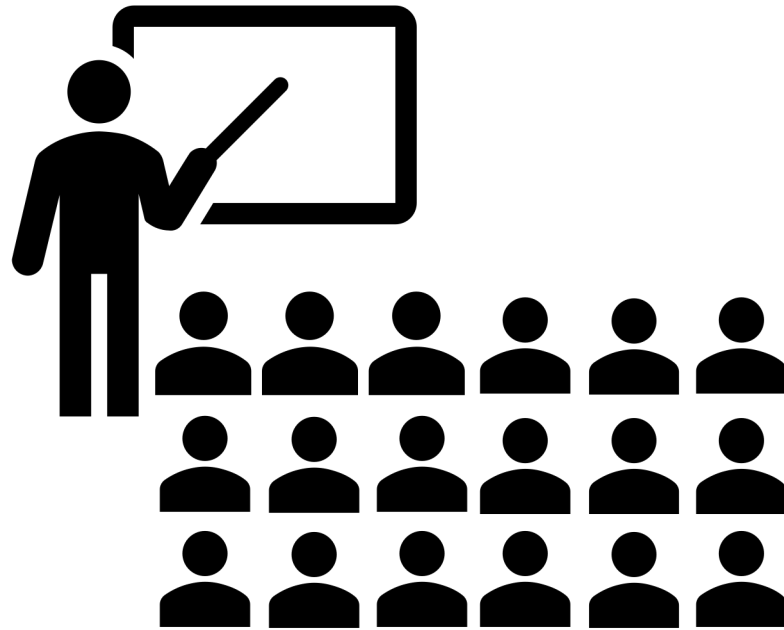


Overview



Real Peers May Not Be Available

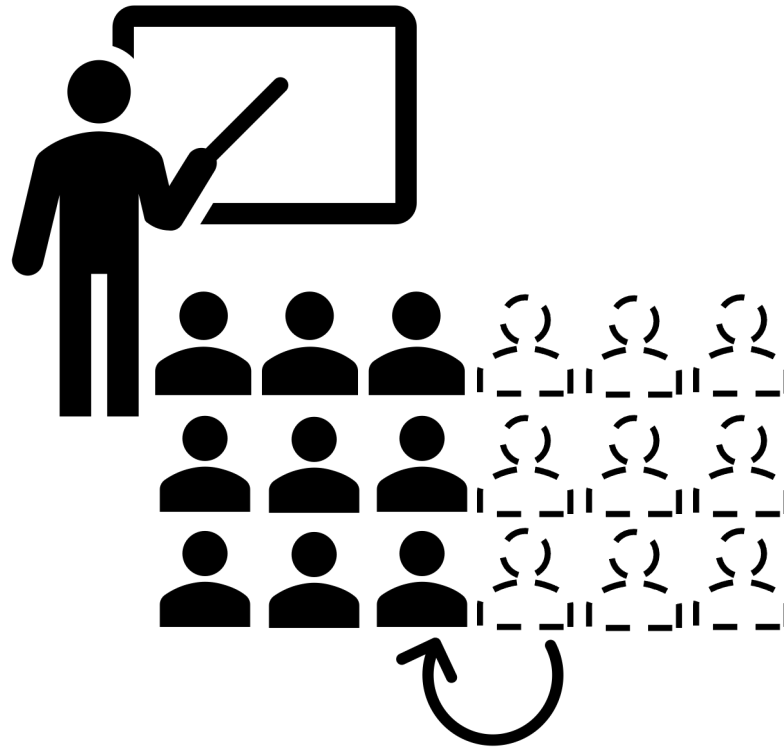
MOOCs (Massive Open Online Courses)



Peer students are not synchronized.

Virtual Peer Feedback: Peer Digital Twin

MOOCs (Massive Open Online Courses)



LLM-powered virtual peers to support real students.

VirtualPeer: Augmenting Student Cognitive Performance with Virtual Peer Feedback

Songlin Xu, Xinyu Zhang. ACM CHI 2026.

Research Question

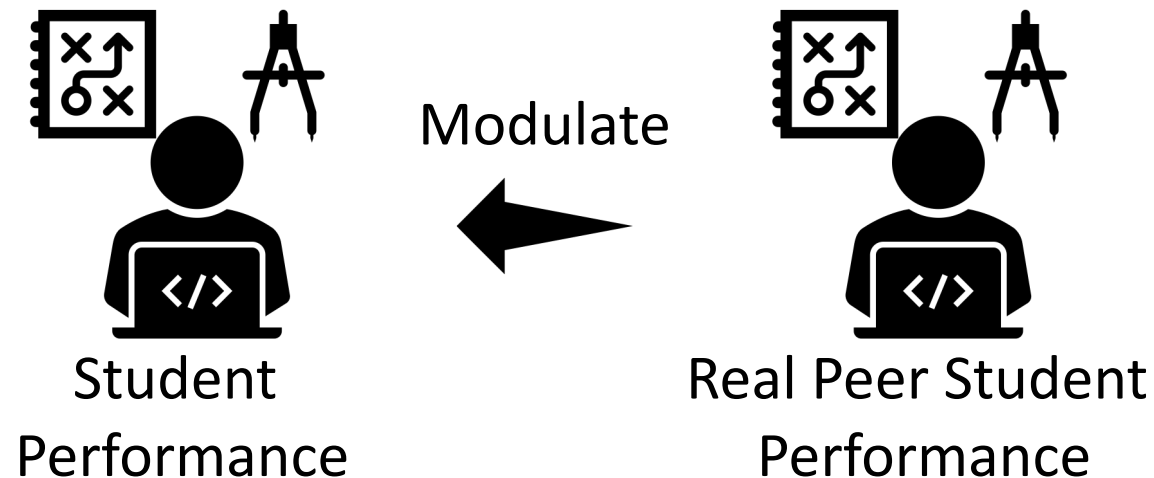
- (a) LLMs mimic real peers to generate responses as peer feedback;
- (b) Augment students' problem solving, in addition to learning (PeerEdu).

VirtualPeer: Augmenting Student Cognitive Performance with Virtual Peer Feedback

Songlin Xu, Xinyu Zhang. ACM CHI 2026.

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- (a) LLMs mimic real peers to generate responses as peer feedback;
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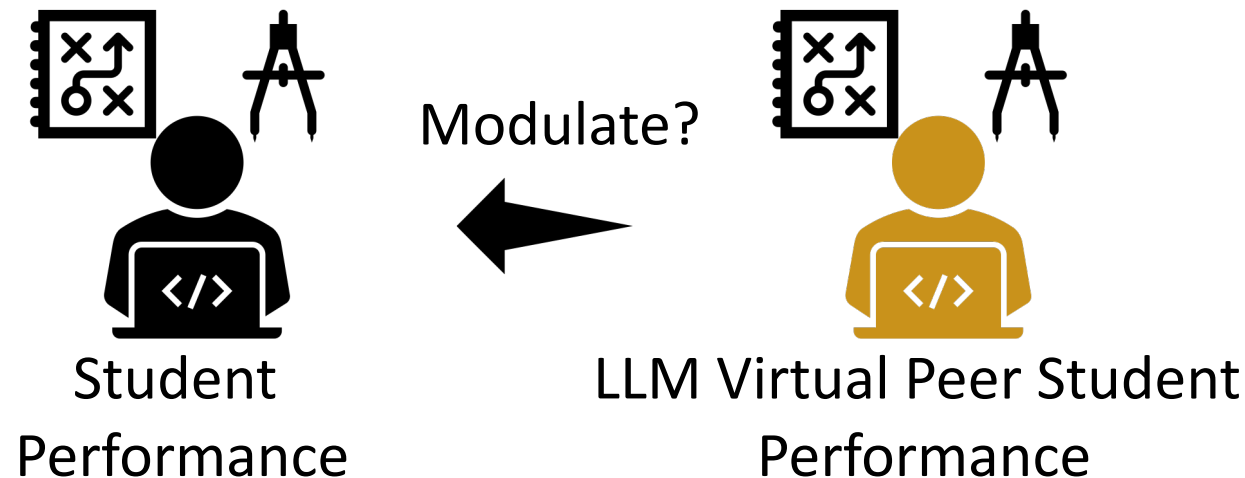


VirtualPeer: Augmenting Student Cognitive Performance with Virtual Peer Feedback

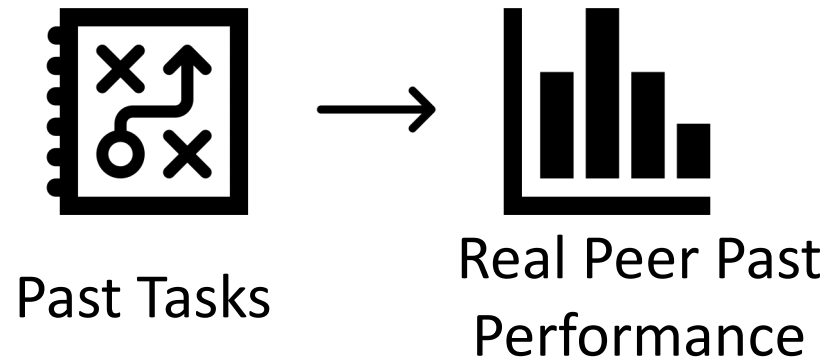
Songlin Xu, Xinyu Zhang. ACM CHI 2026.

Research Question

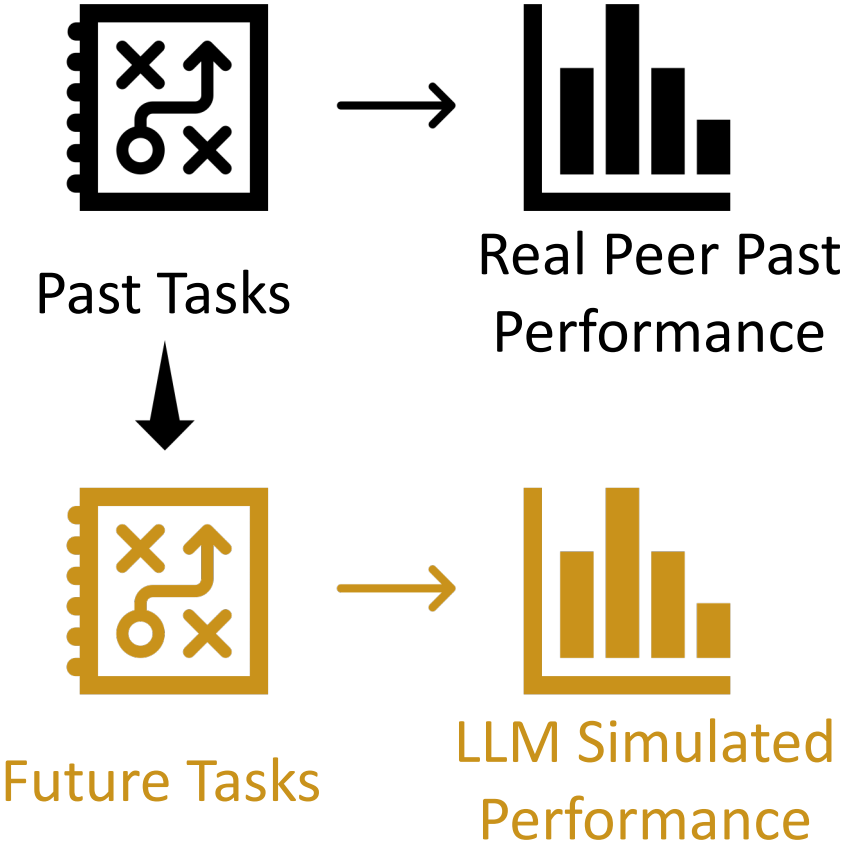
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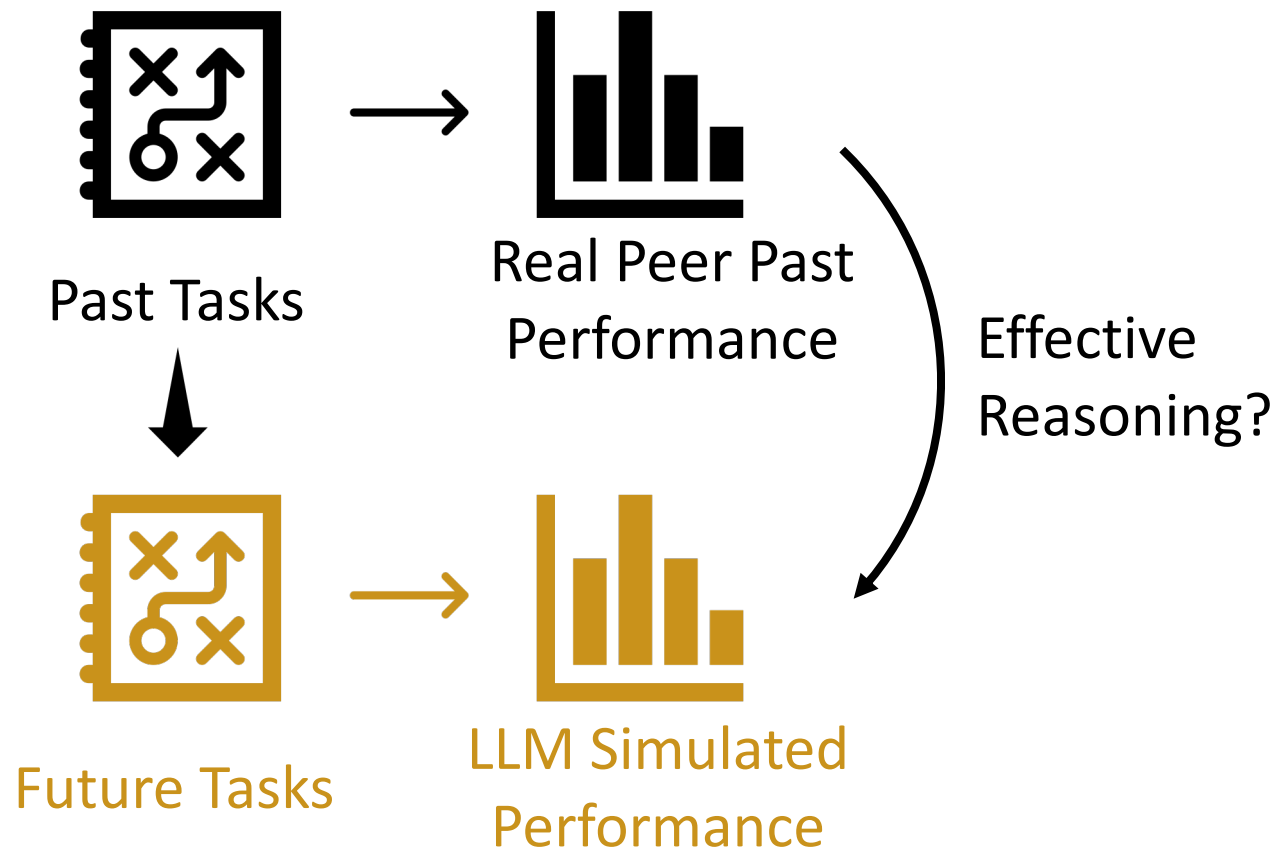
Simulation Scenario



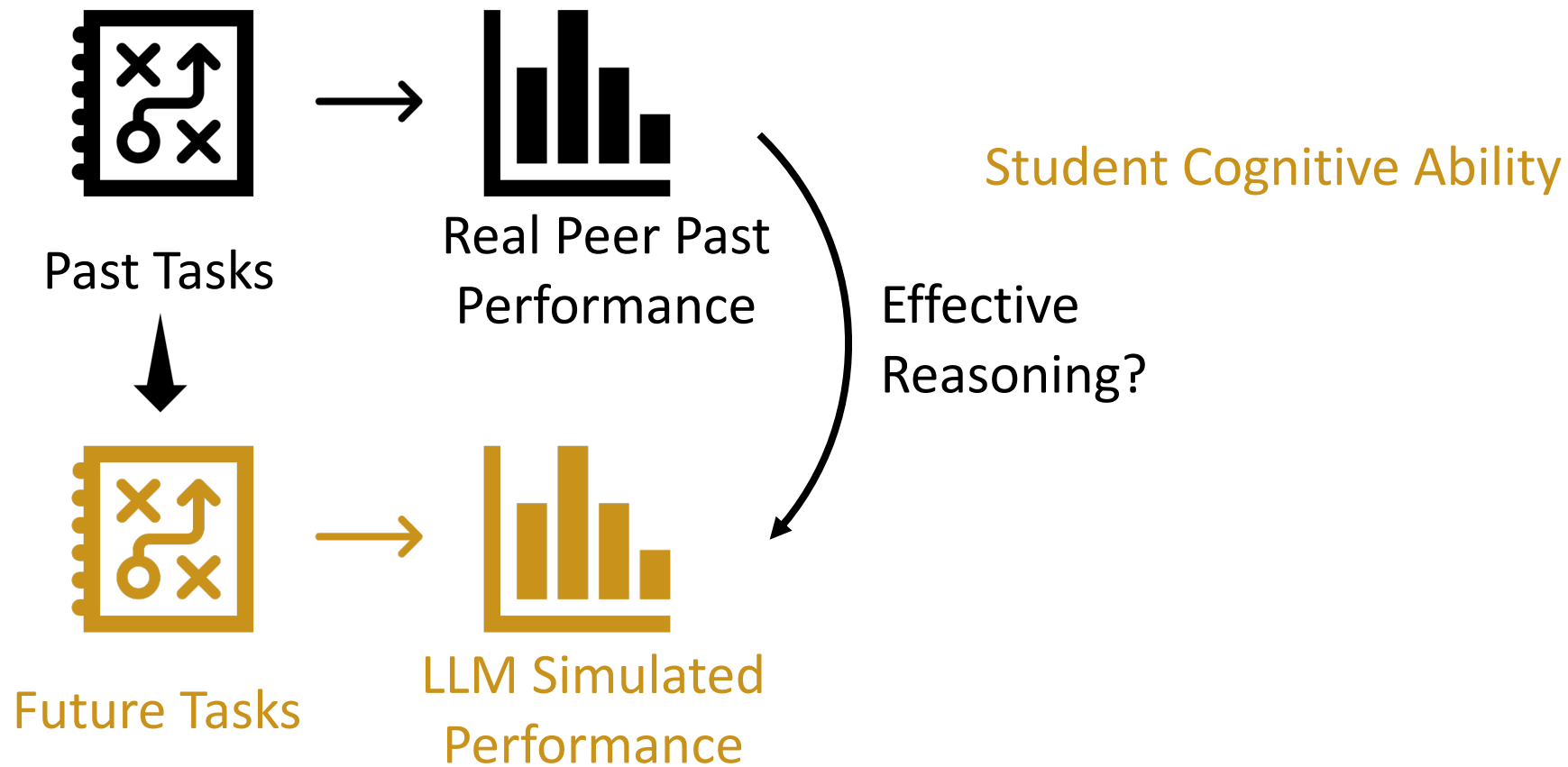
Simulation Scenario



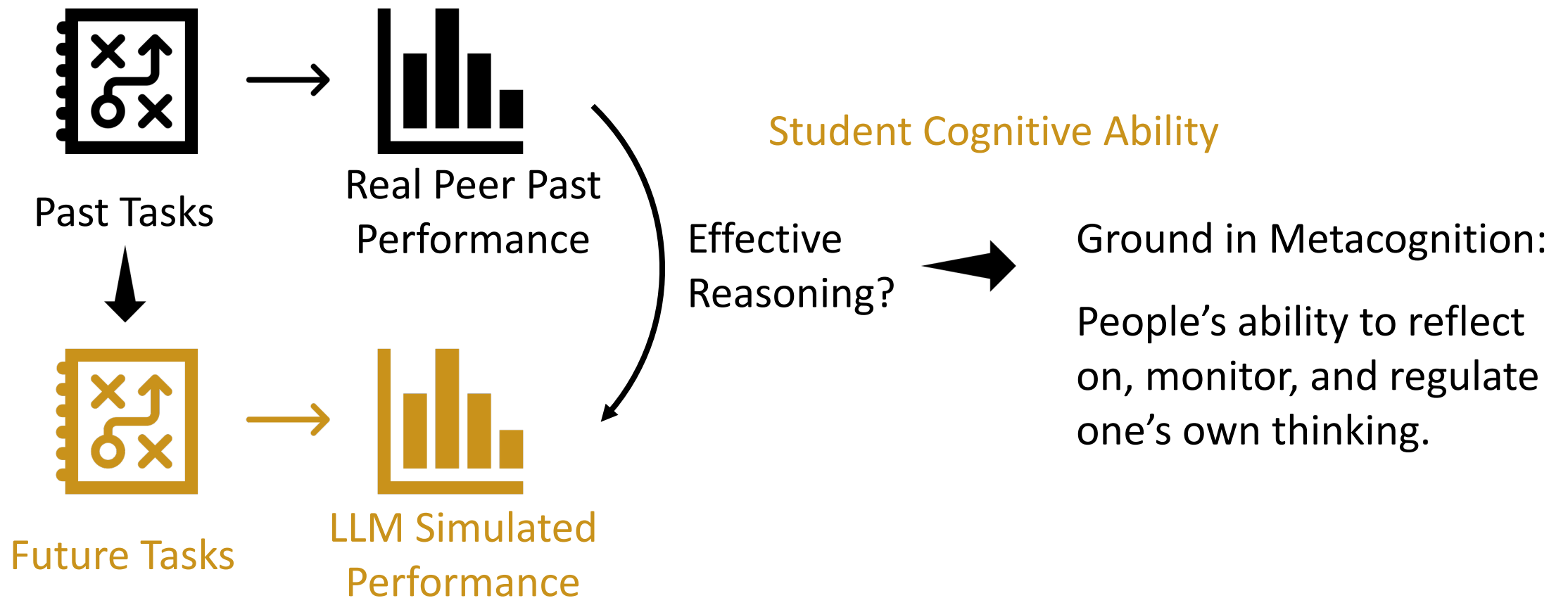
Simulation Scenario



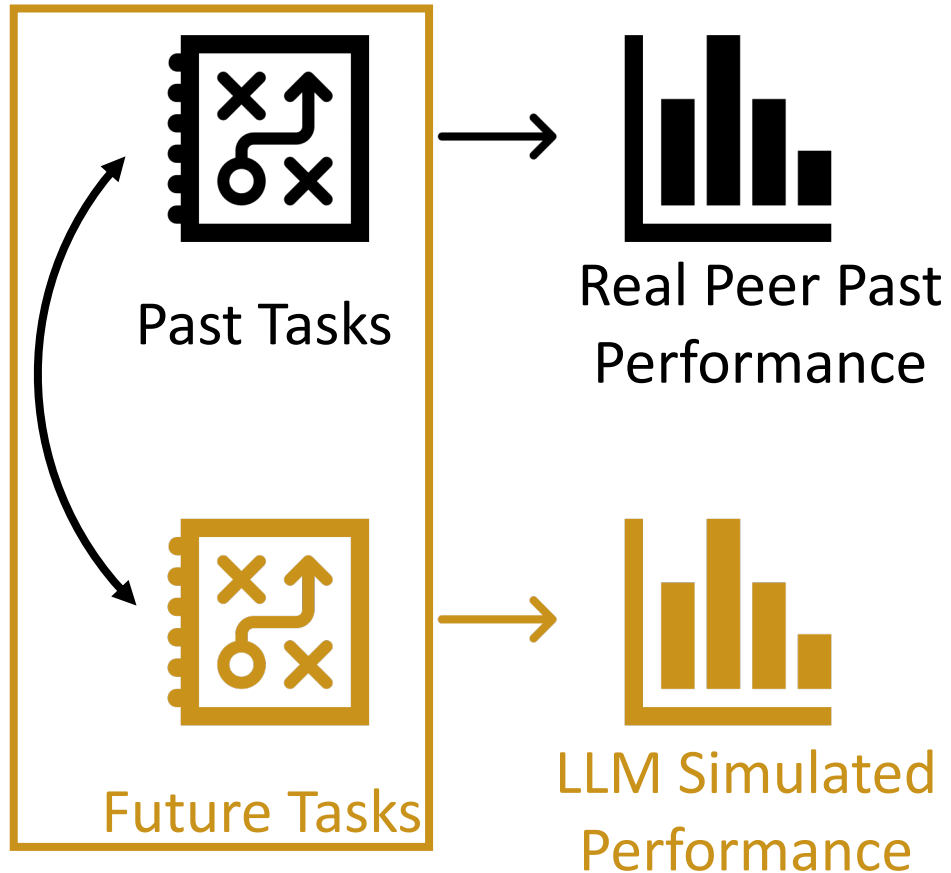
Simulation Scenario



Simulation Scenario

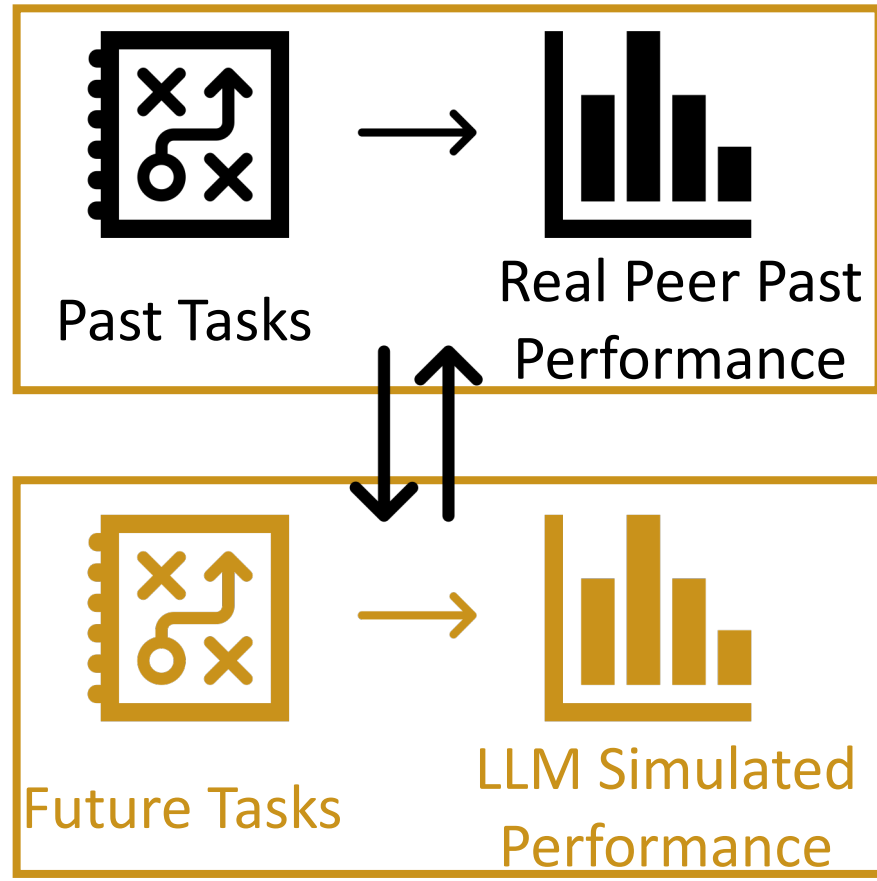


Dual-Layer Metacognitive Refinement



Layer 1:
Metacognitive Strategy in
Student Problem Solving

Dual-Layer Metacognitive Refinement

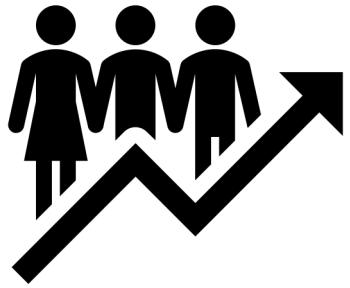


Layer 2:
Metacognitive Strategy in
LLMs' Own Reasoning during Simulation

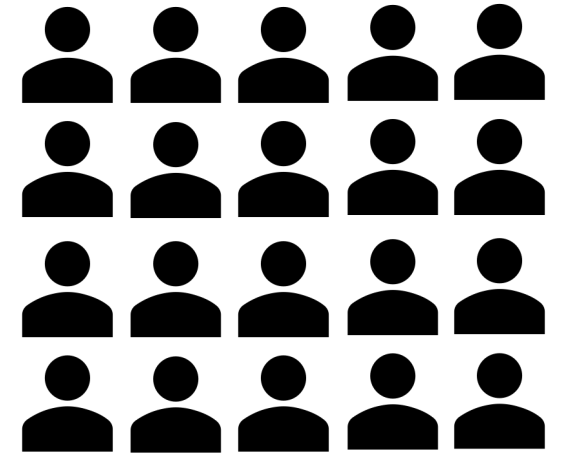
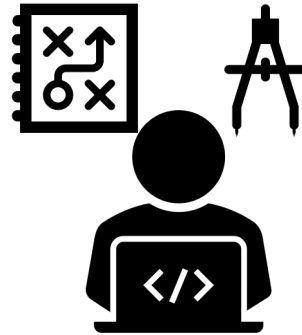
Simulation Accuracy: Ours v.s. Baselines

Mean Absolute Error (MAE)				
Model		Coef.	z	P> z
Intercept		3.225	10.962	0.000
Ours is smaller than {	Standard	0.138	2.524	0.012
	CoT	0.188	3.428	0.001
	Self-Refine	0.578	10.555	0.000
	PoT	0.153	2.794	0.005

Evaluation to Augment Student Performance



+



Virtual Peer Progress

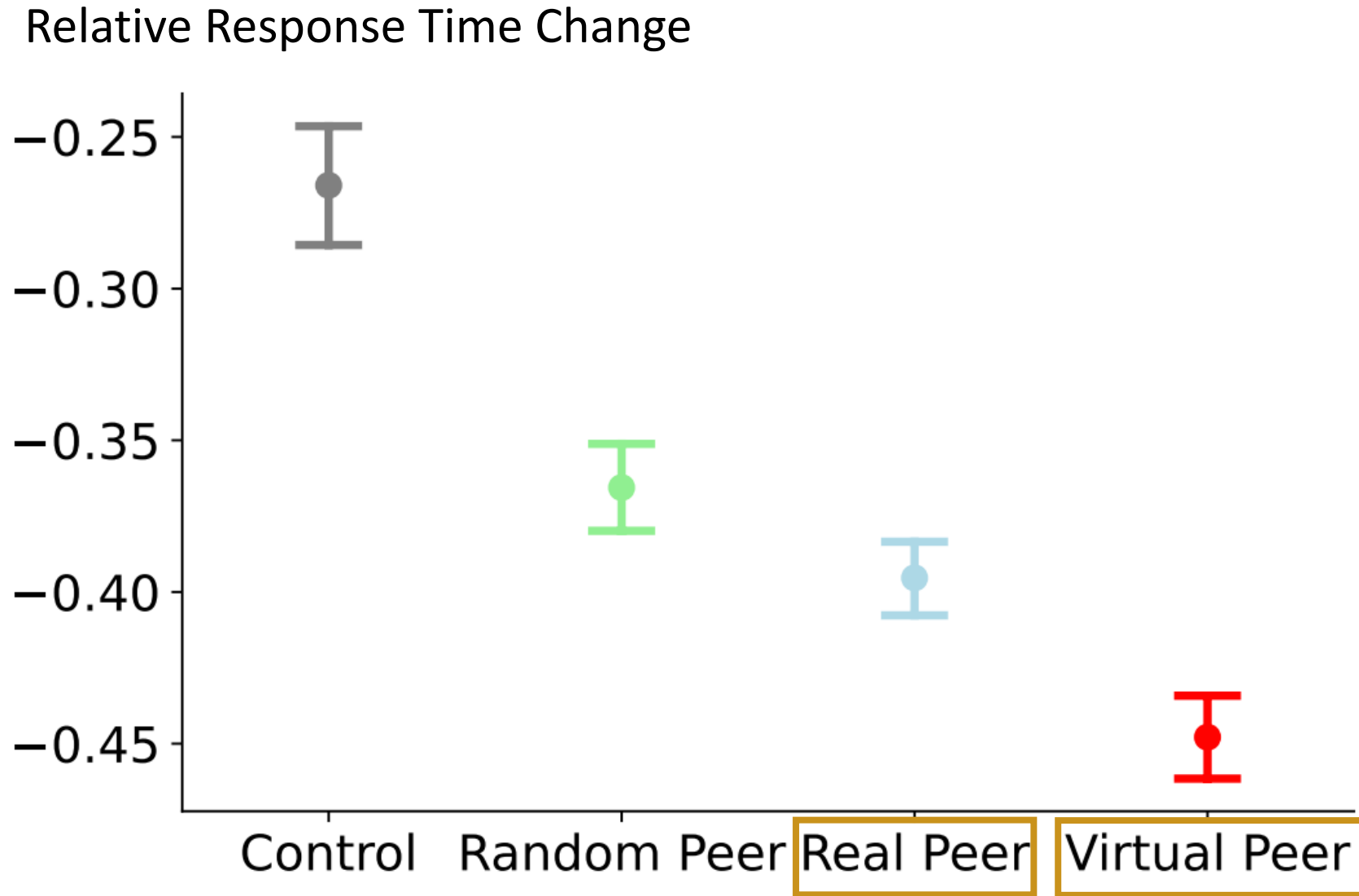
v.s.

Control, Random, Real Peer

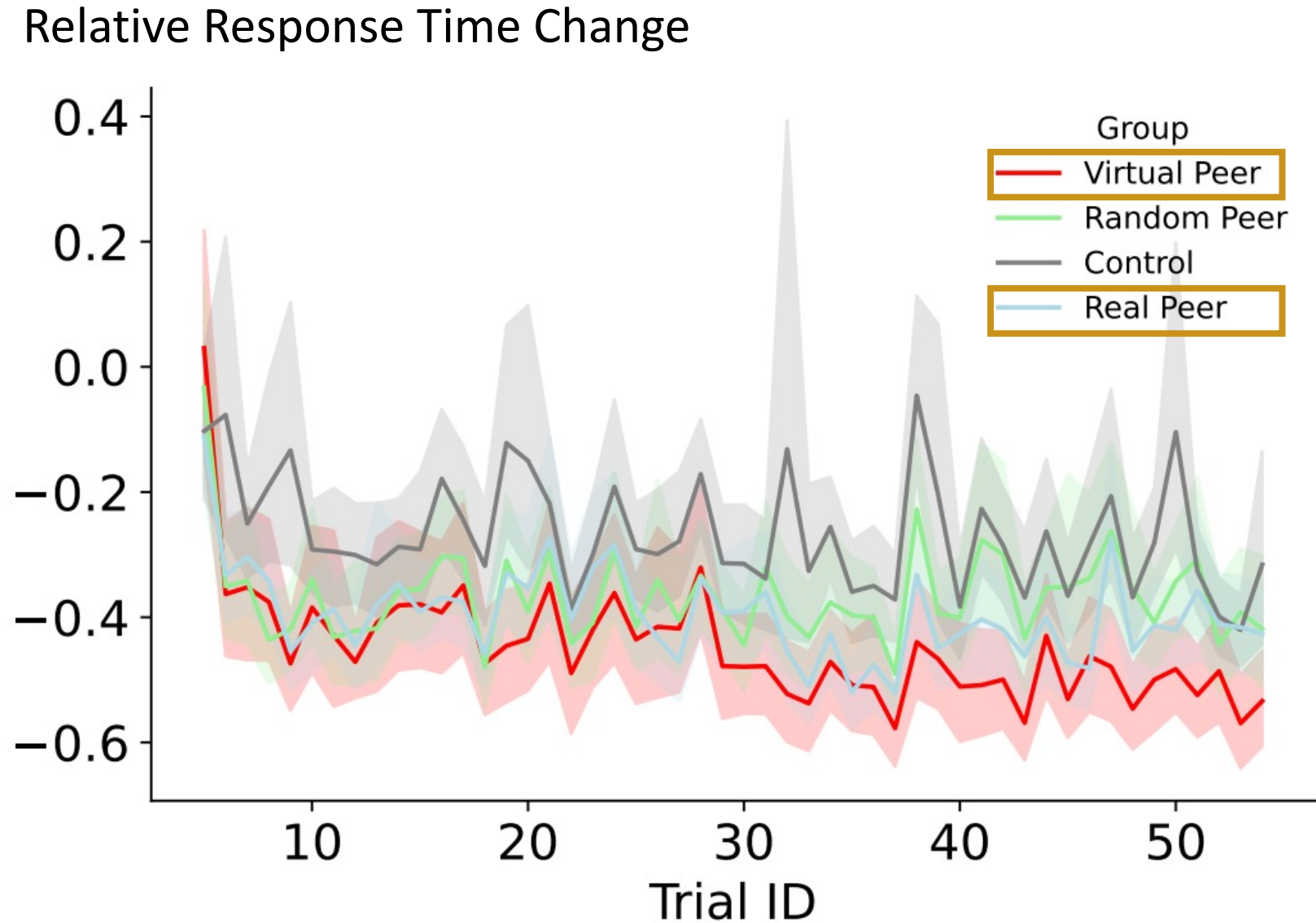
Logical Reasoning Tasks

N = 188 User Study

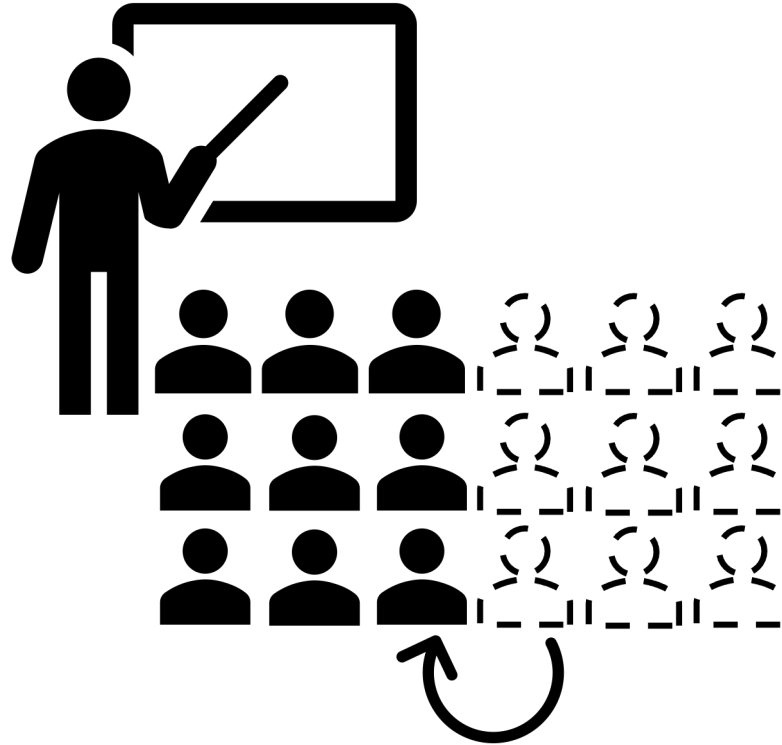
Evaluation: Response to Solve Tasks Correctly



Evaluation: Response Time Change over Time

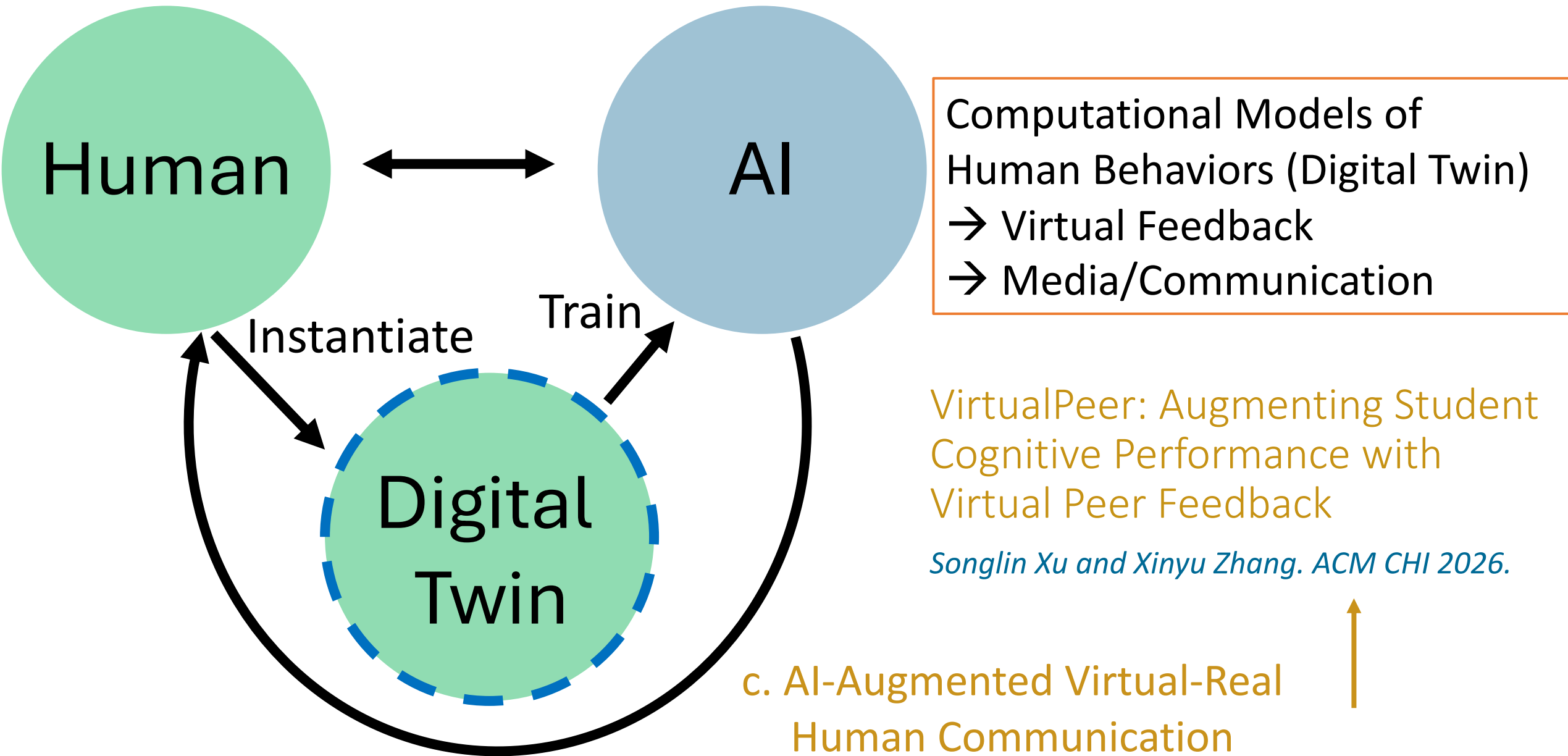


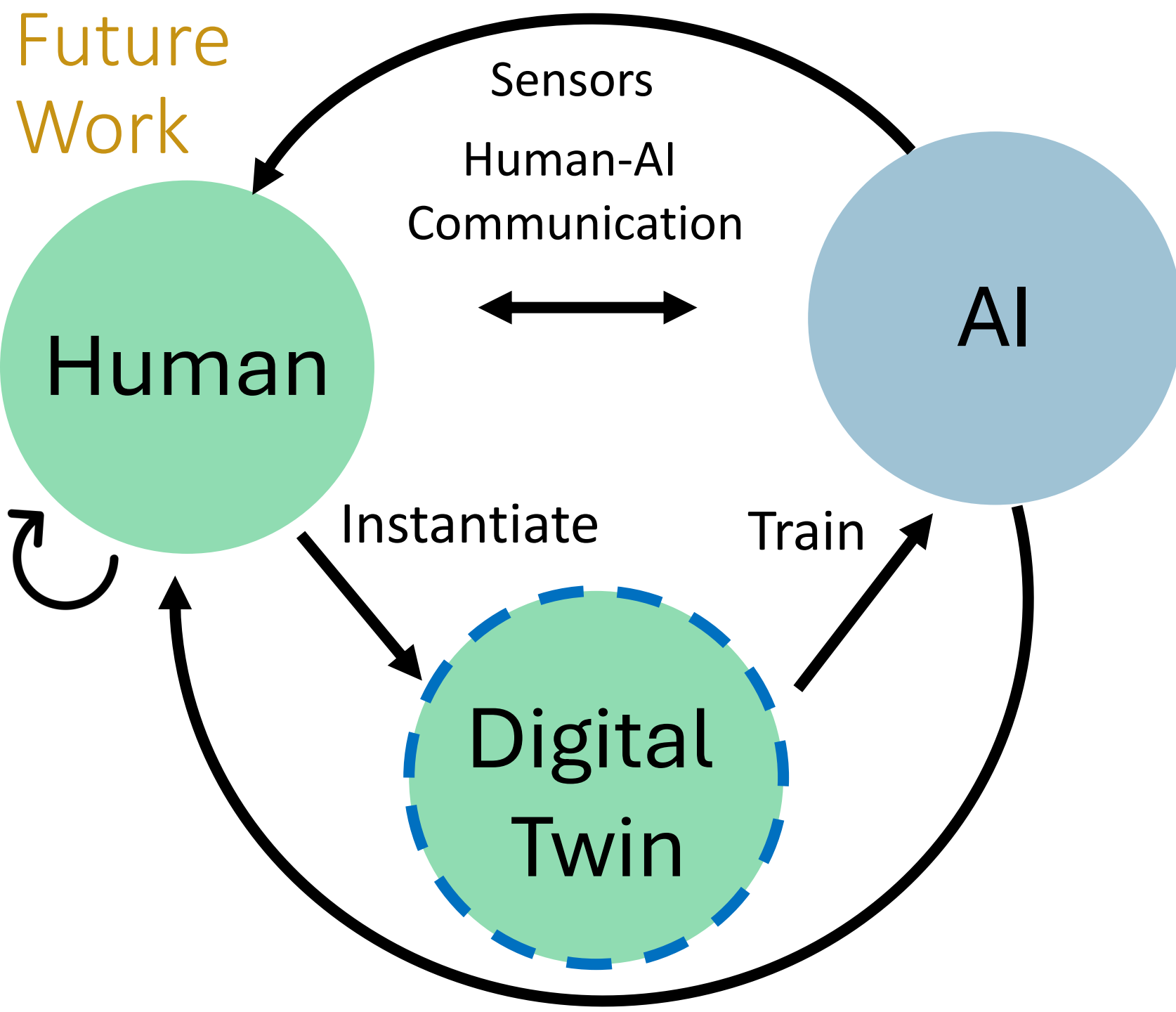
Virtual Peer Feedback: Peer Digital Twin

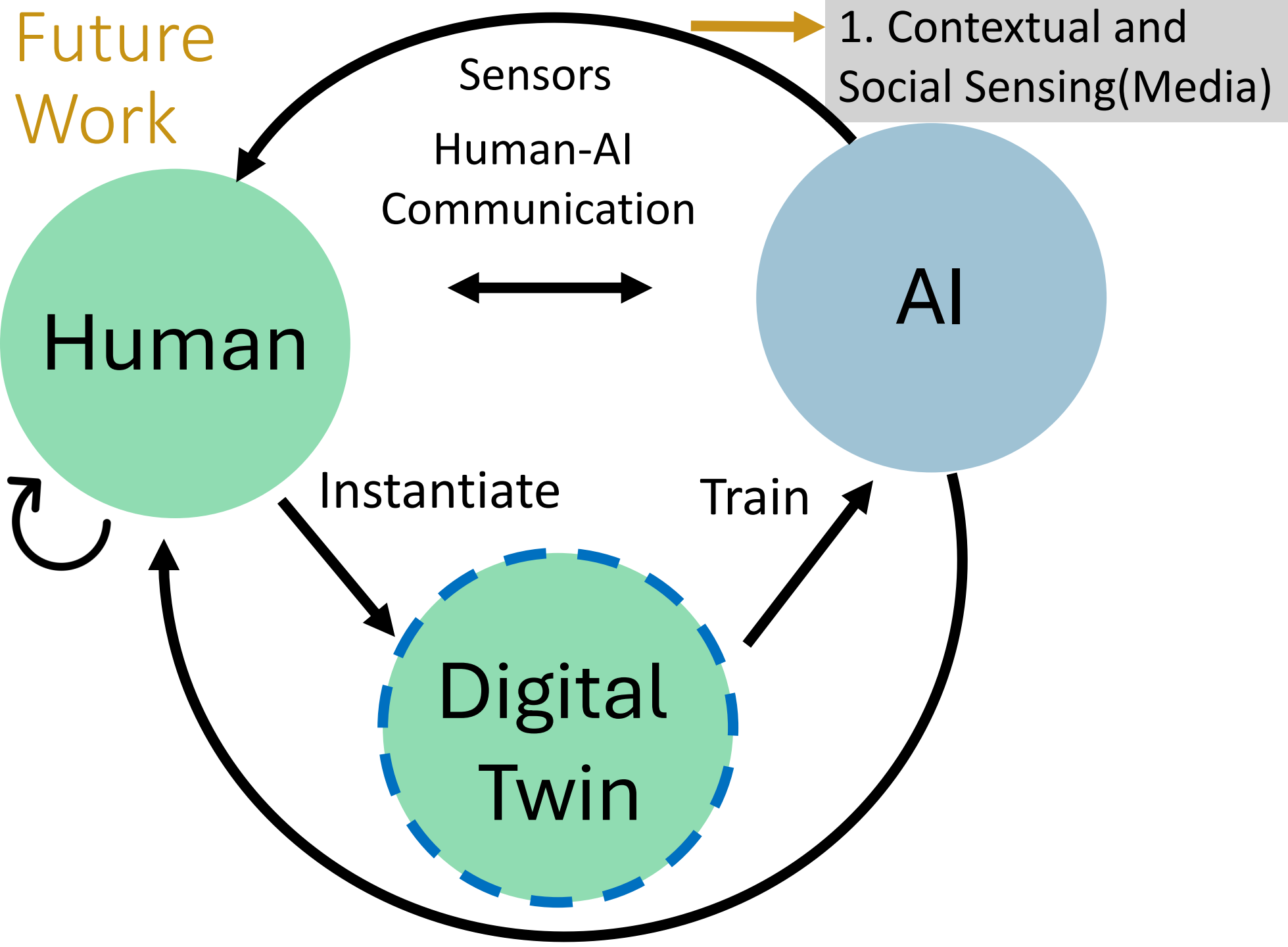


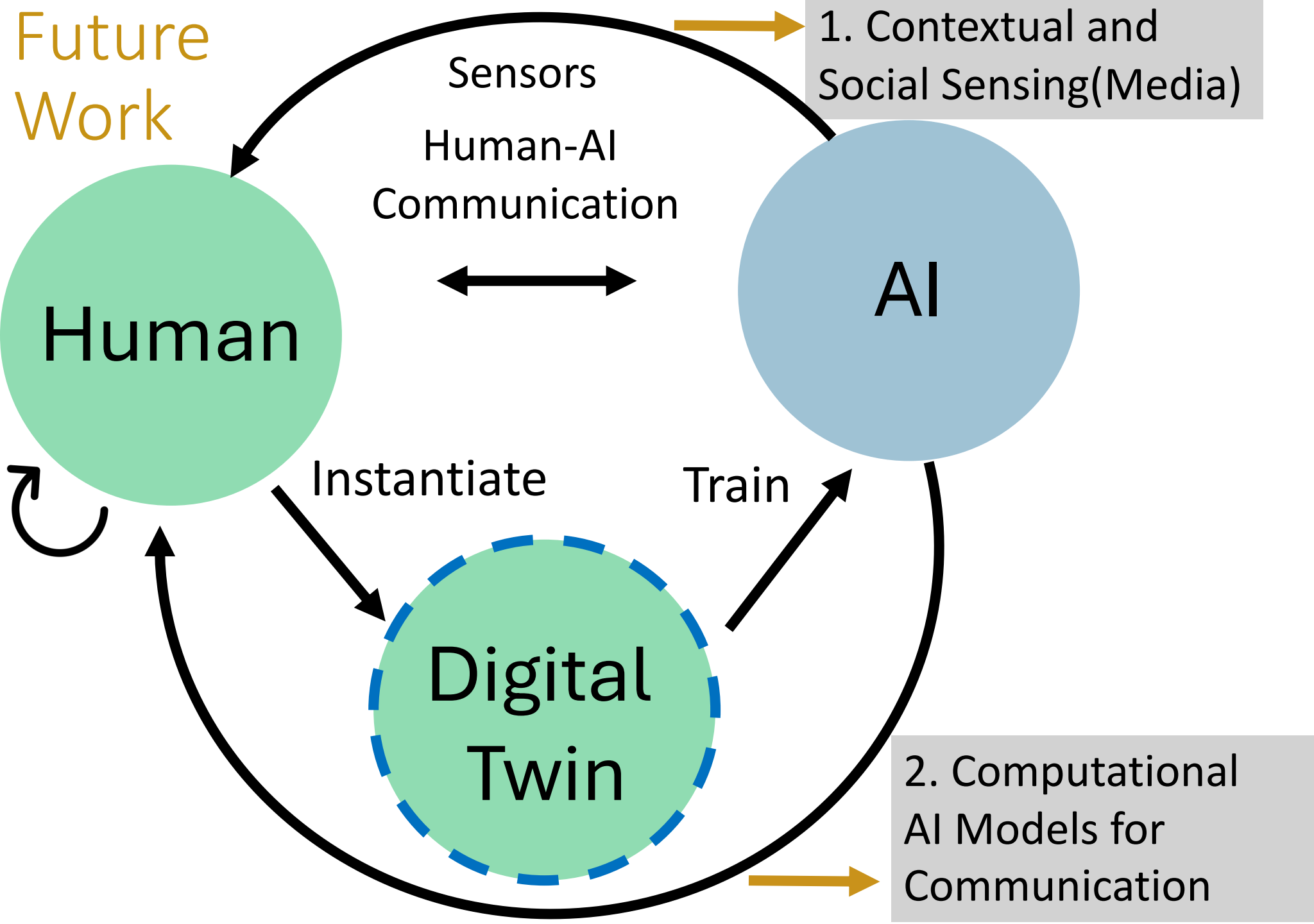
More **accessible** human augmentation when real peers are not available to deliver feedback.

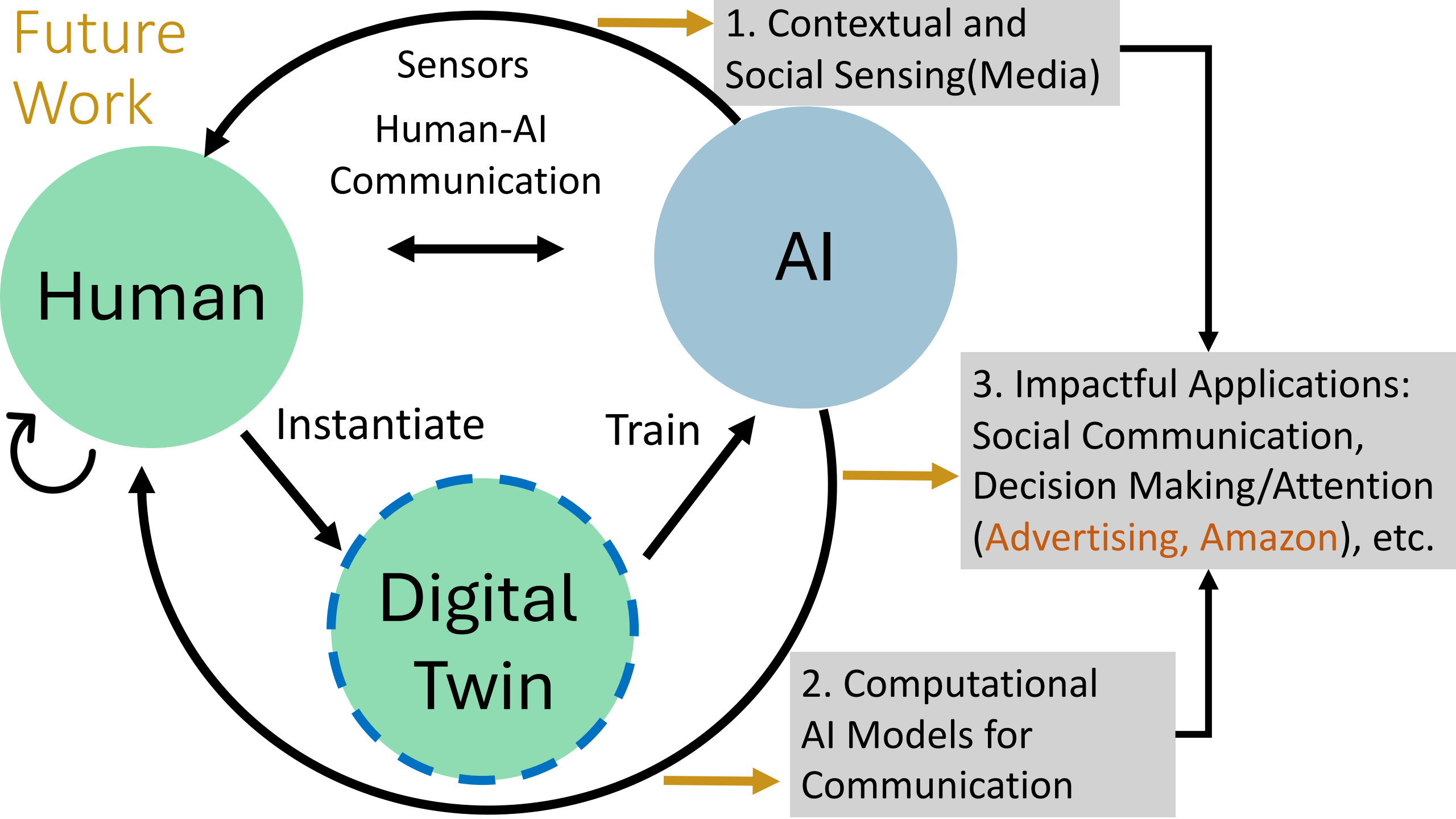
Insight



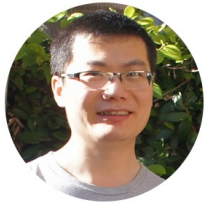








Thanks to my mentors and collaborators



Xinyu Zhang
UC, San Diego



Lianhui Qin
UC, San Diego



Xing-Dong Yang
Simon Fraser University



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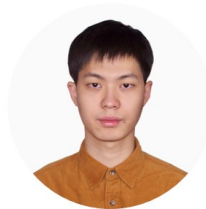
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